Aero-Opt: CFD-Based Aerodynamic Optimisation for Sustainable Engineering

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by

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Abstract

Aviation is a significant contributor to global CO_2 emissions. Fuel burn is proportional to lift-to-drag ratio, and small improvements in the lift-to-drag ratio can result in large savings in emissions over the lifetime of an aeroplane. This thesis uses Ansys Fluent and its Adjoint solver to achieve two goals: To optimise the 4 selected airfoils across 5 cases, and to determine the ideal optimisation pipeline that saves resources and man-hours.

Four airfoils were tested: NACA 0012, NACA 2412, BACXXX, and NASA SC(2)-0714, over five cases in subsonic, transonic, and supersonic flow conditions. They were verified against published data, then optimised for real-world conditions. All five showed improvements in lift-to-drag ratio, with BACXXX completely eliminating the transonic shock wave in one case.

The previous literature has used Ansys Adjoint for select cases, but has not been automated and compared across subsonic, transonic, and supersonic conditions. This study included the development of an automated process using PyFluent, a Python library designed for Ansys Fluent automation, to easily compare parameters using the same boundary conditions.

The projected impact of the improved lift-to-drag ratio was measured based on the Cessna 172 and Boeing 747-400, projecting a significant reduction in CO_2 emissions, saving up to 1.91 million tonnes of CO_2 per year.

The supplementary trials included identifying the best volume mesh method (Polyhedra), ideal y^+ values (in 2D and 3D), the best solver (Coupled), and the ideal turbulence model ($k-\omega$ GEKO). For resource-efficiency, the optimisation was completed in 2D, then verified in 3D. All optimisation was single point. Multi-point optimisation was out of scope and proposed as future work.

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Acknowledgements

Firstly, I, the author of this project, thank God for the strength, wisdom, and resolve required to complete this project with all its difficulties.

Furthermore, I am grateful to my girlfriend for believing in me when I did not.

I would also like to thank my family for their support, and my friends for keeping me company in the library.

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Nomenclature

c – Chord length, m

C_d – Drag coefficient

C_f – Skin friction coefficient

C_l – Lift coefficient

C_s – Speed of sound, m/s

g – Acceleration due to gravity, m/s^2

h – Altitude, m

M – Mach number

M_{air} – Molar mass of air, kg/mol

P – Static pressure, Pa

P_0 – Pressure at sea-level, Pa

R^* – Universal gas constant, $J/(mol * K)$

Re – Reynolds Number

R – Range, m

S – Sutherland constant, K

t – Time, s

T – Temperature, K

T_0 – Temperature at sea-level, K

U_∞ – Free-stream velocity, m/s

u^* – Friction velocity, m/s

V – Flow velocity, m/s

W – Weight, N

y – First-layer height, m

y^+ – Dimensionless wall distance

α – Angle of attack, °

γ – Adiabatic index

λ – Temperature lapse rate, m^2/s^3

μ – Dynamic viscosity, $Pa * s$

μ_0 – Reference dynamic viscosity at T_0, $Pa * s$

ν – Kinematic viscosity, m^2/s

ρ – Density of air, kg/m^3

τ – Wall shear stress, Pa

$2D/3D$ – Two/Three-dimensional

AoA – Angle of attack

BACXXX – Boeing Airfoil

CFD – Computational Fluid Dynamics

CPU – Central Processing Unit

DNS – Direct Numerical Simulation

GEKO – Generalised $k - \omega$ (turbulence model)

GUI – Graphical User Interface

NACA – National Advisory Committee for Aeronautics

NASA – National Aeronautics and Space Administration

RANS – Reynolds-Averaged Navier-Stokes

SA – Spalart-Allmaras (turbulence model)

SC – Supercritical

SST – Shear-Stress Transport (turbulence model)

TSFC – Thrust-Specific Fuel Consumption

1 Introduction

Aviation accounts for roughly $2 - 3\%$ of all global emissions [1]. The fuel consumption required to fly is tied to the aerodynamic drag of an aeroplane, as well as its weight. Minute changes in the drag or the lift coefficient can have a large impact over the aircraft's operational lifetime [2] [3]. Therefore, a good way to decrease emissions from aviation is to reduce the in-flight energy demand.

This project utilises simulations to optimise the aeroplane designs by increasing the lift-to-drag ratio, thus reducing the energy consumption during flight [4]. To limit the scope of the project, the improvement is achieved by studying and optimising the 2D cross-sectional area of the wing, also known as the airfoil. Airfoils are one of the most well-researched parts of the aeroplane, with decades of research available from sources like NACA.

Furthermore, the project aims to optimise not only the airfoils, but the optimisation process itself, by outlining the optimal process to save man-hours and computing power.

CFD, also known as computational fluid dynamics, is a way of simulating fluid flow. Ansys is an industry-standard engineering simulation software. Ansys Fluent is a finite-volume CFD solver, chosen for its reliability and widespread use, but most importantly, because of its built-in Ansys Fluent Adjoint solver, an adjoint-based optimisation tool.

Four airfoils were analysed at subsonic, transonic, and supersonic flow conditions. The airfoils were first tested unmodified against a published paper to verify the validity of the solver setup. Next, the airfoils were tested at selected real-world scenarios to get a baseline performance. Lastly, all cases were optimised using adjoint, with constant-length and variable-length trials.

Since the number of simulations required was large (four airfoils across five cases, a verification trial, a real-world trial, then two optimisation cases each), and since yet more simulations were required to identify the ideal solver settings, PyFluent, a Python API that allows for communication with Ansys

Fluent was used. Using the code, it is possible to compare different solver settings, flow speeds, or boundary conditions. Three scripts with a total of three functions were developed, each completing one part of the CFD optimisation process: creating the mesh, completing the initial simulation, and optimising the design. The functions each exported the appropriate result files and pictures.

2 Literature Review

The literature review presented in this section provides a historical overview and reviews previous studies on airfoil design, physics of flow, and CFD methods.

2.1 Flow Characteristics

2.1.1 Viscosity

Viscosity is the resistance to sliding between the layers of a fluid, or the rate of change in shear strain of a fluid [5]. Viscosity affects the force required to move through a fluid, and moving through high viscosity fluids, like oil, takes more force than through low viscosity fluids, like air. Viscosity can be specified as dynamic viscosity (also known as absolute viscosity), or kinematic viscosity [5].

2.1.2 Drag

Drag is one of the largest sources of energy waste in aerospace applications. It can be categorised into multiple sources:

- ***Skin friction drag**: Generated by viscous shear forces on the body. Taken as zero in inviscid flow [5].*
- ***Pressure drag**: Generated by the forces due to pressure on the airfoil. Can be further broken down into:*
    - ***Induced drag**: Caused by the vortices created at the wingtip [5].*
    - ***Wave drag**: Generated by shock waves during transonic or supersonic flight [5].*
    - ***Form drag**: Created by the low-pressure wake left behind the object. Affected most by the shape of the object, hence the name [5].*

Much of the drag in subsonic airfoils comes from skin friction. For transonic and supersonic airfoils, however, wave drag becomes increasingly prominent [5].

2.1.3 Lift

An airfoil generates lift by relying on Bernoulli's Theorem. On cambered airfoils, the air flowing along the upper surface speeds up and creates an area of low pressure [5]. On symmetrical airfoils, there is no lift at 0 angle of attack (AoA). However, with a slight angle of attack they can generate lift.

2.1.4 No-Slip Condition

The no-slip condition states that due to viscosity, the velocity in a flowing fluid is not uniform. Instead, the layer of a fluid closest to a surface or a wall is stationary, with the velocity increasing away from the zero velocity layer [6].

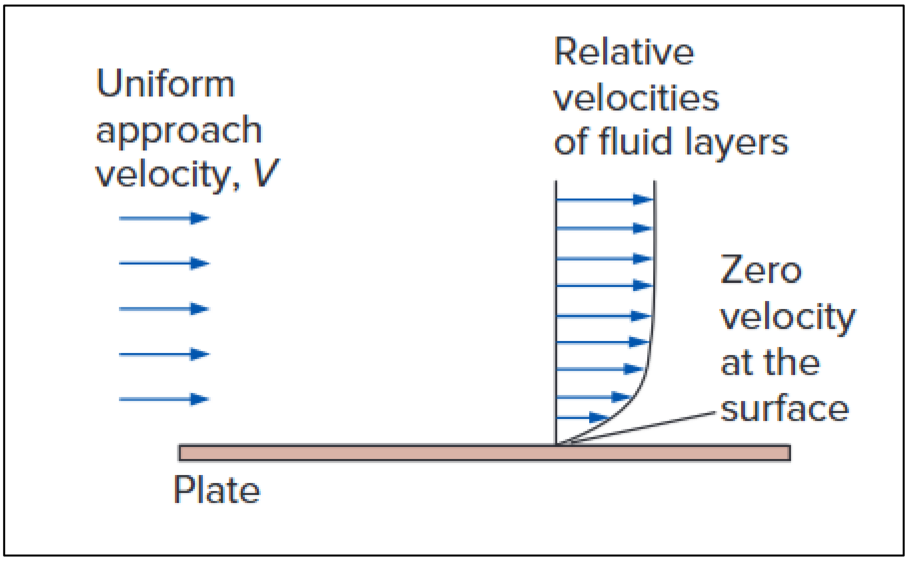

Figure 2-1 - Fluid going over a stationary surface. The relative velocity profile due to the no-slip condition is shown [6]

The flow region closest to the wall is called the boundary layer. Sometimes, when flowing over a curved surface, or when shock waves are present, the boundary layer might detach. This is called flow separation [6].

2.1.5 Laminar vs Turbulent Flow

A laminar flow is one in which the fluid layers do not exchange mass and simply glide smoothly over one another. A turbulent flow is one with more random motion, where the velocity profile is not smooth and there is an exchange of mass between the layers of the fluid [5].

Whether the flow is laminar or turbulent depends on the geometry, surface, roughness, velocity, temperature, and fluid properties. In 1880, Osborne Reynolds derived a formula to characterise the flow regime using a dimensionless number called the Reynolds number [6]. A lower Reynolds number usually signifies laminar flow, with higher numbers signifying turbulent flows.

2.1.6 Mach Number - Subsonic, Transonic, and Supersonic Flows

A flow with constant density is considered incompressible. In nature, there are no completely incompressible flows, but it is possible to model most aerodynamic simulations as incompressible. However, the effects of compressibility play an increasingly important role at higher speeds [7].

The ratio of the flow velocity over the speed of sound in that medium is called the Mach number. A Mach number of less than 1 is considered subsonic, with a Mach number higher than 1 considered supersonic (if the Mach number is equal to 1, the flow can be described as sonic) [7]. However, the flow over the upper surface of an airfoil speeds up. Because of this, the air travelling over the top surface of an airfoil can be supersonic, even if the airfoil itself is travelling at Mach 0.8. Similarly, the airflow on the bottom surface of an airfoil might be subsonic even if the airfoil is travelling at Mach 1.2. A flow where subsonic and supersonic regions are both present is called transonic flow [7].

2.1.7 Subsonic Flow

In subsonic flow, all air is moving slower than Mach 1. At this speed, the air can be approximated as incompressible. An airfoil in subsonic flow has smooth streamlines, with lower pressure on the upper surface, shown in Figure 2-2.

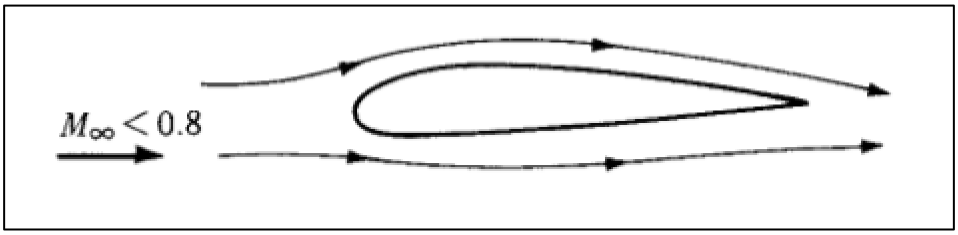

Figure 2-2 - Subsonic flow over an airfoil [7]

2.1.8 Transonic Flow

During transonic flow, as mentioned above, the flow can become locally supersonic, even if the freestream velocity is below Mach 1, causing weak, local shock waves, shown in Figure 2-3. At transonic flows above Mach 1, an unattached shock wave, also known as a bow shock wave is formed, with a weaker shock wave generated at the trailing edge, shown in Figure 2-4 [7].

The critical Mach number is the point at which the flow on the top surface of an airfoil becomes supersonic. At this speed, there is a large increase in drag coefficient. Thinner airfoils usually have a higher critical Mach number, owing to the smaller pressure difference between the upper and lower surfaces [7].

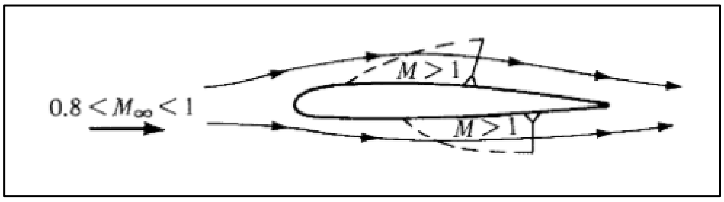

Figure 2-3 - Transonic flow below Mach 1 over an airfoil [7]

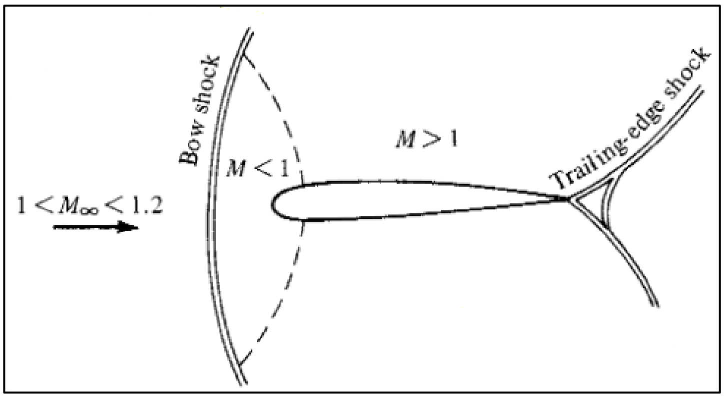

Figure 2-4 - Transonic flow above Mach 1 over an airfoil [7]

2.1.9 Supersonic Flow

In supersonic flow, air is faster than Mach 1 everywhere. At this speed, shock waves are common, causing a massive increase in drag coefficient. The point at which this increase in drag can be observed is referred to as the drag divergence Mach number [7].

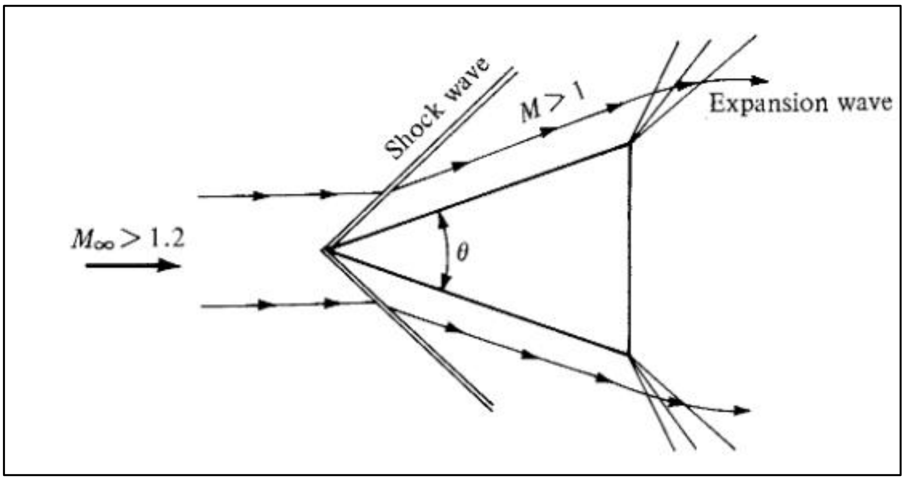

Figure 2-5 - Supersonic airflow around a wedge [7]

2.2 Airfoil Design

An airfoil is defined as a body that, at the appropriate angle to airflow, produces more lift than drag [5]. The most common airfoils are cross-sectional areas of aeroplane wings. These airfoils consist of two curved surfaces. The rounded front (with respect to direction of flow) of the airfoil is referred to as a leading edge, and the rear end as a trailing edge. The shortest line from the leading to the trailing edge is called the chord line or chord length. For standard airfoil coordinates, this is usually given as 1 metre. The thickness is measured normal to the chord line and usually expressed as a percentage of the chord. The airfoil can be symmetrical on the chord line, or it might have camber, which is an asymmetrical curvature through the body of an airfoil. The angle of attack is defined as the angle between the airfoil's chord line and the free stream velocity, as shown in Figure 2-6 [5].

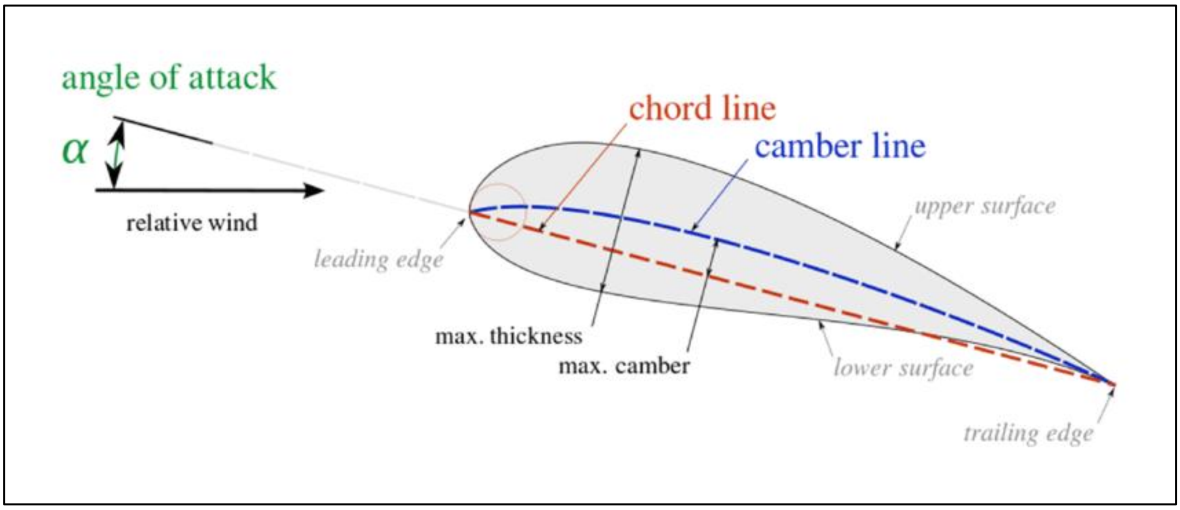

Figure 2-6 - Components of an airfoil [8]

2.2.1 NACA Airfoils

Although the American Wright brothers completed their first flight on 17 December 1903, in the United States, flying was regarded as a dangerous stunt, instead of an engineering achievement [9]. In the period between that first flight in 1903 and the beginning of World War I in 1914, many unique designs were produced, but mostly by amateur hobbyists [9]. Hence, there was little scientific foundation in most American flying machines of the time. By the start of World War I overseas, the lack of progress and organisation in American aeronautics had become increasingly obvious. Finally, on 3 March 1915, the National Advisory Committee for Aeronautics (NACA) was formed to formalise and support ongoing aeronautical research [10].

Although NACA was created at first only as an advisory committee and received little funding, they nevertheless acquired a wind tunnel in 1920, and a variable density wind tunnel in 1922. The latter allowed for control of the density of the air flowing through the wind tunnel, meaning it was possible to scale the results based on Reynolds number (more on Reynolds number below) and simulate the flow conditions more accurately [10].

Finally, by 1929, NACA had developed the now prominent system (first published in 1933 [11]) that represents the cross-section of a wing, also known as an airfoil, using a series of four numbers [10], called the NACA Four-Digit-series airfoils. Later, in 1935, the NACA 5-Digit series were introduced to describe profiles with the maximum camber shifted forward, towards the leading edge [12]. In the late 1930s, the NACA 1-series airfoils appeared, followed by the NACA 6-series, 7-series, and 8-series airfoils.

2.2.1.1 *NACA Four-Digit-Series Airfoils*

The numbering system devised for the NACA four-digit airfoils describes the airfoil geometry. The first digit denotes the maximum value of the camber as a percentage of the chord. The second digit indicates the distance from the leading edge of the airfoil to the location of the largest camber, in tenths of the chord. The final two digits specify the airfoil thickness, in percent of the chord [13]. For example, the NACA 2412 airfoil has 2% camber at 0.4 chord lengths from the leading edge and a thickness of 12% (Figure 2-7). Symmetrical airfoils will have the first two digits equal to zero, since there is no camper, such as the NACA 0012.

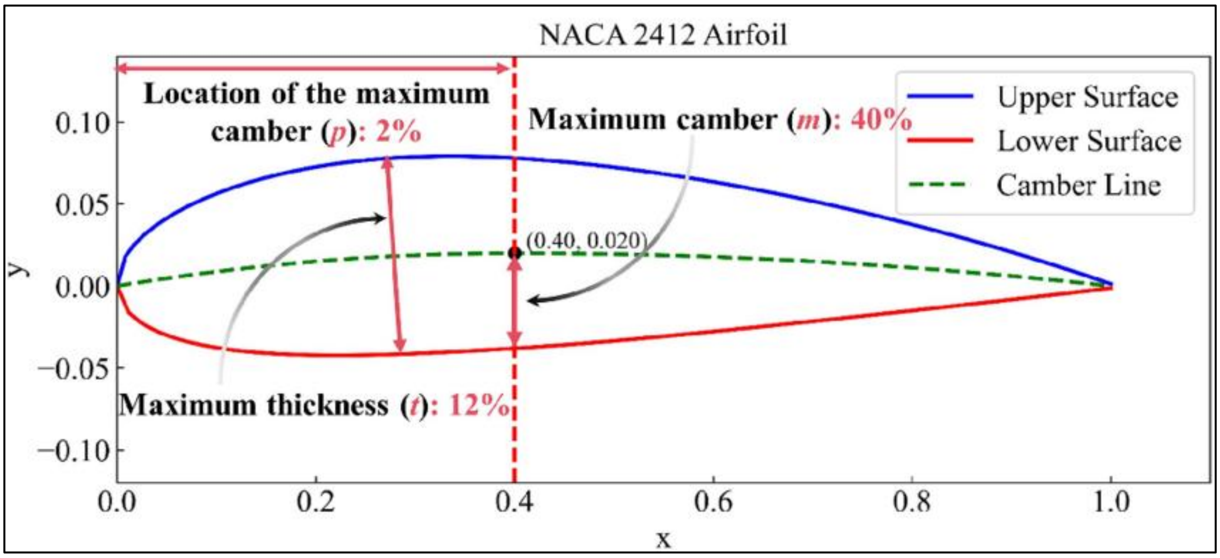

Figure 2-7 - NACA 2412 Airfoil [14]

2.2.1.2 *NACA Five-Digit-Series Airfoils*

The NACA Five-digit series airfoils were developed specifically for higher lift coefficients [13]. They follow a five-digit numbering system; the first digit, when multiplied by three-halves, indicates the design lift coefficient. The next two digits signify the distance, multiplied by two, from the leading edge to the maximum camber, in percentage. Finally, the last two digits indicate the airfoil thickness as a percentage of the chord [13] [15]. For example, the NACA 23012 airfoil has a design lift coefficient of $2 * \frac{3}{2} = 3$ tenths, or 0.3. The maximum camber is at $30 * 0.5 = 15\%$ of the chord length, and the thickness-to-chord ratio is 12%.

2.2.1.3 NACA 1-Series Airfoils

The NACA 1-series airfoils were the first to be designed specifically to increase the critical Mach number of the airfoils. It is also notable that the 1-series airfoils (sometimes also referred to as 16-series airfoils) were also the first to be developed based on airfoil theory, by first specifying the ideal pressure distribution over the chord and then calculating the shape that produced that pressure gradient. Because of this, the 1-series airfoils have relatively low lift coefficient but low drag at high speeds, with an increased critical Mach number [15]. This is achieved by shifting the point of lowest pressure (and highest lift) further back than on other airfoils [7].

NACA 1-series airfoils are comprised of five digits, the first two separated from the last three by a hyphen. The first number specifies the series designation, the second number indicates the distance from the leading edge to the point of minimum pressure, in tenths. The third number is the design lift coefficient, in tenths, and the last two digits show the thickness-to-chord-ratio [13]. For example, the NACA 16-212 is a series 1 airfoil, with the minimum pressure occurring at 0.6 chord lengths, a design lift coefficient of 0.2, and 12% thickness.

2.2.1.4 NACA 6-Series Airfoils

Like the NACA 1-series, the NACA 6-series airfoils were developed by deriving the shape based on the desired pressure distribution. For the 6-series airfoils, the goal was similar; improve the high-speed performance of the airfoils. This was achieved by maintaining an evenly decreasing pressure gradient for as long as possible, delaying the transition from laminar to turbulent flow which was found to significantly reduce skin drag. Hence, the 6-series are known for low drag, but through a smaller range of operating conditions [16]. Notably, the 6-series airfoils are more susceptible to surface imperfections [15].

The NACA 6-Series follows a complicated naming convention; for example, NACA 65_4-415 $a = 0.6$. The first number denotes the series (in this case, the 6-series), the second, 5 specifies the location of the minimum pressure in tenths of the chord, so 0.5 chords for this airfoil. The subscript, 4, shows that the low drag is maintained for lift coefficient values 0.4 above or below the design lift coefficient. The design lift coefficient is shown by the number after the dash, 4, in tenths, so $C_l = 0.4$. The final two numbers show the thickness in percentage of the chord, so 15%. The fraction shown by a is the percentage of the chord, 60% over which the pressure gradient is uniform [13].

2.2.1.5 NACA 7-Series and 8-Series Airfoils

The NACA 7-series airfoils were designed to improve the 6-series and further improve the regions of laminar flow. Similarly to the 6-series airfoils, they exhibit a very low drag coefficient over a limited range of operating conditions [15]. The naming convention includes three digits, a letter, and a further three digits. The first digit denotes the series (7 for the 7-series airfoils), the second number shows the place of lowest pressure on the upper surface of the airfoil, in tenths of the chord. The third number shows the point of lowest pressure on the bottom surface of the airfoil, likewise in tenths of the chord. The final three numbers, like the 6-series airfoils, show the design lift coefficient in tenths and the thickness, in percentage of the chord. The letter is used to distinguish between airfoils that otherwise would have the same designation (if an airfoil has the same lift coefficient, same thickness, same pressure gradient over the upper and lower surfaces, but a different mean-line) [15].

For example, the NACA 747A315 is a 7-series airfoil where the point of lowest pressure on the upper surface is at 0.4 chords, and at 0.7 chords on the lower surface, with a design lift coefficient of 0.3 and a thickness of 15% [13].

The NACA 8-series airfoils were designed for even higher speeds. The naming convention is the same as the NACA 7-series airfoils, but with an 8 as the first digit to specify the series [15].

2.2.2 NASA Supercritical Airfoils

Transonic flow introduces the problem of shock waves. The formation of a shock wave, at best, massively increases drag, and at worst, can damage the airframe. In fact, in 1941, a test pilot was killed when a shock wave from the wings damaged the horizontal stabiliser and sent the plane into an uncontrollable dive [10]. While NACA researched transonic and supersonic flight, the technology to design supercritical airfoils was lagging. In fact, the NACA 1-series and 6-series airfoils (and, to an extent, the 7 and 8 series airfoils) were an attempt to mitigate the effects of shock waves during transonic flight [17].

However, there never came to be a NACA supercritical airfoil series, as due to the rising interest in space travel, the United States was looking to establish a national space programme. NACA had expertise in missile research and a track record of working closely with the military, so on 1 October 1958, the decision was made to establish the National Aeronautics and Space Administration (NASA), and to absorb the existing NACA as the foundation [10]. Hence, supercritical airfoils (airfoils specifically designed for supersonic flows), that were first developed in 1965, were designated as the NASA supercritical airfoils [18].

The NASA supercritical airfoils feature an upper surface with greatly reduced curvature, which results in less acceleration of the flow on the upper surface, thus delaying the formation of the shock wave, as well as reducing its size once it forms. However, a flatter upper surface also results in a smaller pressure difference, and thus a smaller lift coefficient. To help remedy this, the rear part of the lower surface is concave, which makes up for the loss in lift coefficient by channelling the air downwards [18]. A comparison between a NACA 6-series airfoil and a NASA supercritical airfoil can be seen in Figure 2-8.

The early designs were successful in increasing the drag divergence Mach numbers but experienced an increase in drag just before the divergence, associated with the rear of the upper surface. These were designated as phase 1 designs [17]. The designs were later improved and designated as phase 2 and phase 3 designs.

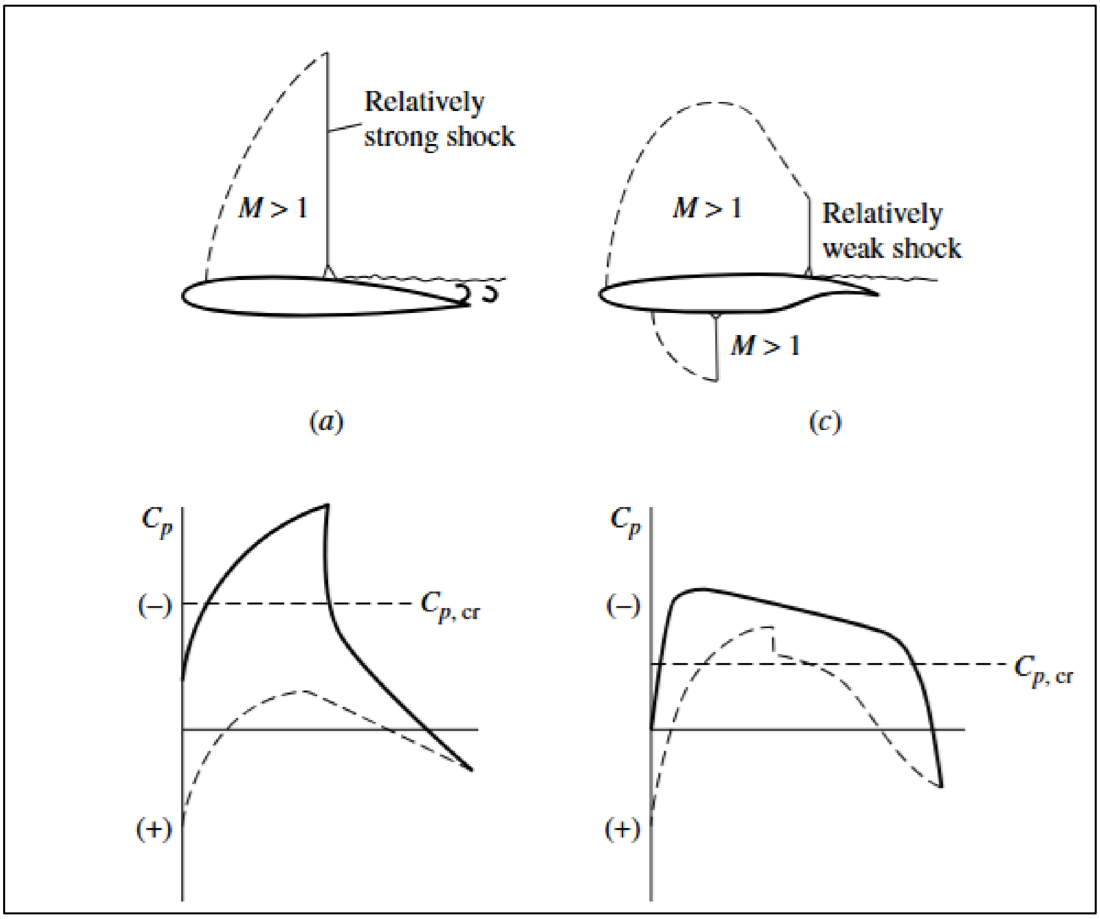

Figure 2-8 - The airflow over a NACA 6-series airfoil (a) compared to a NASA supercritical airfoil (c), with their corresponding pressure distributions [7]

The NASA supercritical airfoil naming scheme is as follows: SC, signifying supercritical, is followed by a number in parenthesis, signifying the phase, then a dash, followed by two digits signifying the design lift coefficient, followed by two digits describing the thickness, as a percentage of the chord. For example, the NASA SC(2)-0714 is a supercritical phase 2 airfoil, with a design lift coefficient of 0.7, and a maximum thickness of 14% [17].

2.2.3 Supersonic Airfoils

While the NASA SC series airfoils were successful in transonic flow, they were ill-suited to full supersonic flow. For supersonic flows, two types of airfoils are suitable: Double-wedge airfoils and biconvex airfoils. The blunt leading edge of transonic airfoils causes the formation of a bow shock at supersonic speeds. Hence, unlike all airfoils mentioned above, these designs feature a sharp leading edge instead [19].

A double-wedge airfoil usually consists of a sharp leading and trailing edge, with a middle section composed of two planes, creating a diamond shape. A biconvex airfoil is composed of two curves.

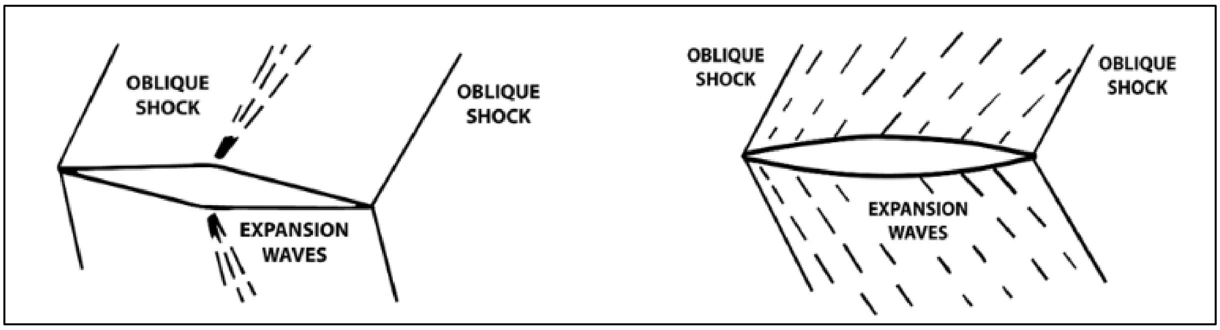

Figure 2-9 - A visualisation of supersonic airflow over a double-wedge airfoil (left) and biconvex airfoil (right) [19]

2.3 Calculating the Properties of Air

To correctly evaluate the behaviour of a body moving through air, certain properties of air must be calculated, such as temperature, pressure, density, and viscosity [5]. However, these properties vary based on altitude and location. To help with this, an arbitrary set of values were defined, known as the international standard atmosphere [20].

2.3.1 Temperature

The temperature of air generally decreases with altitude. The temperature at sea level (0 metres), is taken as $288.15\ K$ [20]. At altitudes above $11\ km$, temperature stops falling and remains constant. The region from sea level to 11 kilometres is called the troposphere. The region above the troposphere is named the stratosphere [5].

To calculate the temperature at altitude [5]:

$$T = T_0 - \lambda h \qquad \text{Eqn. 1}$$

Where:

- T *is the temperature at the desired altitude*
- $T_0 = 288.15\ K$ *is the temperature at sea level [21]*
- h *is the altitude, or height above sea level*
- $\lambda = 0.0065$ *is the temperature lapse rate per metre, taken as [20]*

2.3.2 Pressure

To calculate the pressure based on altitude, the standard atmospheric model is used [21]:

$$P = P_0 \left(\frac{T_0}{T_0 - \lambda h}\right)^{\frac{gM_{air}}{-\lambda R^*}} \qquad \text{Eqn. 2}$$

Where:

- P *is the pressure at altitude*
- $P_0 = 101{,}325\ Pa$ *is the pressure at sea level*
- $T_0 = 288.15\ K$ *is temperature at sea level*
- $g = 9.80665\ m/s^2$ *is acceleration due to gravity*
- $M_{air} = 0.0289644\ kg/mol$ *is molar mass of air*
- $R^* = 8.31432\ J/mol * K$ *is the universal gas constant [21]*

2.3.3 Density

To calculate density based on pressure [21]:

$$\rho = \frac{M_{air}P}{TR^*} \qquad \text{Eqn. 3}$$

Where:

- ρ *is the density of air*
- P *is the pressure at altitude, in Pascal, calculated above*

2.3.4 Speed of Sound

The speed of sound through a medium changes based on the temperature of that medium [21]:

$$C_s = \sqrt{\gamma \frac{R^*}{M} T} \qquad \text{Eqn. 4}$$

Where:

- C_s *is the speed of sound*
- $\gamma = 1.4$ *is the adiabatic index, also known as the heat capacity ratio (a dimensionless quality) [21]*
- T *is the temperature at altitude, in Kelvin, calculated above*

2.3.5 Viscosity

Viscosity is calculated using Sutherland's viscosity formula, which is an estimation of viscosity based on temperature of the fluid [22]:

$$\mu = \mu_0 \left(\frac{T}{T_0}\right)^{\frac{3}{2}} \left(\frac{T_0 + S}{T + S}\right) \qquad \text{Eqn. 5}$$

Where:

- μ *is the dynamic viscosity of air at the specified temperature*

- $T_0 = 288.15\,K$ *is the temperature at sea level, where the dynamic viscosity was experimentally measured*
- $\mu_0 = 1.7894 * 10^{-5}\ Pa/s$ *is the measured dynamic viscosity at* T_0
- *T is the current temperature*
- *S=110 K is the Sutherland constant [7]*

Kinematic Viscosity is defined as the ratio of dynamic viscosity to the density, so $\nu = \mu/\rho$, where ν is the kinematic viscosity and ρ is the density at the specified temperature [7].

2.3.6 Reynolds Number

Reynolds number is calculated automatically using CFD software. However, it is useful to calculate it manually to verify the results; flows with similar Reynolds numbers should yield the same drag and lift coefficients [7].

To calculate the Reynolds number for flow over an airfoil:

$$Re = \frac{\rho V c}{\mu} \qquad \text{Eqn. 6}$$

Where:

- Re *is Reynolds number (dimensionless)*
- ρ *is the density of air*
- c *is the chord length of the airfoil*
- μ *is the dynamic viscosity of the air [7]*

2.4 Simulating Airflow

Computational Fluid Dynamics (CFD) serve as a digital wind tunnel for this project. The reliability of the results depends on the validity of the inputs given. The calculation of reference values is discussed above. This section discusses the selection of appropriate CFD settings.

2.4.1 Turbulence Models

Modelling turbulent flow is an incredibly difficult task, as turbulence is one of the more complicated occurrences in physics. This stems partially from the fact that turbulent flows have multiple scales: The chord length of an airfoil is in metres, and when modelling 3D airframes, can be up to 10^2 metres, while the turbulent vortices can be as small as 10^{-5} metres, depending on RE. Because of this, direct numerical simulations (DNS) are limited to extremely small scales and small time frames [23].

To solve this issue, the concept of Reynolds-Averaged Navier-Stokes (RANS) equations were conceptualised. Instead of calculating individual flows and then averaging them, these equations first average the flow and then calculate the time-dependent variables. The downside of this method is the elimination of turbulence phenomena from the model. To remedy this, turbulence models are used to retroactively calculate the turbulence to correct the simulation results [23]. Ansys uses the RANS equations for analysis.

2.4.2 Wall y^+

As shown in Figure 2-1, flow profile near the surface or a wall is highly curved. However, the flow is always modelled as linear throughout a cell of the simulation. Thus, modelling the curve accurately requires very fine cells close to the wall, as shown in Figure 2-10 [24].

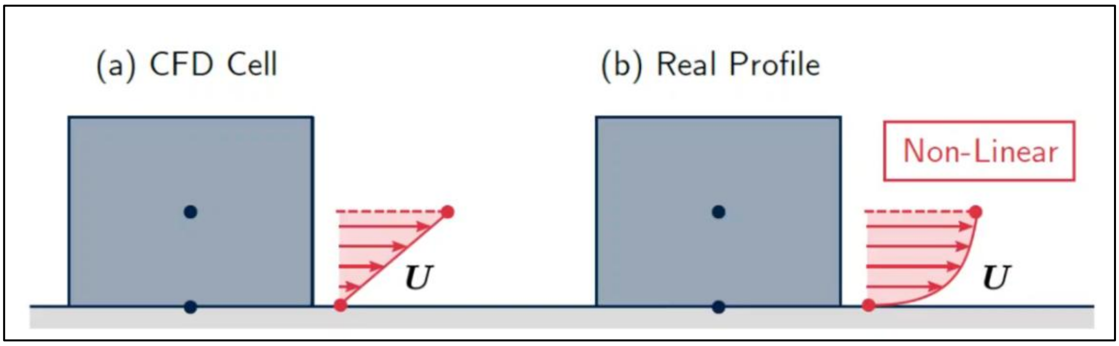

Figure 2-10 - The velocity profile generated in CFD, compared to the real velocity profile [25]

Wall y^+ is a dimensionless measure denoting the distance of the first cell from the wall for 2D flow analysis [26]. Lower y^+ values are generally considered more accurate, and a goal when creating the mesh is $y^+ < 1$. Even if coarser meshes are acceptable, values between 5 and 30 (referred to as the buffer region) can produce poor results. It is therefore recommended to maintain y^+ values either below 5 or above 30, depending on the type of flow and wall function used [23].

To calculate the first layer height necessary to achieve a desired wall y^+ [27]:

$$y = \frac{y^+ \mu}{u_* \rho} \quad \text{Eqn. 7}$$

Where:

- *y is the first layer height,*
- *μ is dynamic viscosity, calculated in Eqn. 5*
- *u_* is friction velocity, a non-dimensional parameter used for calculating y^+* [27]
- *ρ is density of air, calculated using Eqn. 3*

To calculate u_*:

$$u_* = \sqrt{\frac{\tau}{\rho}} \quad \text{Eqn. 8}$$

Where τ is the wall shear stress, a tangential force exerted on the body by the skin drag of the fluid [27]. It is calculated using [28]:

$$\tau = \frac{C_f \rho \, U_\infty}{2} \quad \text{Eqn. 9}$$

Where:

- *U_∞ is the velocity of free-stream air, in m/s (calculated by $U_\infty = MC_s$, where M is the Mach number and C_s is the speed of sound at altitude, calculated via Eqn. 4)*

- *C_f is the skin friction coefficient, a dimensionless coefficient describing the amount of friction exerted on the surface of the airfoil by air.*

To estimate C_f [28]:

$$C_f = \left(2\log_{10}(Re) - 0.65\right)^{-2.3} \qquad \text{Eqn. 10}$$

2.4.2.1 y^+ Insensitive Wall Treatments

y^+ insensitive wall treatments are a type of wall function that can accurately model flows across a range of y^+ values, including values in the buffer region.[1] Some turbulence models in Ansys support y^+ insensitive wall treatments.

2.4.3 Choosing a wall function

Ansys Fluent has multiple turbulence models; the Spalart-Allmaras model is designed for aerodynamic simulations and can handle attached flows or mild flow separation. Since it is a one-equation model, it offers performance benefits over the two-equation models. Standard, realisable or RNG $k - \varepsilon$ models, as well as the $k - \omega$ model, are considered legacy models and are not recommended. $k - \omega$ SST is recommended for higher Re flows with flow separation. The generalised $k - \omega$ (GEKO) model is recommended in all applications. Based on the inputs, it can be tuned to resemble other turbulence models, and is y^+ insensitive [23].

2.4.4 Pressure-Based vs Density-Based Solvers

Pressure-based solvers are generally used for incompressible (or very slightly compressible) flows. They also offer performance benefits over density-based solvers. Density-based solvers are used for compressible flows [29]. However, the Ansys Fluent User Guide [30] states that both approaches are

[1] It is important to clarify that y^+ insensitive wall treatments do not negate the need for fine cells near the boundary layer. For all y^+ values, the wall treatment still requires the appropriate number of cells inside the boundary layer in order to work [23]. Thus, although the y^+ is no longer required to be less than one, the first layer height is still limited.

now applicable to a wide range of flows, although the density-based solver might still be more accurate for high-speed compressible flows.

2.5 Adjoint Solver

When attempting to optimise the aerodynamics of an airfoil, a design may be tested and improved manually over several cycles of CFD analysis followed by a wind tunnel test. However, this method is limited, as it is not guaranteed to ever identify the best possible design (if this design is identified by chance, there is no way to verify that it is the best). Adjoint-based optimisation is designed specifically to tackle this challenge. The adjoint method is a numerical model that calculates the sensitivity of a function (such as lift or drag) based on the variations in the shape of the model. Therefore, adjoint can be used to find the optimal shape of the airfoil based on given operating conditions [29] [30].

2.5.1 Pros and Cons of Adjoint-Based Airfoil Optimisation

While adjoint optimisation can improve the design, it is important to acknowledge the limitations of the method. Drela [33] outlines the main issue with single-point design optimisation, namely that the off-design performance of the airfoil worsens drastically. Especially at higher geometric degrees of freedom, the solver develops minute changes in the surface, exploiting the fluid flow to create a more favourable lift-to-drag ratio. However, the small changes introduced only work at the sampled operating conditions, and fail if the slightest changes in the conditions are introduced. [33] partially solves this by applying more design points (up to 6), which, although it does not eliminate the issue, nevertheless improves the off-design performance at the cost of processing time. In fact, [33] states that increasing the geometric degrees of freedom appears to require a corresponding increase in the number of operating conditions sampled. [33] concludes that while airfoil optimisation is a good tool for identifying the areas of improvement, it does not eliminate the need for manual inspection of the results.

2.6 Boundary Conditions & Verification

2.6.1 NACA 0012

Ladson [34] tested the low-speed aerodynamic characteristics of the airfoil in a wind tunnel at varying Reynolds Numbers and angles of attack. Hasan et al. [35] simulated and compared the NACA 0012 and NACA 2412 airfoils over a range of angle of attacks. Both recorded a stall angle at just past 16° AoA. [34] recorded a drag coefficient at 0° AoA of $C_d = 0.00634$ at $RE = 5.95 * 10^6$. [35] simulated a drag coefficient at 0° AoA of $C_d \approx 0.0095$[2] at $RE = 6 * 10^6$. For both [34] and [35] the lift coefficient was negligible at 0° AoA. The full comparison between the simulation and experiment can be seen in Table 2-1.

Table 2-1 - Comparison between Ladson [34] and Hasan [35]

AoA (°)	C_d Ladson	C_d Hasan	C_l Ladson	C_l Hasan
0	0.00634	0.009	-0.0078	0
4	0.00725	0.01	0.4325	0.38
8	0.00803	0.0135	0.874	0.85
12	0.01262	0.018	1.296	1.25
16	0.02903	0.0285	1.5616	1.6

2.6.2 NACA 2412

Hasan et al. [35] simulated the NACA 2412 airfoil at $RE = 6 * 10^6$, and recorded $C_l \approx 0.21$ and $C_d \approx 0.095$ at 0° AoA. Gowda [36] simulated and compared the NACA 2412 and NACA 4412 airfoils at $RE = 2 * 10^6$, and recorded similar values.

2.6.3 Boeing BACXXX

Abdel-Raheem et al. [37] simulated the BACXXX airfoil at various Mach speeds, altitudes, and angles of attack. However, the reliability of this paper is questionable, as certain lift values recorded are unreasonably high. For example, at an altitude of $h = 9{,}000\ m$ and Mach speed of M=0.85, at 2° AoA, the recorded coefficient values were $C_l = 4.5$ and $C_d = 0.85$, both far outside of the expected range.

[2] Estimation based on a chart. Exact values are not given in table form in the report.

2.6.4 NASA SC(2)-0714

Sri et al. [38] simulated the NASA SC(2)-0714 at various Mach speeds and angles of attack. At M=0.8 and AoA of $\alpha = 0°$, $C_l = 0.068$ and $C_d = 0.0087$ were recorded. At M=1.2 at the same angle of attack, the values were $C_l = 0.013$ and $C_d = 0.024$.

2.7 Breguet Range Equation

The Breguet range equation is used to find the range of an aeroplane [4]:

$$R = \frac{V\frac{L}{D}}{TSFC}\ln\left(\frac{W_{start}}{W_{end}}\right) \qquad \text{Eqn. 11}$$

Where:

- R *is the range*
- V *is velocity*
- $TSFC$ *is thrust-specific fuel consumption*
- W_{start} *is the weight at take-off (weight of aircraft and fuel)*
- W_{end} *is the weight at landing*
- L/D *is the lift-to-drag ratio*

Assuming constant range, velocity, and thrust-specific fuel consumption, the equation can be rearranged to find $\frac{W_{start}}{W_{end}}$:

$$\frac{W_{start}}{W_{end}} = e^{\frac{R*TSFC}{V(L/D)}} \qquad \text{Eqn. 12}$$

Since W_{start} is the sum of W_{end} and fuel:

$$\frac{W_{fuel} + W_{end}}{W_{end}} = e^{\frac{R*TSFC}{V(L/D)}} \Rightarrow \frac{W_{fuel}}{W_{end}} + 1 = e^{\frac{R*TSFC}{V(L/D)}} \Rightarrow W_{fuel} = W_{end}\left(e^{\frac{R*TSFC}{V(L/D)}} - 1\right) \qquad \text{Eqn. 13}$$

With Eqn. 13, it is possible to estimate the weight of fuel required for flight.

If estimating the range based on the number of flight hours ($R = t * V$, where t is the flight time, in seconds), the speed cancels out:

$$W_{fuel} = W_{end}\left(e^{\frac{t*TSFC}{(L/D)}} - 1\right) \qquad \text{Eqn. 14}$$

3 Equipment & Methods

3.1 Equipment

All simulations in this project were done on an HP ENVY x360 Convertible 15-es0xxx, equipped with an 11th Gen Intel(R) Core(TM) i7-1165G7 @ 2.80GHz, alongside 16 GB of RAM and 8 logical processors. The detailed specification of the laptop and the operating system can be seen in Table 8-1 below. Along with the computer, the details of the software used are relevant. All mesh files, as well as all simulations were completed using Ansys Fluent release 2025 R2. The version remained unchanged through the entire process. Ansys Fluent was controlled by using the Ansys PyFluent API version 0.37.1, running in Python version 3.11.9. The scripts were run in Microsoft PowerShell version 7.5.4.

3.2 Methods

To analyse multiple airfoil designs and evaluate their drag and lift coefficients, computational fluid dynamics (CFD) models were created. However, a CFD model requires a 3D model, and for fluid simulations, the part (in this case, airfoil) itself is not modelled. Instead, the domain around it is modelled and meshed, with the part itself treated as a wall when conducting the simulation.

3.2.1 Getting the Airfoil Coordinates & Pre-processing

Airfoil coordinates were obtained online [39] and pre-processed using Excel into a format readable by SolidWorks. The airfoil was then imported into SolidWorks as an XYZ curve, shown in Figure 3-1. All curves were imported on the XY plane, with the positive X going from the leading towards the trailing edge, and the positive Y direction going from the lower towards the upper surface (the axes shown in Figure 3-1 are constant throughout the trials). For 3D models, the domain was extruded by $0.1\ m$ on the Z plane.

The airfoil coordinates used are available in Appendix B: Airfoil Coordinates.

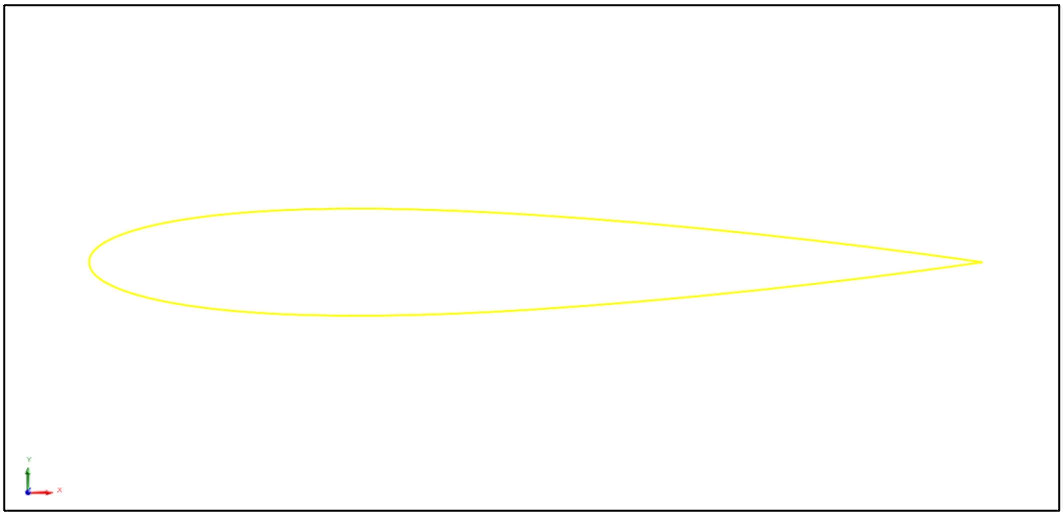

Figure 3-1 - Initial XYZ curve of a NACA 0012 airfoil in SolidWorks

3.2.2 Creating the Domain in SolidWorks

After, the domain around the airfoil was created. All airfoil coordinates are given assuming a chord length of 1 unit of length, with the leading edge located at coordinates (0; 0) [40]. Since the curve was imported into SolidWorks with the units set to metres, all airfoil coordinates drawn had a chord length of 1 m. The domain extended 10 chord lengths ahead of the leading edge and 20 metres behind[3]. It was drawn using a spline curve, with the first and last points at coordinates $(-20;\ 15)$ and $(-20;\ -15)$, and the middle point at $(10;\ 0)$. The result is shown in Figure 3-2.

For 3D models, this domain was then extruded at a thickness of 0.1 m. For 2D models, a surface plane was created. A split line was used to split the upper and lower surfaces of the airfoils, in order to graph the upper and lower surface of the airfoils independently.

[3] Accounting for a chord length of 1 m, the domain extended 19 m behind the trailing edge.

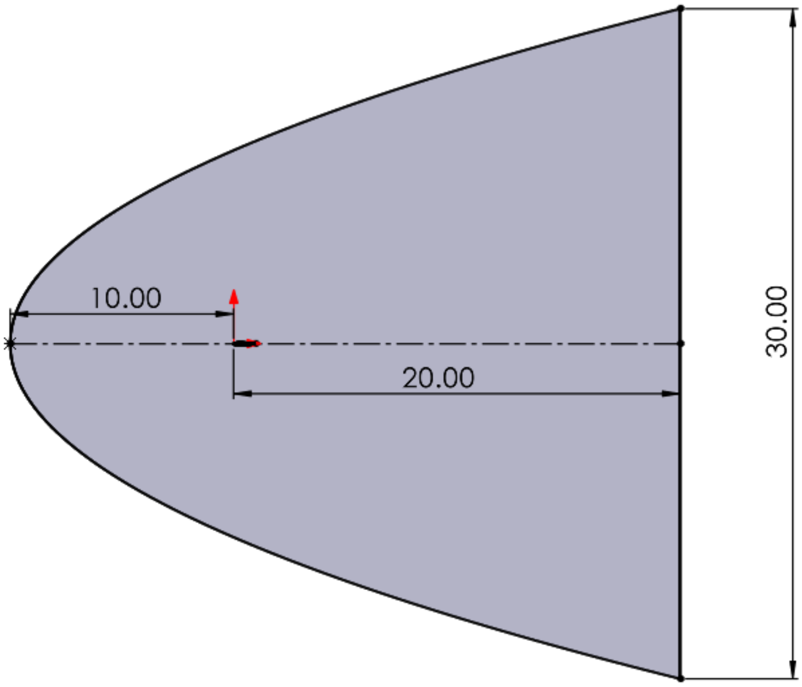

Figure 3-2 - Initial domain in SolidWorks

3.2.3 Ansys SpaceClaim 2025 R2

The SolidWorks file was exported as a $.STEP$ file and imported into Ansys SpaceClaim. SpaceClaim was used to name the different surfaces of the domain. For 3D domains, the upper and lower surfaces of the airfoil were named “upper” and “lower”, respectively, the surfaces on the XY plane of the domain were named “sym1” and “sym2”, (symmetry 1 and symmetry 2), and the surface created by the extrusion was named “farfield”. For 2D domains, the surface on the XY plane was named “surface”, and its edges were designated as “farfield”. The upper and lower parts of the curve were named “upper” and “lower”. Finally, the SpaceClaim geometry file ($.scdocx$) was saved.

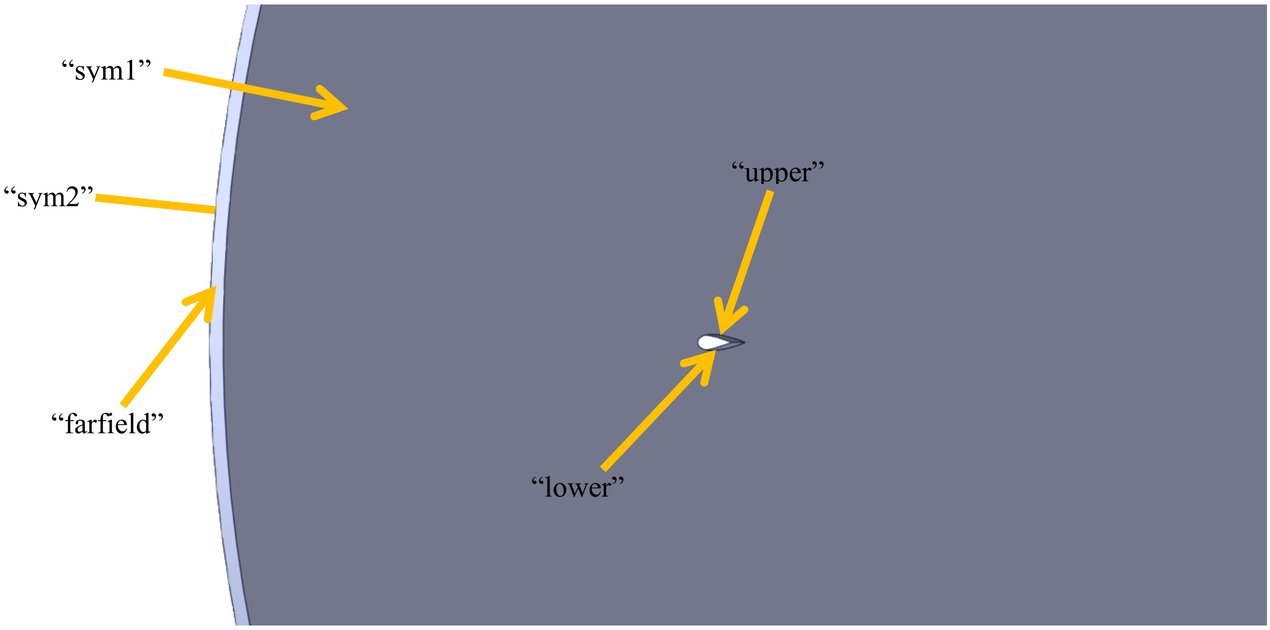

Figure 3-3 - The domain of a NACA 0012 airfoil in Ansys SpaceClaim

3.2.4 Meshing Setup

After naming the surfaces, the geometry files were imported into Ansys Fluent Meshing. The standard process for watertight geometry meshing was followed. For all airfoils, a local sizing was added on the upper and lower surfaces with an element size 1% of chord length ($10\ mm$). Boundary layers were also added, based on the y^+ value (calculated using Eqn. 7 - Eqn. 10). The minimum surface size of the mesh was set to 20 times larger than the first boundary layer height. The default mesh type used was polyhedral, based on 4.6.2 Mesh Type Trial.

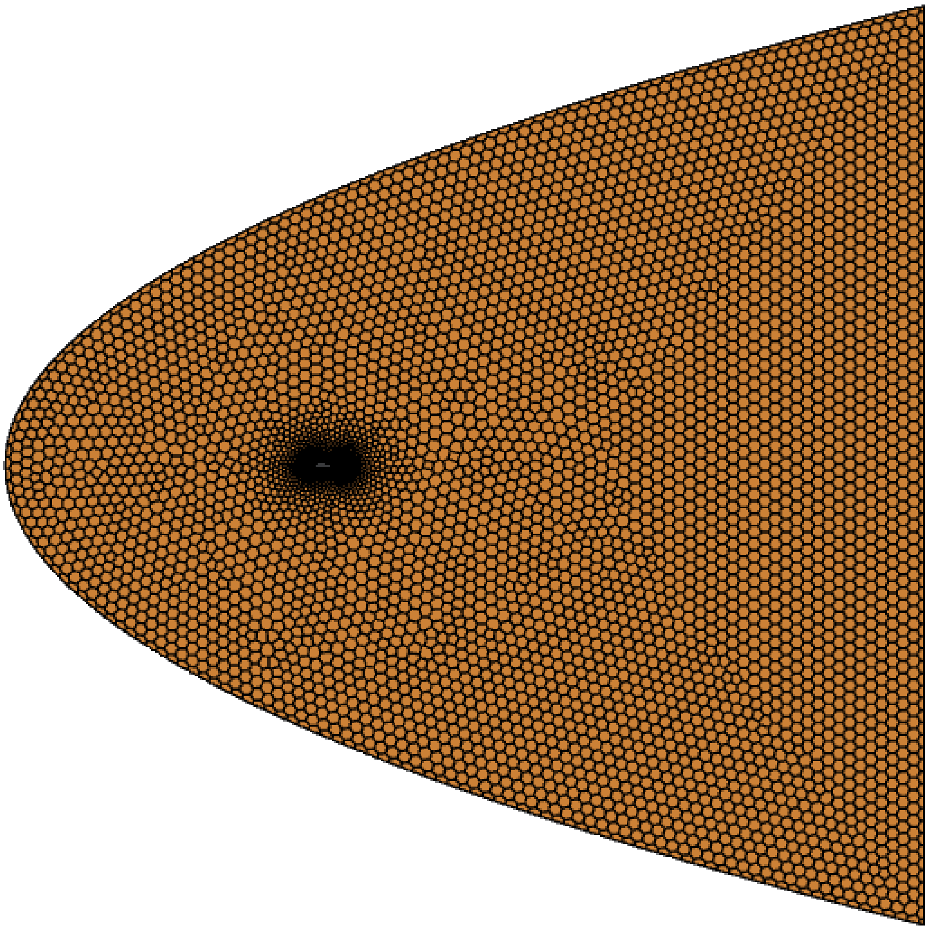

Figure 3-4 - The mesh structure

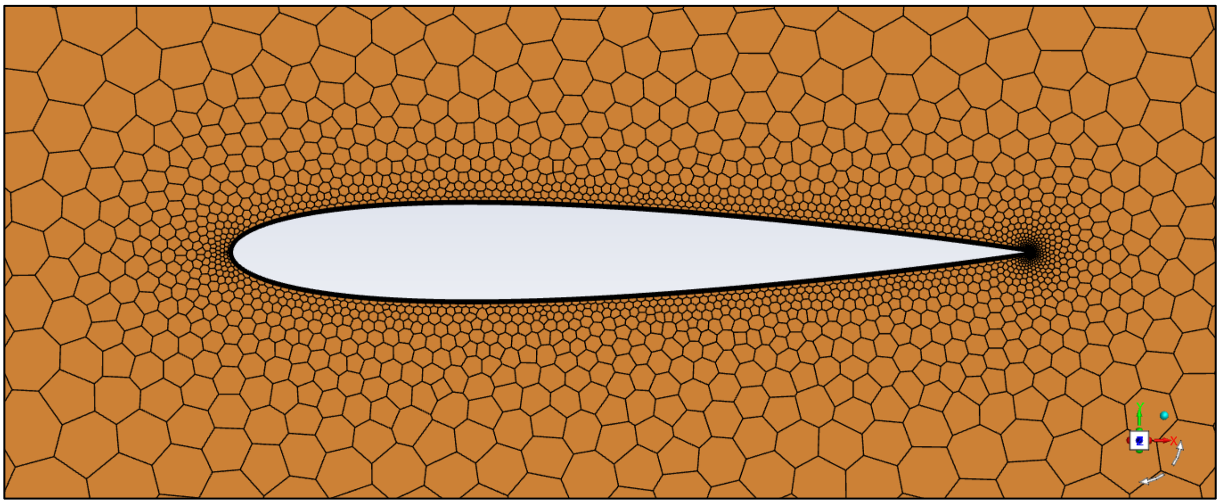

Figure 3-5 - Detailed view of the mesh structure at the airfoil

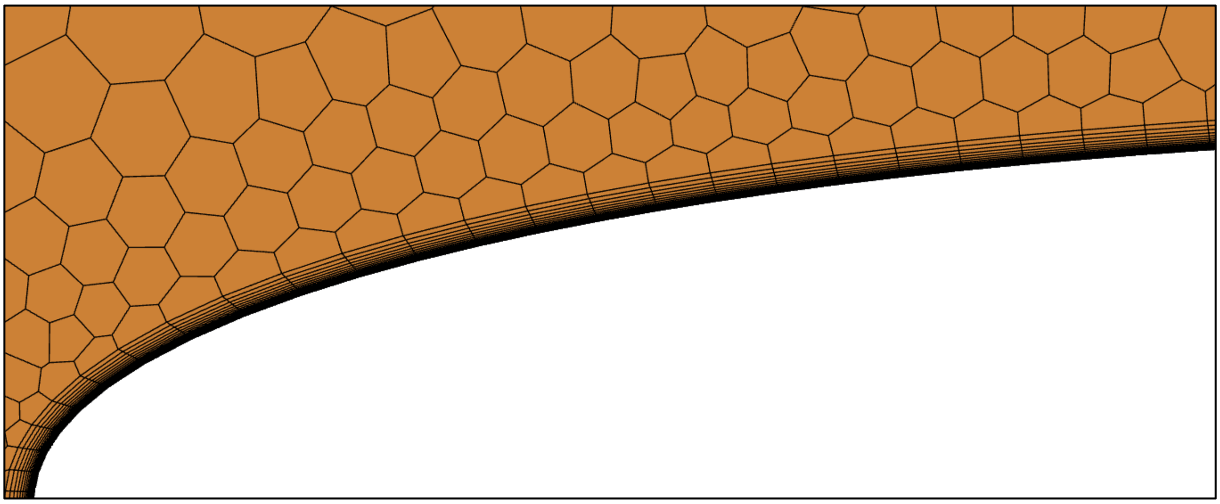

Figure 3-6 - Detailed view of the boundary layers around the airfoil

3.2.5 Solver Setup

The solver was set up in the same order each time:

1. *Core count and precision (single vs double) were specified before launching the session*
2. *The following properties of air were calculated:*
    a. *Temperature, from Eqn. 1*
    b. *Pressure, from Eqn. 2*
    c. *Density, from Eqn. 3*
    d. *Speed of sound, from Eqn. 4*
    e. *Dynamic and kinematic viscosity, from Eqn. 5*
    f. *Reynolds number, from Eqn. 6*
3. *Velocity in the x and y directions was calculated based on the angle of attack*
4. *Solver type was set (pressure-based vs density-based)*
5. *Ideal gas law was enabled*
6. *Sutherland viscosity model was enabled*
7. *The energy equation was enabled*
8. *The operating (gauge) pressure was set to* $0\ Pa$
9. *The turbulence model was selected and tuned*

10. *The following solution methods were specified:*
    a. *Transient mode was enabled, if needed*
    b. *The Gradient, pressure, momentum, energy, density, turbulent kinetic energy discretisation schemes were set*
    c. *High-order term relaxation was enabled, if needed*
    d. *Pseudo-time stepping was enabled, if needed*
11. *Report definitions (for drag and lift) were created and configured*
12. *Reference values were specified, from values calculated in step 2*
13. *Convergence criteria were specified, if needed*
14. *Report-based convergence monitoring for the drag and lift reports was added*
15. *The solution was initialised*
16. *The iteration count and pseudo-time step scale were set. For transient solves, time step size, time step count, and iterations per time step were set*
17. *The calculation was started*
18. *Once the calculation was completed, the mass flow through the farfield boundary was calculated and saved*
19. *The cumulative plot was generated and saved*
20. *The Mach number contours were generated and saved*
21. *The pressure contours were generated and saved*
22. *The* y^+ *plot was generated and saved*
23. *The pressure distribution plot was generated and saved*
24. *The residual plot was saved*
25. *The case and data files (.cas and .dat, respectively) were saved*
26. *The Fluent session was closed*
27. *The outputs generated from the drag and lift reports, in the form of a .out file, were plotted and saved*

3.2.6 Adjoint Solver Setup

The initial setup for adjoint solver closely mirrors the process described in Solver Setup above. However, the Ansys adjoint solver introduces certain limitations. Namely, density-based and transient solvers are unsupported, so the solver type must be pressure-based and steady-state (although pseudo-time-step is available). Furthermore, the turbulence model must be $k - \omega$. While $k - \omega$ SST is also accessible, only $k - \omega$ GEKO was used in this trial as it is the only one that supports enabling turbulence effects.

All optimisation was initially done in 2D to save processing time. When aiming for a $y^+ = 1$, the first layer height is so small that minute changes in the mesh causes cells with zero volume, also known as left-handed faces. Because of this, the optimisation was done with coarse 2D meshes with a larger first layer height (thus a larger y^+). For each case, two runs were completed: one where the chord length was maintained, and one where the solver was free to increase/decrease the length. The resulting displacement files were made into a 3D domain, and the test was re-run in 3D.

For optimisation, the .cas and .dat files generated above were imported into the solver. Then:

1. *Core count and precision (single vs double) were specified before launching the session*
2. *If the boundary conditions were to be changed, the atmospheric properties were recalculated (as discussed in step 2 in Solver Setup).*
3. *Velocity in the x and y directions was calculated based on the angle of attack*
4. *Solver type was set to pressure-based since Ansys Adjoint only supports pressure-based solves*
5. *The turbulence model was set to $k - \omega$ GEKO*
6. *Ansys Adjoint solver was activated*
7. *Temporary iteration count and pseudo-time-step sizes were set*
8. *Drag and lift observables were defined in Adjoint solver*
9. *A lift-to-drag observable was created and set as ratio: lift over drag*

10. *The optimisation target was set*
11. *The target change was set*
12. *Convergence criteria for Adjoint solver was set*
13. *Adjoint solver settings were configured:*
    a. *Best match (Adjoint solver mirrors flow solution settings)*
    b. *Adjoint energy*
    c. *Adjoint turbulence*
    d. *Adjoint preconditioning*
14. *The design tool was used to create a grid over modifiable zones, using 30x30 points*
15. *The initial boundary box was created*
16. *If the length was to be maintained, the boundary box was set to preserve chord length while allowing for vertical deformation. Otherwise, the boundary box was expanded to allow changes in all directions.*
17. *Report definitions (for drag and lift) were created and configured*[4]
18. *If boundary conditions were changed, they were re-specified. Then the case was re-initialised and re-solved*
19. *Mach and pressure contours were defined*
20. *The main optimisation loop was started:*
    a. *The design change was calculated*
    b. *Adjoint solver was run*
    c. *The mesh was morphed (according to the adjoint solver)*
    d. *The mesh was re-meshed to improve quality*
    e. *The current observable value was evaluated on the new mesh*
    f. *The steady state solver was run until the solution converged again*
    g. *The orthogonal quality of the mesh was noted*

[4] Although the drag and lift reports are already in most of the .cas.h5 files (created in step 11 in Solver Setup), they are created again in the code, to ensure files without drag and lift reports are compatible.

    h. *The mesh, Mach contours, and pressure contours were shown and screenshotted*

21. *The loop was repeated a predetermined number of times or until the orthogonal quality of the mesh was too low (depending on the solve)*
22. *The original solver settings were restored*
23. *A screenshot of the residuals was saved*
24. *The solution was re-initialised and solved with full iterations*
25. *A screenshot of the final residuals was saved*
26. *The mass flow through the farfield boundary was calculated and saved*
27. *The cumulative plot was generated and saved*
28. *The y^+ plot was generated and saved*
29. *The pressure distribution plot was generated and saved*
30. *The case and data files ($.cas.h5$ and $.dat.h5$, respectively) were saved*
31. *A displacement file ($.\mathrm{dis}$) was saved, containing the new, optimised coordinates*
32. *GIFs were generated from the mesh, Mach number contour, and Pressure contour screenshots*

3.2.7 Notes On Setup

For both the initial solve and adjoint solver cases, several assumptions were made to simplify the solve and reduce calculation time. Firstly, the standard atmosphere model was used for calculating properties of air, as discussed in 2.3 Calculating the Properties of Air. Furthermore, the altitude, velocity, and angle of attack were all assumed to be constant. Thus, all calculated properties of air (temperature, pressure, density, speed of sound, dynamic and kinematic viscosity) were constant. Specific heat capacity was also assumed to be constant. Absolute pressure was used by setting the gauge pressure to $0\ Pa$.

$30x30$ design points were used when optimising in order to avoid the issue outlined in 2.5.1 Pros and Cons of Adjoint-Based Airfoil Optimisation.

3.2.8 PyFluent

PyFluent is a python library that connects to Ansys Fluent through code [41]. Several scripts were written using the PyFluent library in this project to automate the workflows. Initially, this was done to optimise simulation time by testing whether having the graphical user interface (GUI) on or off affected simulation speeds. However, as shown in section 4.6.3 and discussed in section 5.2.3, enabling or disabling the GUI had no measurable impact on simulation times. Regardless, the automation of all workflows enabled repeated testing, and hence easy comparisons between different parameters or solver setups. It also eliminated human error while setting up solves, as all inputs were done automatically. All simulation results, unless specified, were obtained by running the scripts.

The project included three main scripts, each relying on three separate functions: The meshing, solution, and optimisation functions. All three scripts read a table from an Excel file, which specifies all inputs. Any inputs left blank are assigned default values (see Table 8-16, Table 8-17, and Table 8-18).

To determine whether to launch a 2D or 3D Fluent session, the scripts look at the airfoil input (hence, all two-dimensional solves include _2D at the end of the airfoil name). Output files are named using the following scheme: {airfoil}_{nickname}_{identifier}_{timestamp}. The airfoil and nickname are inputs, specified in the input table and used to differentiate runs from each other. The identifiers are hardcoded values that specify the file's purpose. The timestamp is added to certain files and folders to avoid name duplication.

All times shown in Results are timed using the python scripts. The timer starts upon reading the inputs and stops once the output files are saved.

3.2.8.1 PyFluent Meshing

To mesh a geometry the $mesh()$ function was written. The only input required is the SpaceClaim file ($.scdocx$) location. A list of all other inputs, as well as the default values used in meshing, can be seen in Table 8-16. The script can also handle geometries with trailing edges. The $mesh()$ function outputs three files:

- *The mesh file ($.msh.h5$)*
- *A text file of the inputs used ($.txt$)*[5]
- *A text file of the log, which shows the console output of Ansys Fluent from start to finish ($.txt$)*

3.2.8.2 PyFluent Solution

To get the drag and lift coefficients of an airfoil, the $solve()$ function was written. The $solve()$ function follows the steps outlined in 3.2.5 Solver Setup. The only required input of the $solve()$ function is the mesh file ($.msh.h5$) location. However, the boundary conditions (altitude, AoA, Mach number) were always manually specified. The outputs of the solve() function are:

- *The case and data files ($.cas.h5$ and $.dat.h5$, respectively)*
- *A text file of the inputs used ($.txt$)*
- *A text file of the log, which shows the console output of Ansys Fluent from start to finish ($.txt$)*
- *The drag and lift report files ($.out$)*
- *The drag and lift report plots, plotted using Matplotlib, based on the report files ($.png$)*
- *The cumulative force file ($.xy$)*
- *The mass flow file, showing the total mass flow through the farfield boundary ($.flp$)*
- *A screenshot of the Mach contours ($.png$)*
- *A screenshot of the pressure contours ($.png$)*
- *A screenshot of the y^+ plot, plotted in Fluent ($.png$)*
- *A screenshot of the pressure plot, plotted in Fluent ($.png$)*

[5] All inputs files include only non-default values. All default values can be seen in Appendix D: Default Values.

- *A screenshot of the residuals (.png)*

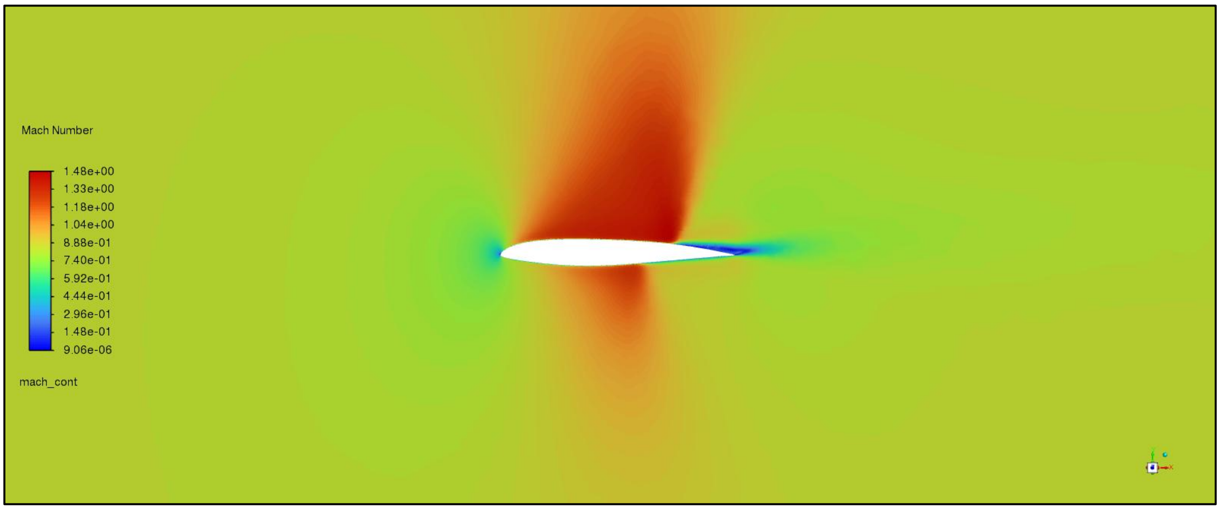

Figure 3-7 - Generated Mach contours of a BACXXX at Mach 0.85

3.2.8.3 PyFluent Optimisation

The $optimize()$ function uses Ansys Adjoint Solver to optimise for certain values. The function takes a case file ($.cas.h5$) location as an input[6] and imports the case and data. It then follows the steps outlined in 3.2.6 Adjoint Solver Setup. The outputs of the $optimize()$ function are:

- *A folder containing screenshots (.png) of the Mach number contours from each iteration*
- *A folder containing screenshots (.png) of the pressure contours from each iteration*
- *A folder containing screenshots (.png) of the mesh from each iteration*
- *The final drag and lift reports of the optimised airfoil (.out)*
- *The cumulative force file of the optimised airfoil (.xy)*
- *The final case and data files (.cas.h5 and .dat.h5, respectively)*
- *A plot showing the residuals of the final solve (.png)*
- *A text file of the inputs used (.txt)*

[6] The data file ($.dat.h5$) is also required. However, it is assumed that the data file has the same name and file location as the case file, so only one input is required.

- *A text file of the log, which shows the console output of Ansys Fluent from start to finish* $(.txt)$
- *Three animated GIFs, showing the change in mesh, pressure contour, and Mach number* $(.gif)$
- *The mass flow file, showing the total mass flow through the farfield boundary* $(.flp)$
- *A text file listing the value of the target observable per iteration* $(.txt)$
- *A file containing the coordinates of the modified mesh* $(.dis)$
- *A screenshot of the* y^+ *plot, plotted in Fluent* $(.png)$
- *A screenshot of the pressure plot, plotted in Fluent* $(.png)$

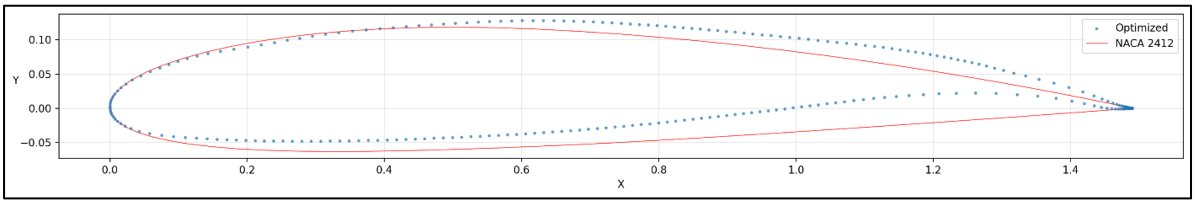

Figure 3-8 - The optimised coordinates (from the $.dis$ file) of a NACA 2412, plotted

3.2.9 Airfoil Selection and Boundary Conditions

Four airfoils were used in this project with the following designations:

- *NACA 0012: A symmetrical NACA 4-digit airfoil*
- *NACA 2412: An airfoil used on the Cessna 172 [42]*
- *BACXXX: An airfoil used on the Boeing 747-400 [37]*
- *NASA SC (2)-0714: A NASA SC series supersonic airfoil*

The NACA 0012 was chosen as a symmetrical airfoil was useful for testing angle of attack simulations, as well as for optimising the lift-to-drag ratio using adjoint. The NACA 2412 was selected as it is used on an aeroplane flying at subsonic speeds [42]. The BACXXX was selected for its use on a transonic aeroplane [43]. The NASA SC(2)-0714 was selected for testing supersonic applications. Since the NACA 2412 and BACXXX are used on commercial aircraft, the cruise speed and cruise altitude of the aircraft were used as a reference for the boundary conditions. For the NACA 0012, the altitude and speed were chosen arbitrarily. For the NASA SC(2)-0714, the speed and altitude were based on the McDonnell Douglas F/A-18 Hornet.[7] The boundary conditions for each airfoil are shown in Table 3-1.

Table 3-1 - The boundary conditions tested for each airfoil

Designation	**NACA 0012**	**NACA 2412**	**BACXXX**	**NASA SC(2)-0714 Transonic**	**NASA SC(2)-0714 Supersonic**
Altitude, h (m)	2,675.2	4,114.8	10,576.6	11,581.8	13,715.3
AoA, α (°)	0.00	0.00	2.00	2.00	0.00
Chord Length, l (m)	1.00	1.49	8.33	3.51	3.51
Airspeed, M (Mach)	0.72	0.20	0.85	0.85	1.80
Density, ρ (kg/m^3)	0.940	0.809	0.384	0.338	0.253
Temperature, T (K)	270.8	261.4	219.4	212.9	199.0
Pressure P, (Pa)	73,049.0	60,717.9	24,178.3	20,625.2	14,474.4
Speed of Sound at altitude (m/s)	329.8	324.1	296.9	292.5	282.8
Airspeed, v (m/s)	236.1	63.8	252.1	248.6	509.0
Tangential Velocity, v_x (m/s)	236.1	63.8	251.9	248.4	509.0
Dynamic Viscosity, μ $(Pa * s)$	1.70E-05	1.66E-05	1.44E-05	1.40E-05	1.32E-05
Kinematic Viscosity, ν (m^2/s)	1.81E-05	2.05E-05	3.74E-05	4.15E-05	5.22E-05
Reynolds Number, Re	1.30E+07	4.64E+06	5.61E+07	2.10E+07	3.42E+07
Skin Friction Coefficient, C_f	2.48E-03	2.90E-03	2.02E-03	2.31E-03	2.16E-03
Wall Shear Stress, τ	6.50E+01	4.78E+00	2.46E+01	2.41E+01	7.09E+01
Friction Velocity, u_*	8.31E+00	2.43E+00	8.01E+00	8.45E+00	1.67E+01
First Layer Height, y (m)	2.18E-06	8.43E-06	4.67E-06	4.91E-06	3.12E-06

3.2.10 Data Analysis and Results Verification

Since the simulations were completed using the Python script, and hence without any observation, to ensure their validity, the file outputs were inspected. First, the mass flow rate was inspected. A

[7]The McDonnell Douglas F/A-18 Hornet does not use the NASA SC(2)-0714 airfoil [44]. The aircraft's cruise speed and altitude were simply used as a reference for a real-world application.

converged solution should have a minuscule total mass flow rate (mass flow in = mass flow out). Any simulations with a mass flow rate above 0.01 kg/s were investigated.

Next, the residual plot was inspected to visually confirm the convergence. A residual plot with decreasing values (shown in Figure 3-9) was expected. The validity of solves with irregular residual plots (shown in Figure 3-10) was manually verified.

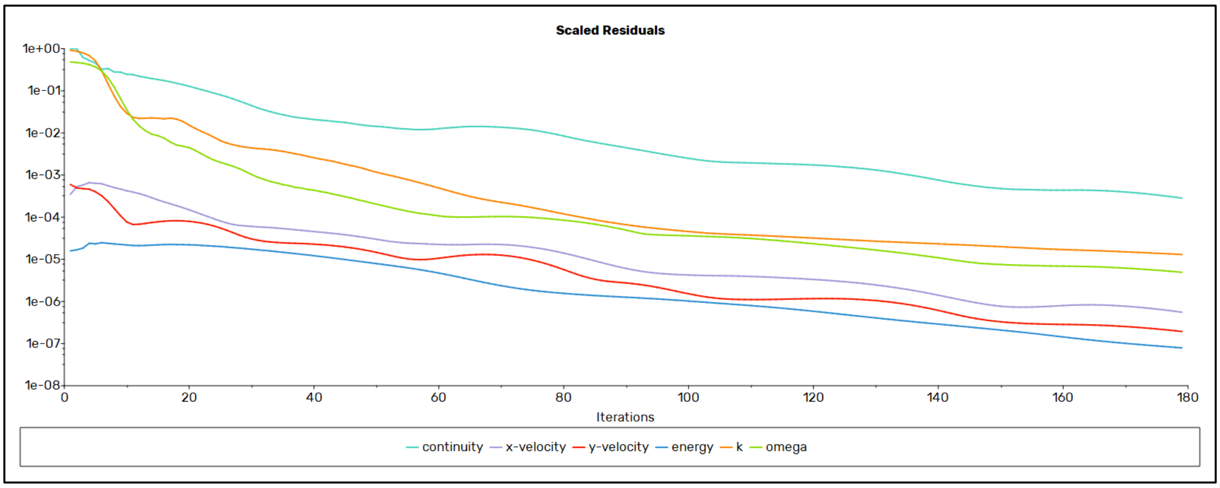

Figure 3-9 - A converging residual plot

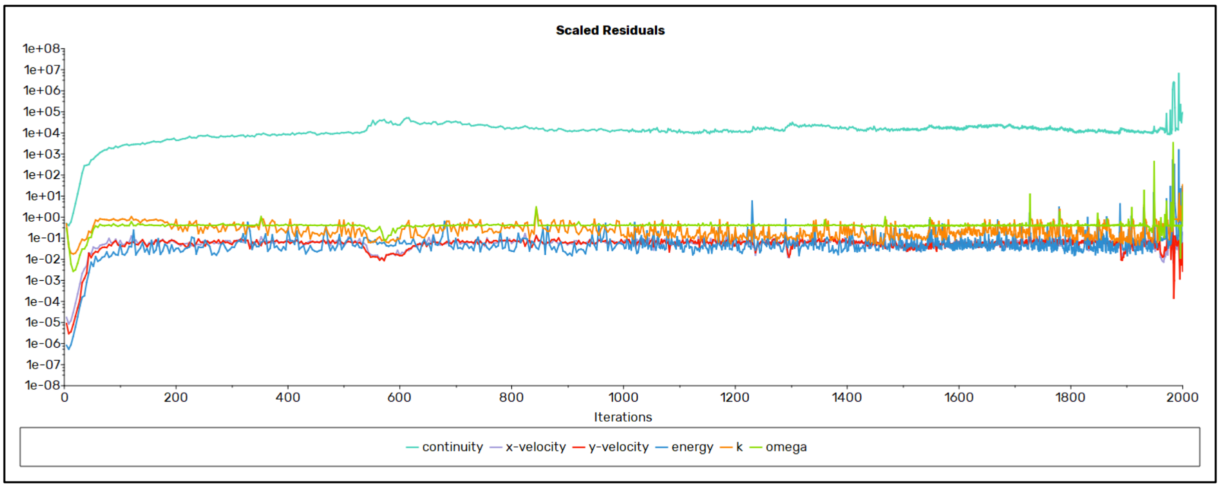

Figure 3-10 - A non-converging residual plot

To verify the convergence of the drag and lift reports, the graph of the outputs was inspected. A converged drag/lift report (shown in Figure 3-11) was expected. An irregular report (Figure 3-12) suggested an incorrect solve.

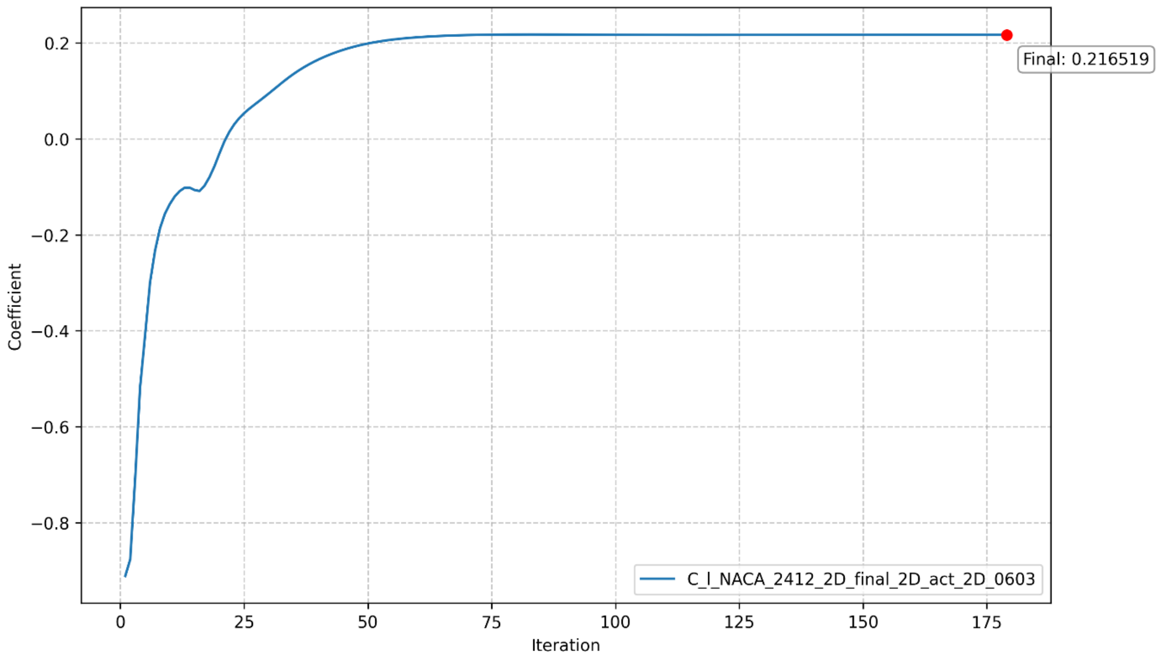

Figure 3-11 - An example of a converged lift coefficient

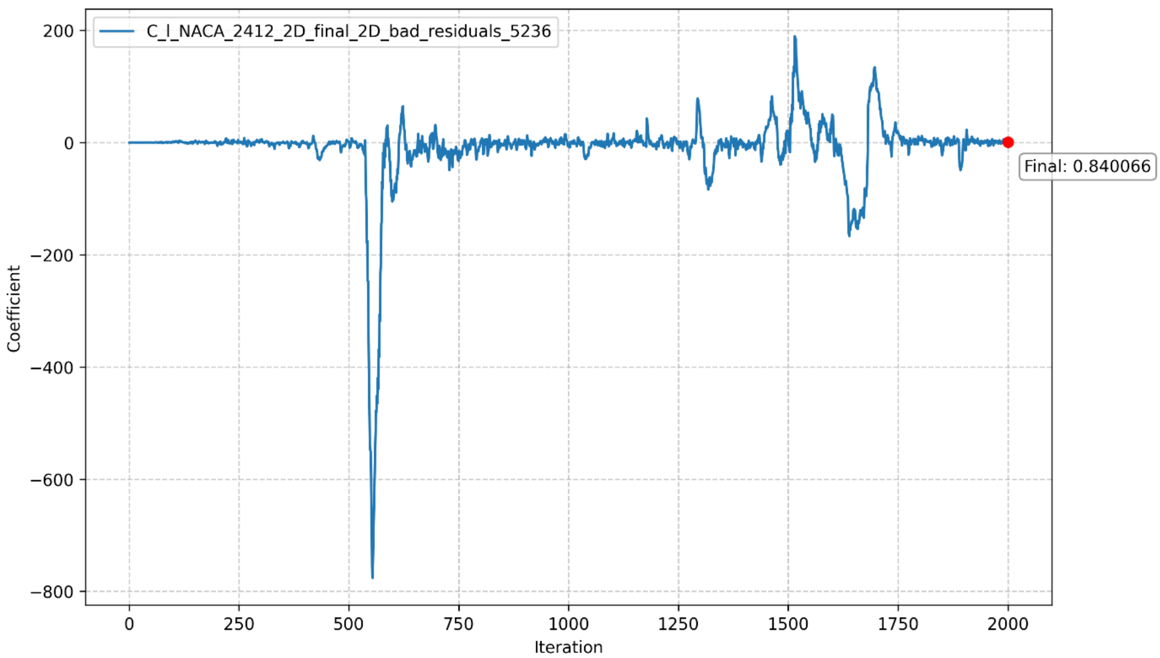

Figure 3-12 - An example of a non-converging lift coefficient

To verify the validity of the solver settings, all airfoils were tested in two cases: The test case, using the boundary conditions shown in Table 3-1, and a reference case, which recreated the boundary conditions of studies with similar Reynolds numbers. The drag and lift coefficients of the reference studies were compared to the drag and lift coefficients recorded using Ansys to establish the reliability of the solver settings.

4 Results

4.1 NACA 0012

The NACA 0012 airfoil was tested in 2D and 3D, first at the boundary conditions used in [35], then at the selected values shown in Table 3-1. For this and all other airfoils, the reference runs were nicknamed ref_3D and ref_2D, and the final tests were nicknamed act_3D and act_2D.

The quality of the meshes generated is shown in Table 4-1. The 3D mesh had a maximum skewness of 0.64 and minimum orthogonal quality of 0.15. The 2D mesh had a minimum orthogonal quality of 0.139.

The results of the verification and final trials can be seen in Table 4-2. As shown, the 2D trials show a higher lift-to-drag ratio when compared to 3D trials. The comparison between recorded and theoretical results (from [34], [35]) is shown in

Table 4-3. The overall results are similar to Hasan [35]. Ladson [34] shows a very similar lift coefficient, but a much lower drag coefficient, thus corresponding to a higher lift-to-drag ratio.

The Mach contours are shown in Figure 4-1. As expected, the flow accelerates over the upper surface and, as shown in Figure 4-2 and Figure 4-3, creates a low pressure area, generating lift.

Table 4-1 - NACA 0012 mesh quality

Variable	NACA_0012	NACA_0012_2D
Maximum Skewness	0.64	
Minimum Orthogonal Quality	0.15	0.138759
Face Count	63,968	
Cell Count	678,703	24,336
Time Taken (s)	336.28	308.75
Mesh File Size (bytes)	61,533,698	650,541

Table 4-2 - Boundary conditions and results of the NACA 0012 test and reference cases

Variable	**Test Results**		**Reference Results**	
Nickname	act_3D	act_2D	ref_3D	ref_2D
Dimensions	3D	2D	3D	2D
Altitude (m)	2,675.17	2,675.17	0.00	0.00
AoA (°)	1.55	1.55	4.00	4.00
Chord Length (m)	1.00	1.00	1.00	1.00
Mach Number	0.716	0.716	0.258	0.258
C_d	0.01018	0.00887	0.01001	0.00958
C_l	0.23587	0.24282	0.43443	0.43862
Lift-to-drag Ratio	23.18	27.37	43.38	45.77
Iterations Needed	133	518	119	2,000
y^+	1.40	1.40	1.00	0.95
Cell Count	678,703	24,336	678,703	24,336
Time Taken (s)	1,329.90	215.50	1,408.57	566.09
Time per iteration (s)	10.00	0.42	11.84	0.28
Reynolds Number	1.30E+07	1.30E+07	6.01E+06	6.01E+06

Table 4-3 - NACA 0012 results compared to [34], [35]

Variable	**ref_3D**	**ref_2D**	**Hasan**	**Ladson**	**Hasan Δ%**[8]	**Ladson Δ%**
C_d	0.01001	0.00958	0.009	0.00634	10.12%	36.68%
C_l	0.43443	0.43862	0.38	0.4325	12.53%	0.44%
Lift-to-drag Ratio	43.38	45.77	42.22	68.22	2.68%	-57.24%

[8] Percentage difference is calculated as the absolute difference between the recorded drag coefficient in 3D and the reference value, divided by the reference value.

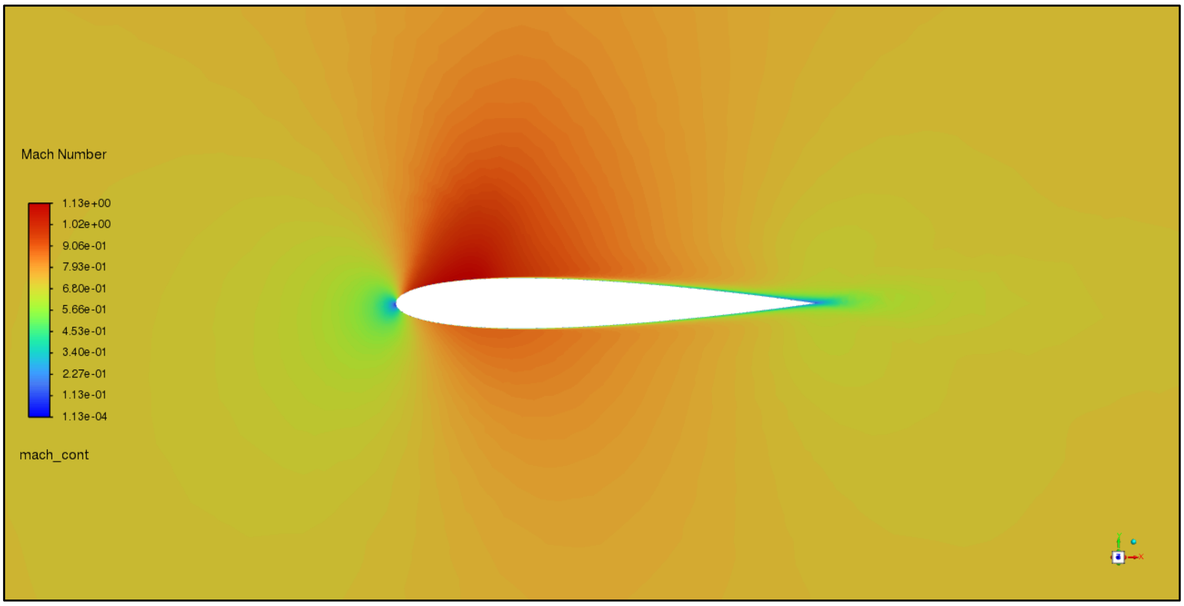

Figure 4-1 - Mach contours of the initial NACA 0012 design

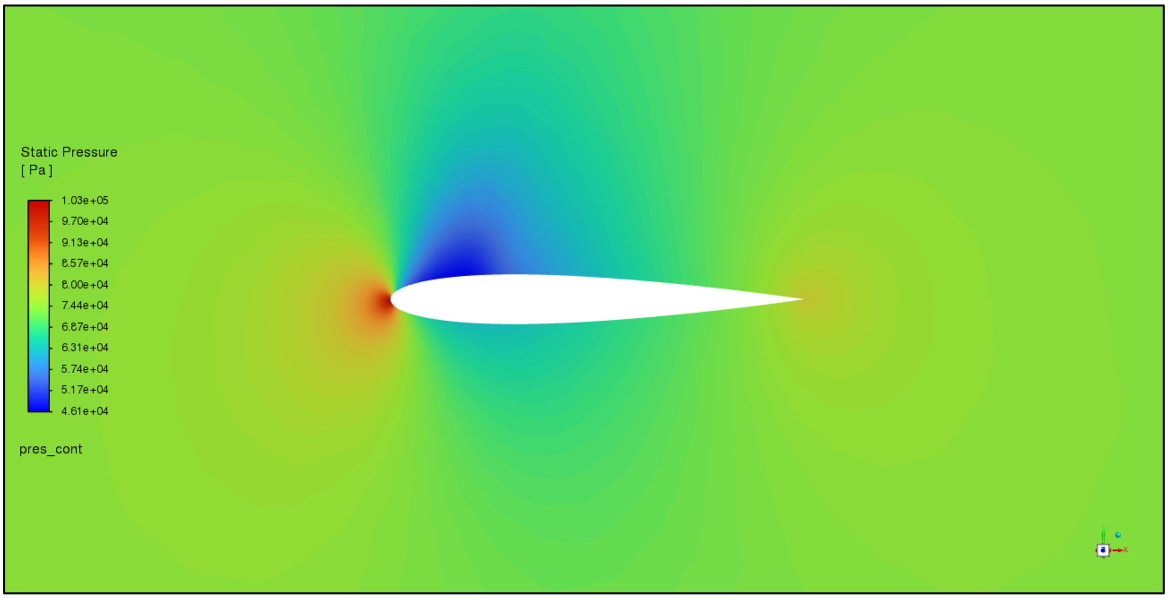

Figure 4-2 - Pressure contours of the initial NACA 0012 design

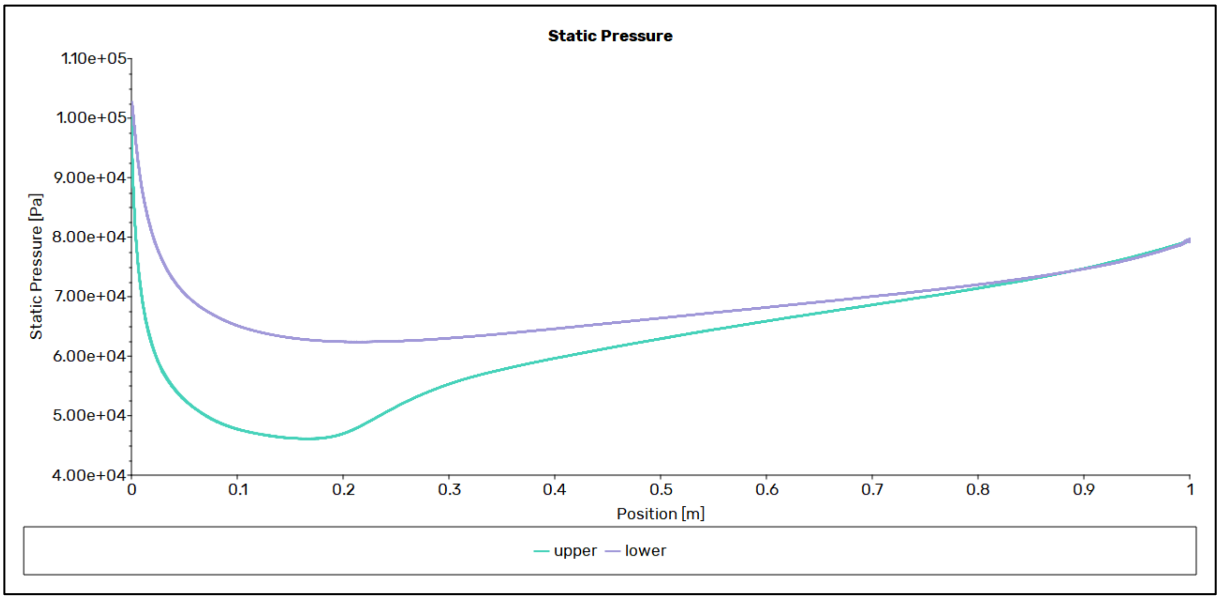

Figure 4-3 - Pressure plot of the initial NACA 0012 design

4.1.1 Optimisation

As shown in Table 4-4, both the variable and constant length trials improved the lift-to-drag ratio by 44%. This was achieved by increasing the lift coefficient by $51\% - 52\%$, at a cost of 5% increase in the drag coefficient.

Figure 4-4 and Figure 4-5 compare the initial and optimised coordinates. As shown, the airfoil profiles are reduced in thickness, and have gained a small camber, with the location of maximum thickness moving towards the trailing edge. The trailing edge of both optimised designs also features a slight downward curve, increasing the drag and lift coefficients.

All optimised airfoil coordinates are available in Appendix C: Optimised Coordinates.

Table 4-4 - Optimised NACA 0012 drag and lift coefficients

NACA 0012	C_d	C_l	Lift-to-drag Ratio
Original	0.01017664	0.23586677	23.17727364
Constant Length	0.01071617	0.35804555	33.41170866
Variable Length	0.01069893	0.35868351	33.52517588
Δ% Constant	5.30%	51.80%	44.16%
Δ% Variable	5.13%	52.07%	44.65%

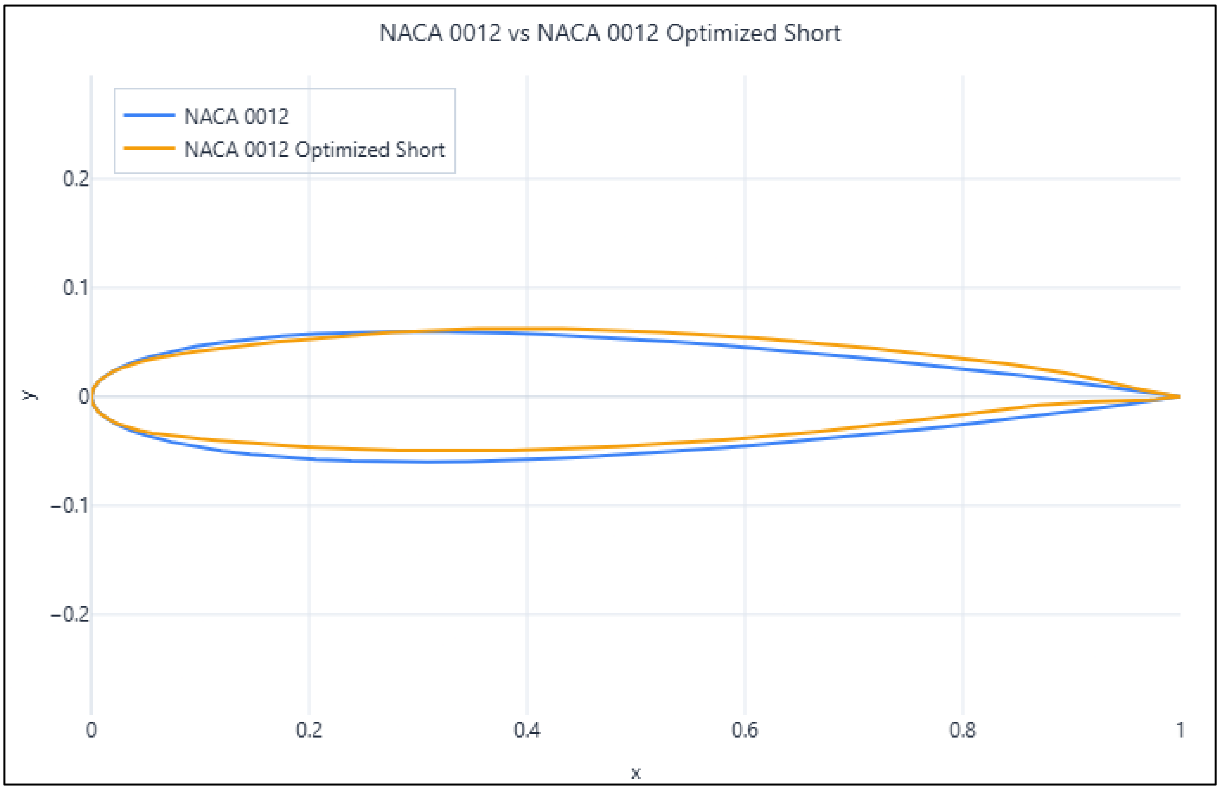

Figure 4-4 - Initial vs optimised NACA 0012 coordinates (constant length)

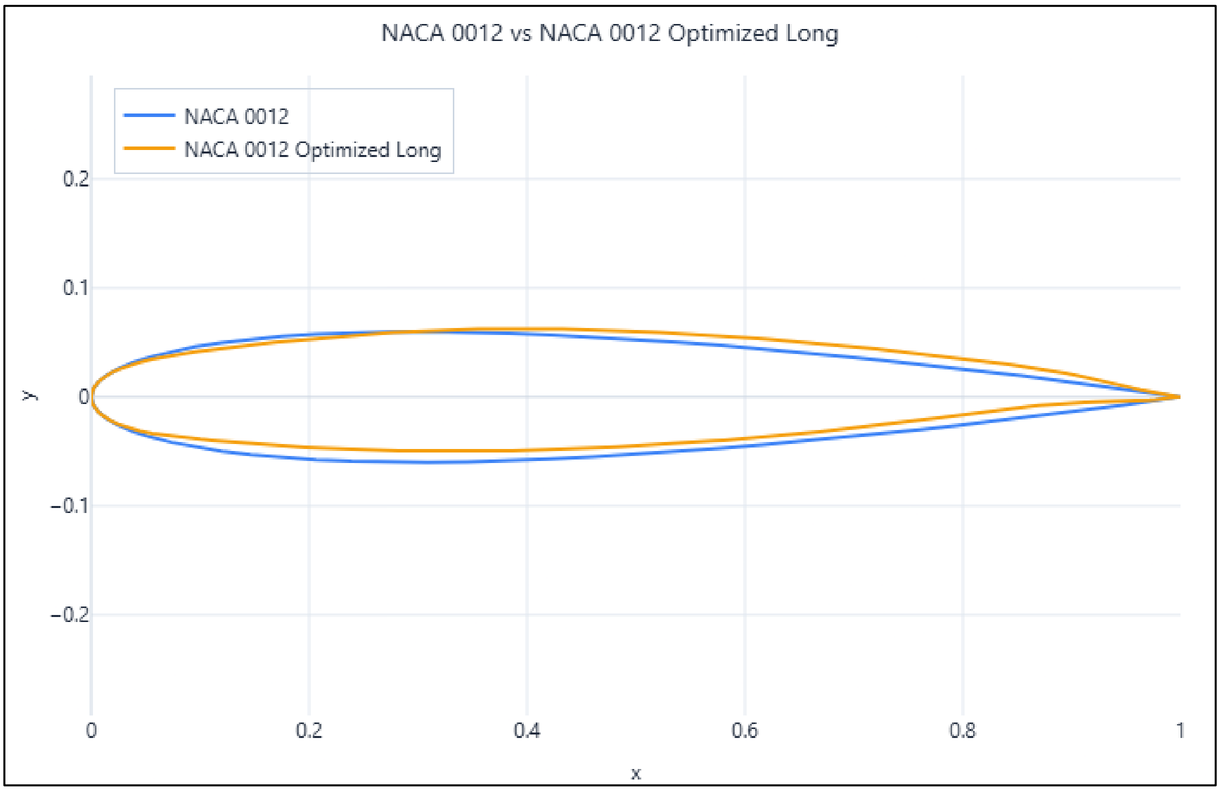

Figure 4-5 - Initial vs optimised NACA 0012 coordinates (variable length)

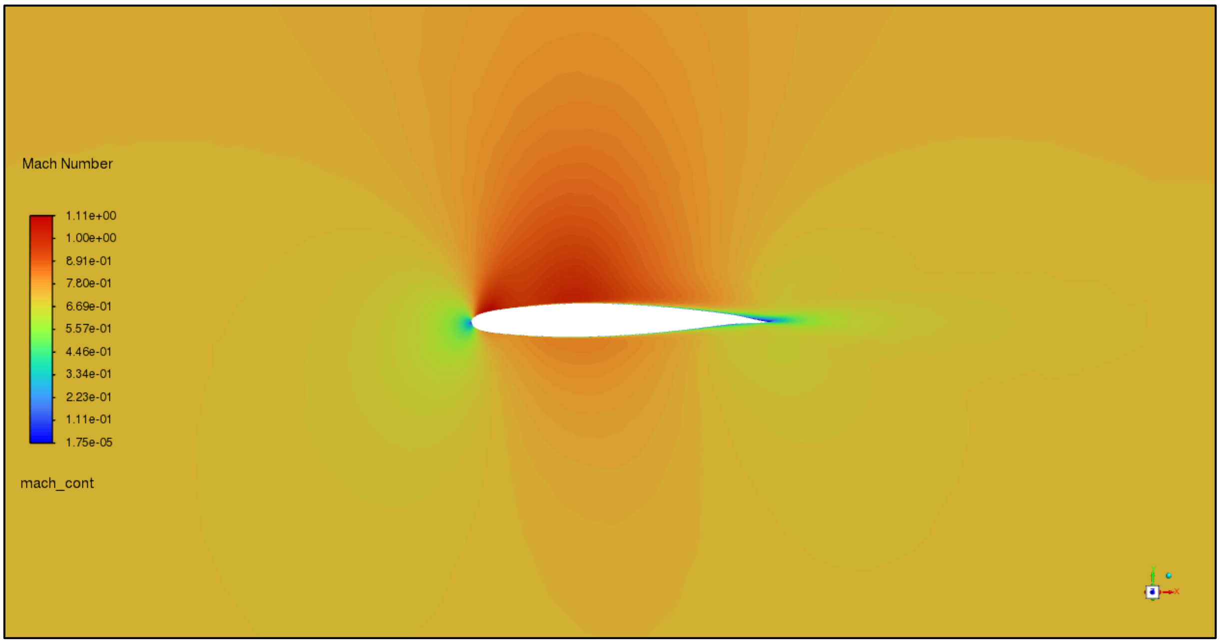

Figure 4-6 - Mach contours of the optimised NACA 0012 design (constant length)

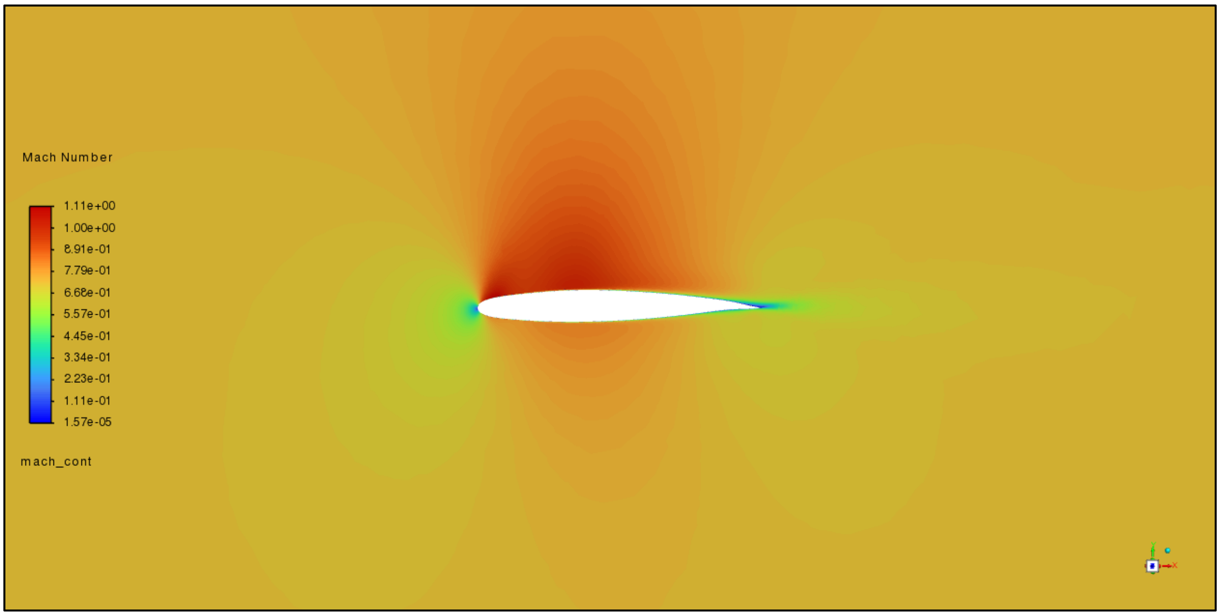

Figure 4-7 - Mach contours of the optimised NACA 0012 design (variable length)

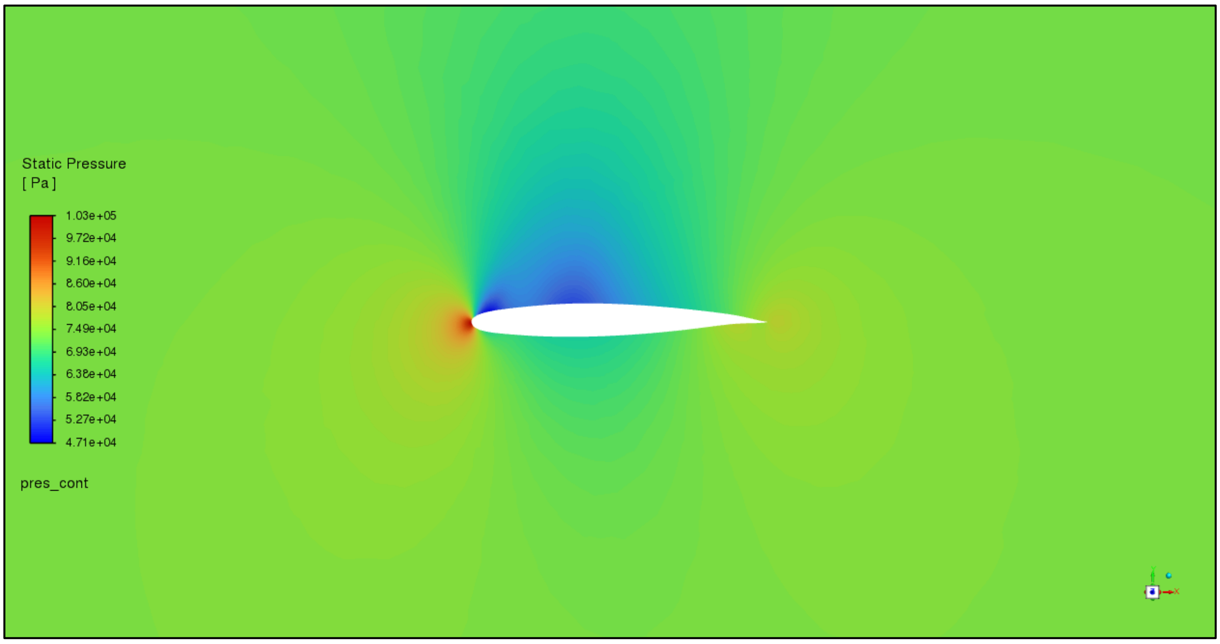

Figure 4-8 - Pressure contours of the optimised NACA 0012 design (constant length)

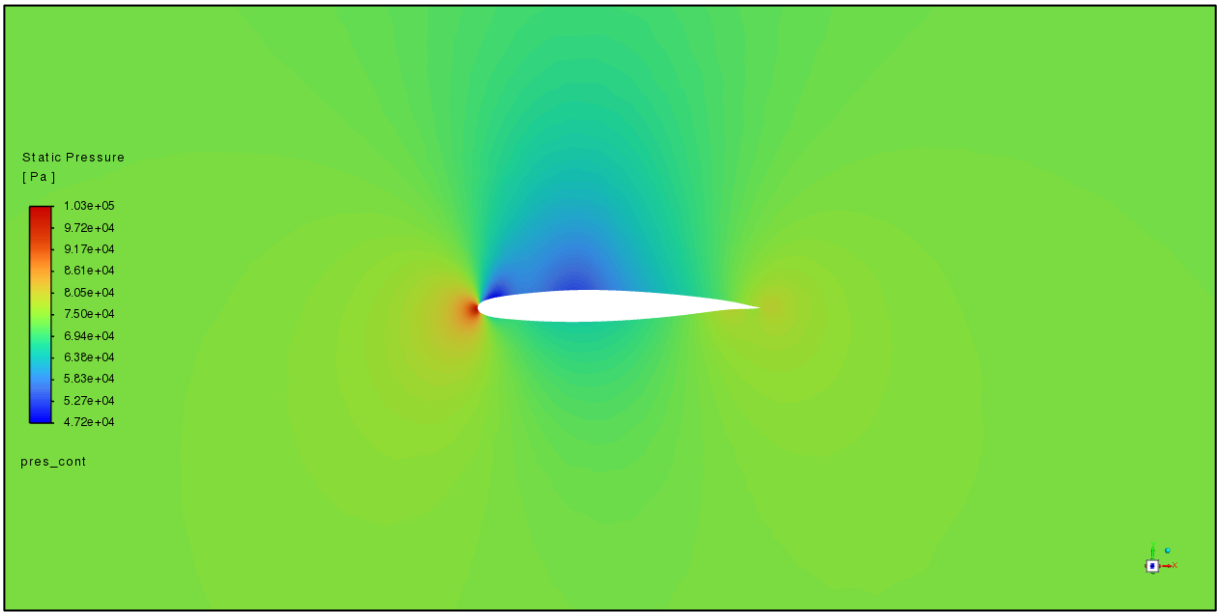

Figure 4-9 - Pressure contours of the optimised NACA 0012 design (variable length)

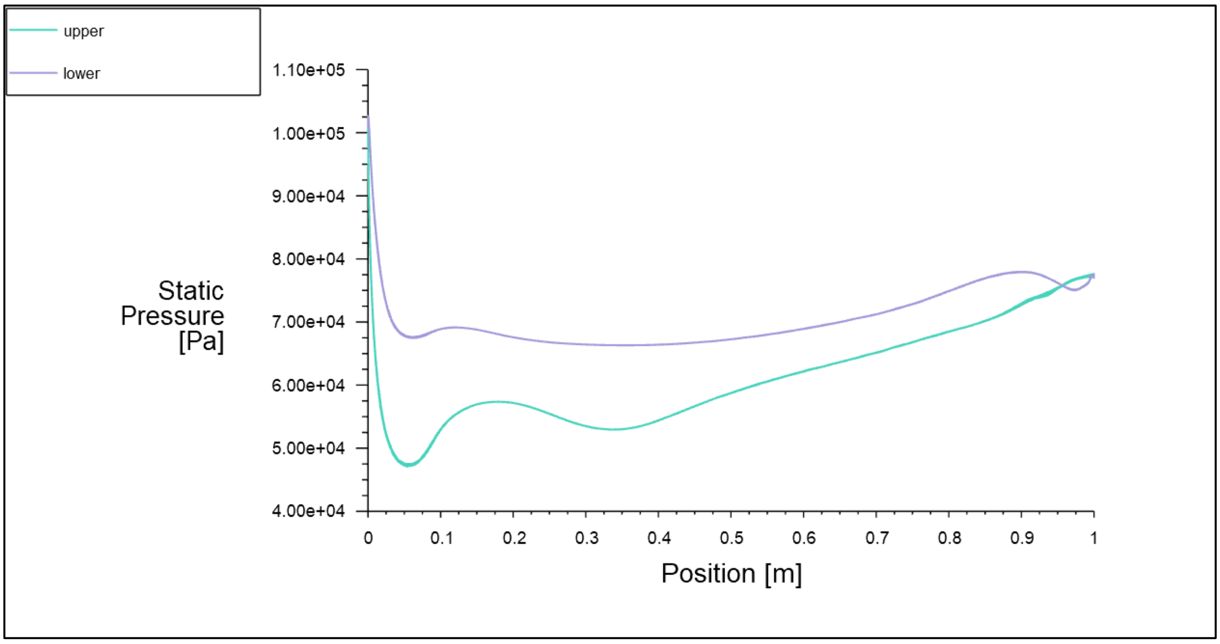

Figure 4-10 - Pressure plot of the optimised NACA 0012 design (constant length)

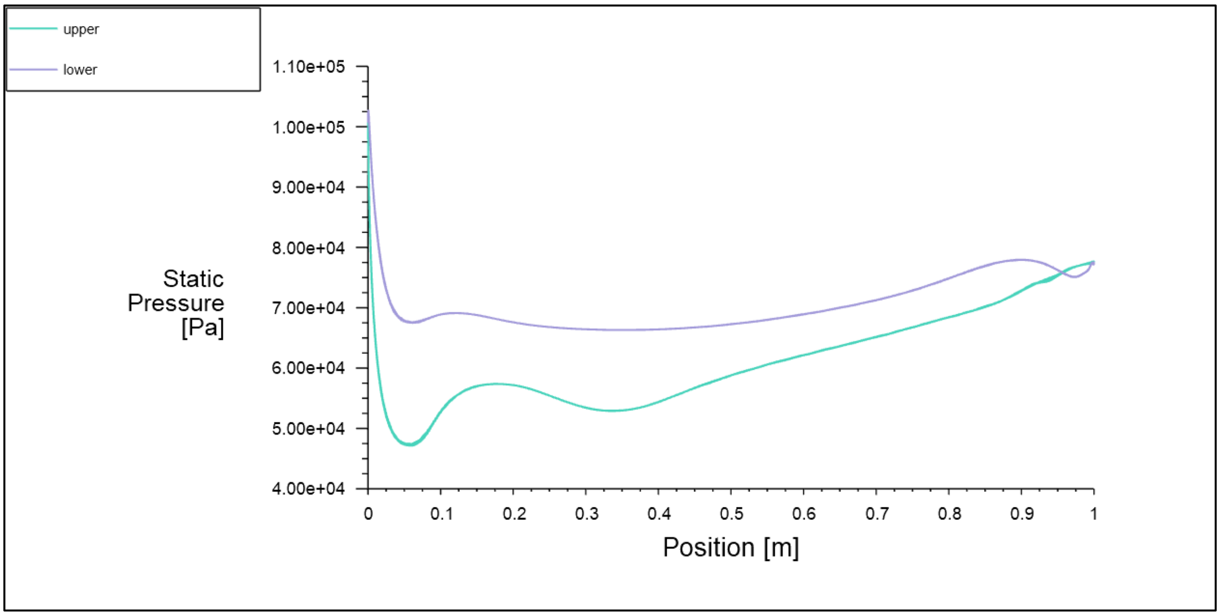

Figure 4-11 - Pressure plot of the optimised NACA 0012 design (variable length)

4.2 NACA 2412

Similarly, the NACA 2412 airfoil was tested in 2D and 3D, first at the boundary conditions used in [35], then at the selected values shown in Table 3-1.

As seen in Table 4-5, the mesh quality and skewness were satisfactory, with both 2D and 3D meshes having a minimum orthogonal quality above 0.1. The results of both the verification and test cases, in 2D and 3D, can be seen below. Similarly to results from NACA 0012 (in Table 4-2), the 2D results show a larger lift-to-drag ratio.

The comparison between recorded and theoretical results (from [35]) is shown in Table 4-6. There is good correlation between the recorded results and [35], suggesting the use of correct solver settings.

The Mach and pressure contours are displayed in Figure 4-12 and Figure 4-13, respectively, showing smooth flow. Figure 4-13 and Figure 4-14 clearly show the pressure difference between the upper and lower surfaces.

Table 4-5 - NACA 2412 mesh quality

Variable	NACA_2412	NACA_2412_2D
Maximum Skewness	0.64	
Minimum Orthogonal Quality	0.15	0.138332
Face Count	64,170	
Cell Count	679,485	24,254
Time Taken (s)	362.62	300.24
Mesh File Size (bytes)	62,675,351	649,453

Table 4-6 - Boundary conditions and results of the NACA 2412 test and reference cases

Variable	**Test**		**Reference**	
Nickname	act_3D	act_2D	ref_3D	ref_2D
Dimensions	3D	2D	3D	2D
Altitude (m)	4,114.80	4,114.80	0.00	0.00
AoA (°)	0.00	0.00	0.00	0.00
Chord Length (m)	1.49	1.49	1.00	1.00
Mach Number	0.197	0.197	0.258	0.258
C_d	0.00913	0.00875	0.00890	0.00843
C_l	0.19987	0.21663	0.20174	0.21993
Lift-to-drag Ratio	21.89	24.77	22.67	26.09
Iterations Needed	373	179	82	91
y^+	0.55	0.58	0.70	0.70
Cell Count	679,485	24,254	679,485	24,254
Time Taken (s)	2,072.00	71.46	497.98	73.69
Time per iteration (s)	5.55	0.40	6.07	0.81
Reynolds Number	4.65E+06	4.65E+06	6.01E+06	6.01E+06

Table 4-7 - NACA 2412 results compared to [35]

Variable	**ref_3D**	**ref_2D**	**Hasan**	**Δ%**
C_d	0.00890	0.00843	0.009	-1.14%
C_l	0.20174	0.21993	0.21	-4.09%
Lift-to-drag Ratio	22.67	26.09	23.33	-2.92%

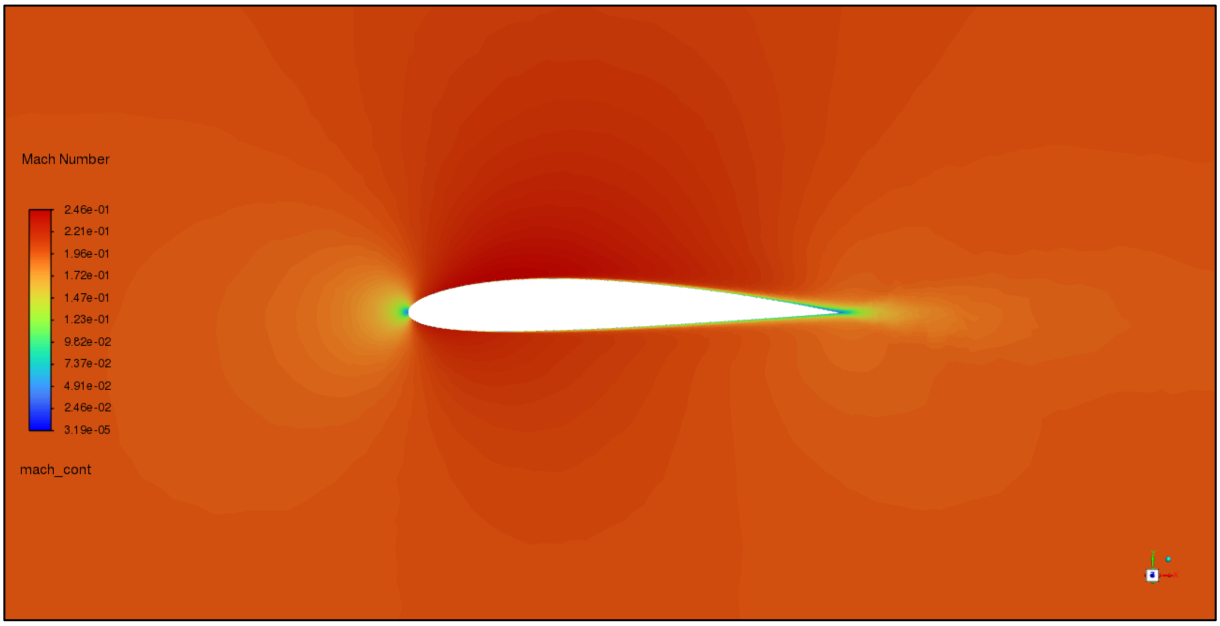

Figure 4-12 - Mach contours of the initial NACA 2412 design

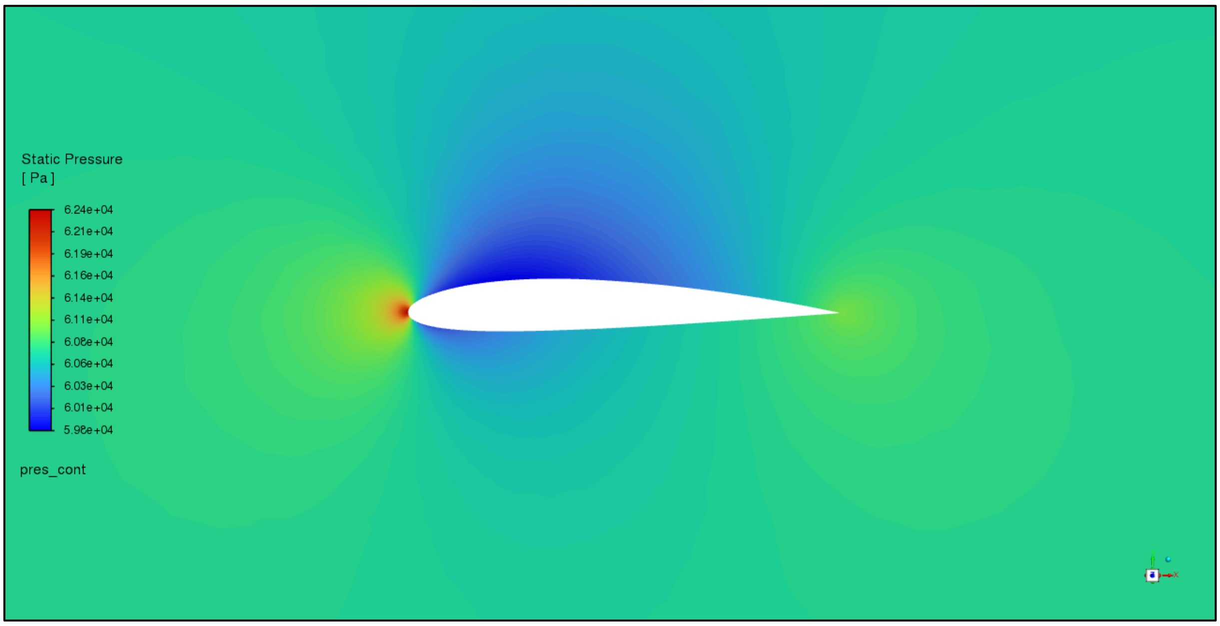

Figure 4-13 - Pressure contours of the initial NACA 2412 design

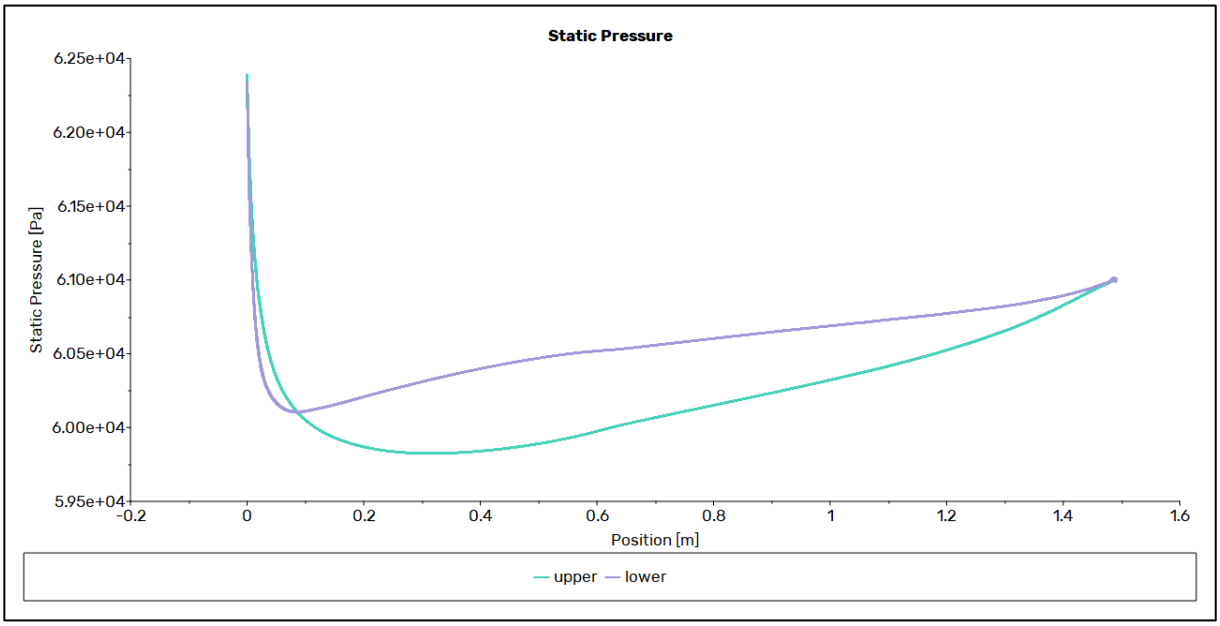

Figure 4-14 - Pressure plot of the initial NACA 2412 design

4.2.1 Optimisation

Table 4-8 shows the improved drag and lift coefficients. Both the constant and variable length trials showed large improvement. Both the lift and drag coefficients increased, but the lift coefficient showed larger changes, improving the lift-to-drag ratio.

Figure 4-15 and Figure 4-16 show the comparison of the initial and optimised designs. As seen, the constant length trial developed a curved trailing edge, creating a high-pressure point and generating lift. This is clearly visible in Figure 4-19 and Figure 4-21. The variable length trial has a smaller chord length, a sharper leading edge, and a trailing edge curving downwards.

Table 4-8 - Optimised NACA 2412 drag and lift coefficients

NACA 2412	C_d	C_l	Lift-to-drag Ratio
Original	0.00913238	0.19987044	21.88590926
Constant Length	0.01035371	0.48186869	46.54067866
Variable Length	0.01489452	0.82690445	55.51736142
Δ% Constant	13.37%	141.09%	112.65%
Δ% Variable	63.10%	313.72%	153.67%

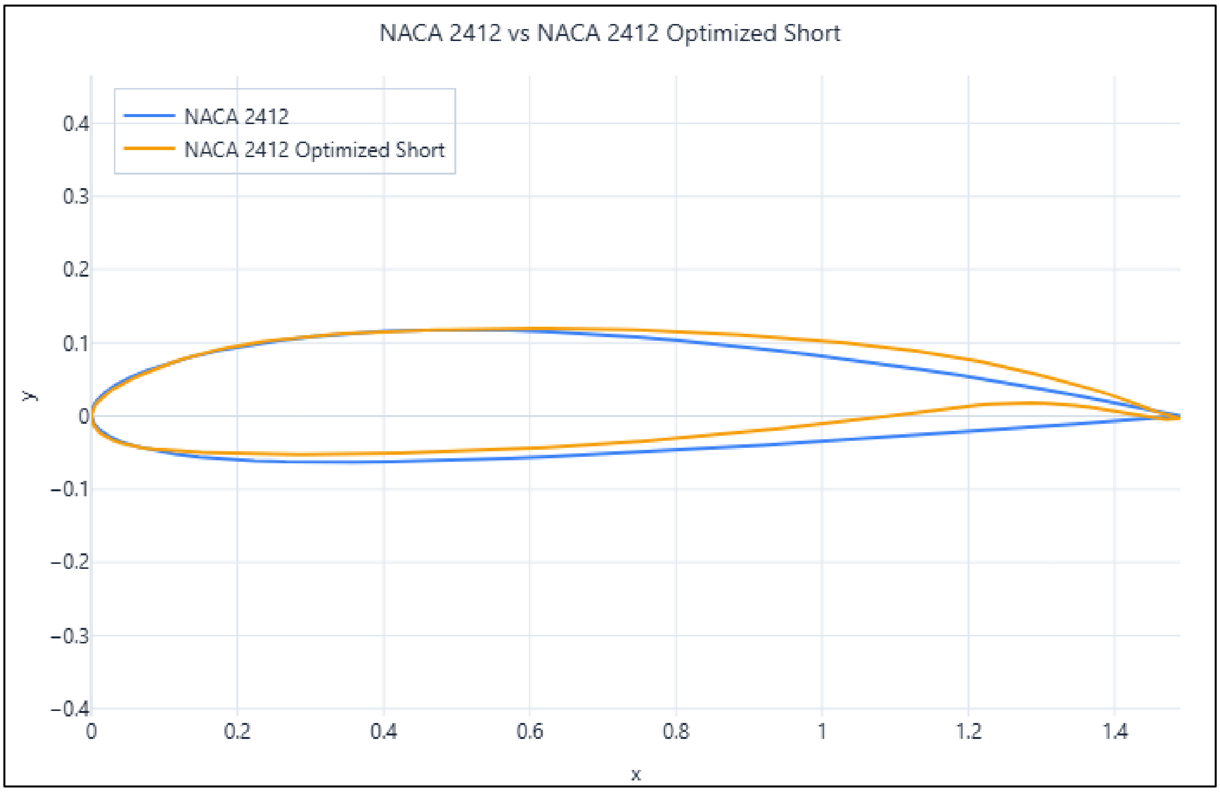

Figure 4-15 - Initial vs optimised NACA 2412 coordinates (constant length)

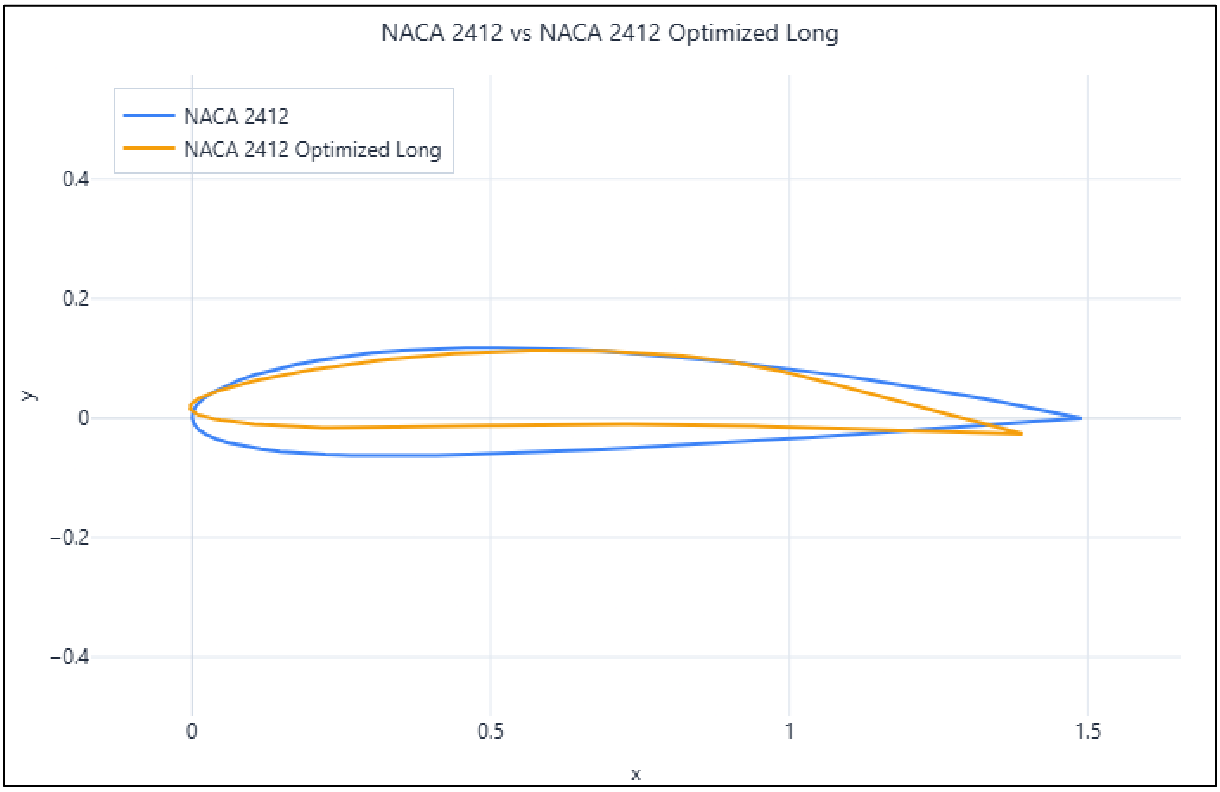

Figure 4-16 - Initial vs optimised NACA 2412 coordinates (variable length)

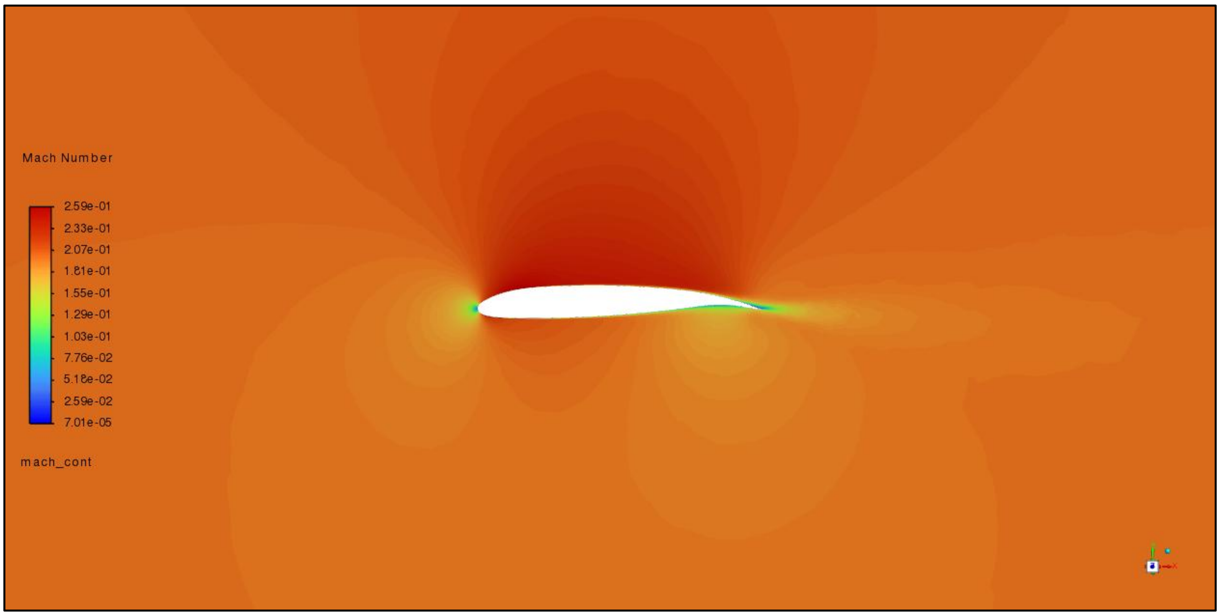

Figure 4-17 - Mach contours of the optimised NACA 2412 design (constant length)

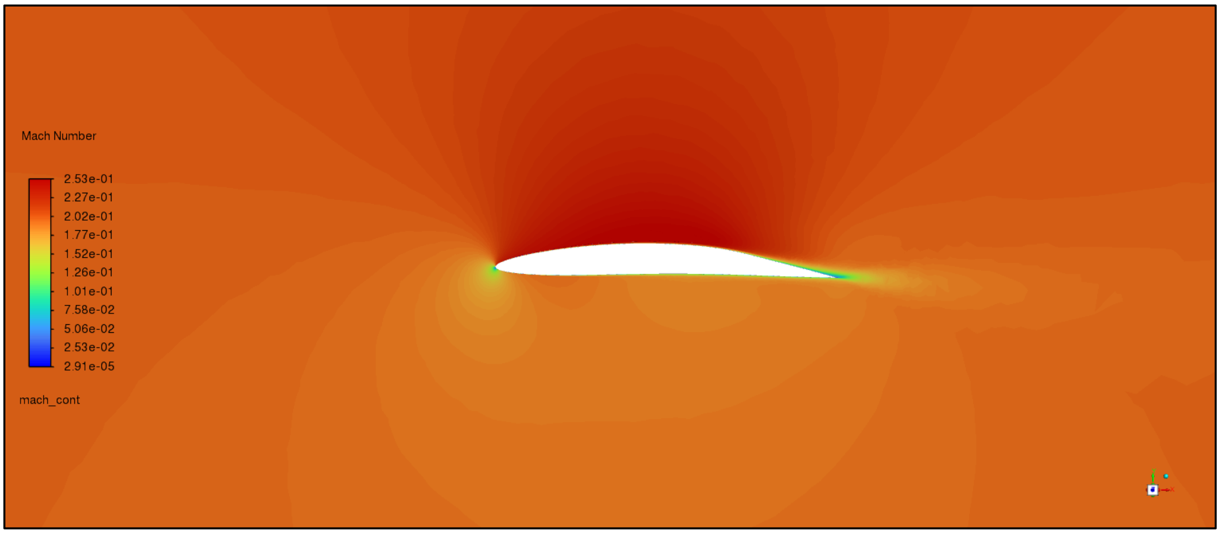

Figure 4-18 - Mach contours of the optimised NACA 2412 design (variable length)

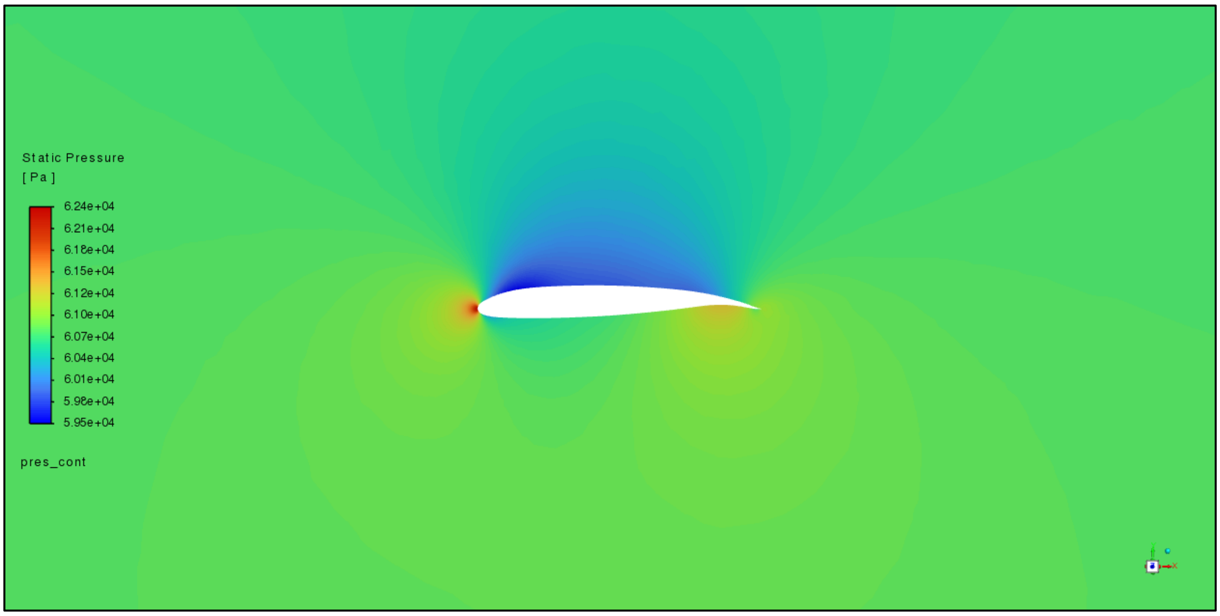

Figure 4-19 - Pressure contours of the optimised NACA 2412 design (constant length)

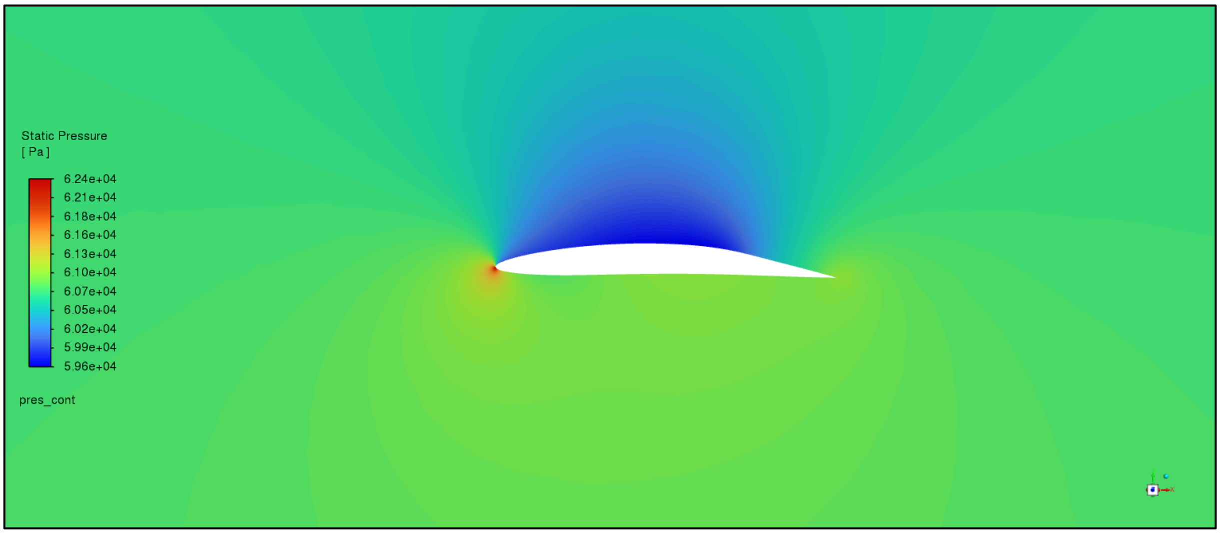

Figure 4-20 - Pressure contours of the optimised NACA 2412 design (variable length)

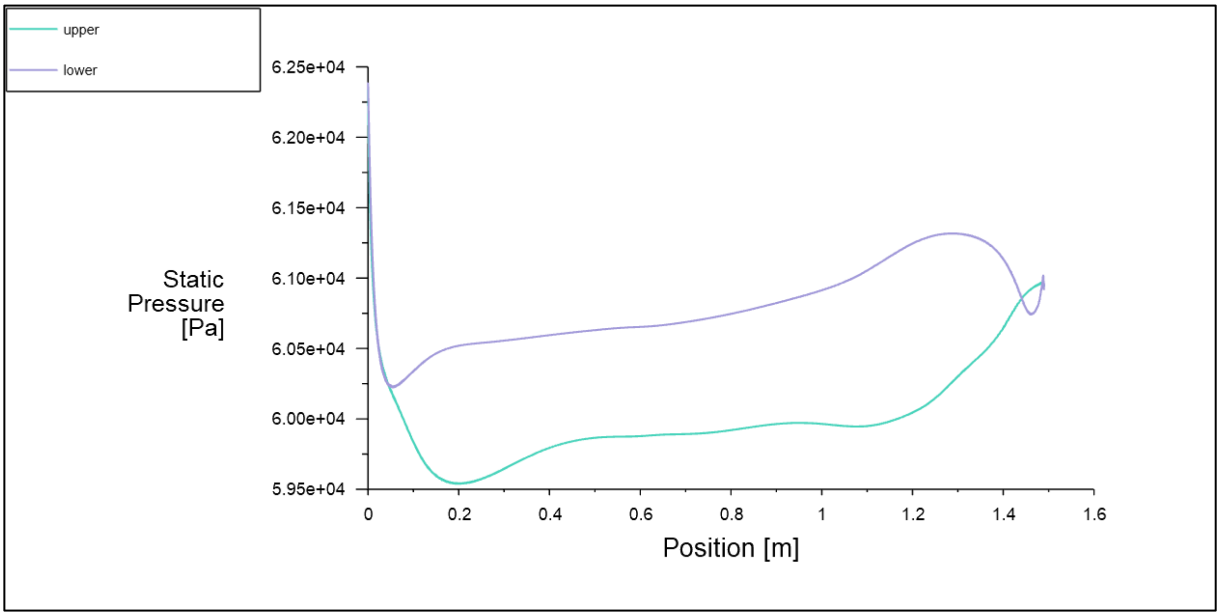

Figure 4-21 - Pressure plot of the optimised NACA 2412 design (constant length)

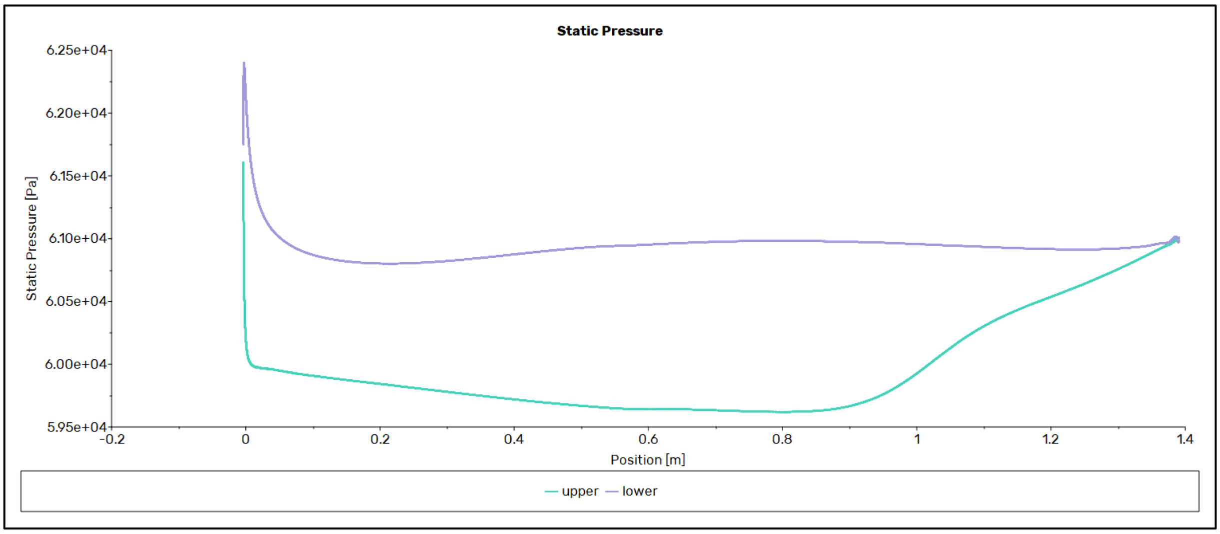

Figure 4-22 - Pressure plot of the optimised NACA 2412 design (variable length)

4.3 BACXXX

The quality of the meshes is shown in Table 4-9. The 3D mesh has a maximum skewness of 0.64. The minimum orthogonal quality of both the 2D and the 3D meshes is above 0.1.

The results of the verification and final trials are displayed in Table 4-10. The lift-to-drag ratio of the 2D trials is higher than that of the 3D trials. The comparison between results is shown in Table 4-11. There is a large difference between the trials.

The Mach contours are shown in Figure 4-23, and the pressure contours are shown in Figure 4-24. Both show the formation of a transonic shock wave. The location of the shock wave is also visible on Figure 4-25.

Table 4-9 - BACXXX mesh quality

Variable	BACXXX	BACXXX_2D
Maximum Skewness	0.64	
Minimum Orthogonal Quality	0.213	0.234809
Face Count	113,260	
Cell Count	1,361,805	20,970
Time Taken (s)	2333.83	148.63
Mesh File Size (bytes)	125,781,263	545,173

Table 4-10 - Boundary conditions and results of the BACXXX test and reference cases

Variable	Test		Reference	
Nickname	act_3D	act_2D	ref_3D	ref_2D
Dimensions	3D	2D	3D	2D
Altitude (m)	10,576.56	10,576.56	9,000.00	9,000.00
AoA (°)	2.00	2.00	2.00	2.00
Chord Length (m)	8.33	8.33	14.34	14.34
Mach Number	0.850	0.850	0.850	0.850
C_d	0.05844	0.08122	0.05838	0.07862
C_l	0.36741	0.61110	0.36604	0.60108
Lift-to-drag Ratio	6.29	7.52	6.27	7.65
Iterations Needed	750	500	750	1,000
y^+	0.40	0.40	0.73	0.73
Cell Count	1,361,805	20,970	1,361,805	20,970
Time Taken (s)	5,904.74	210.83	11,826.47	481.45
Time per iteration (s)	7.87	0.42	15.77	0.48
Reynolds Number	5.62E+07	5.62E+07	1.16E+08	1.16E+08

Table 4-11 - BACXXX results compared to [37]

Variable	ref_3D	ref_2D	Abdel-Raheem	Δ%
C_d	0.05838	0.07862	0.85	-1356.10%
C_l	0.36604	0.60108	4.5	-1129.38%
Lift-to-drag Ratio	6.27	7.65	5.29	15.57%

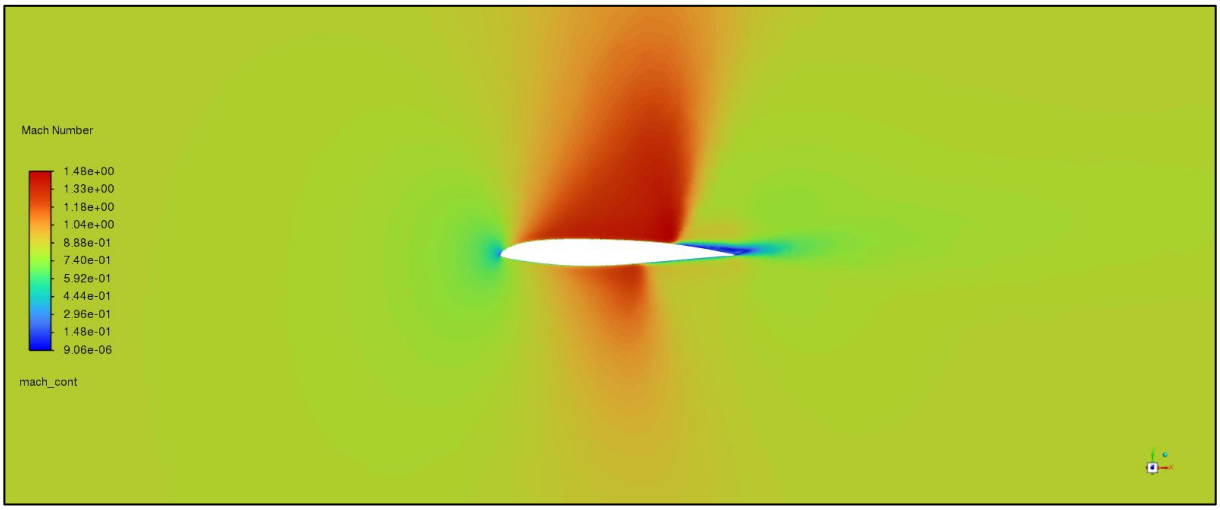

Figure 4-23 - Mach contours of the initial BACXXX design

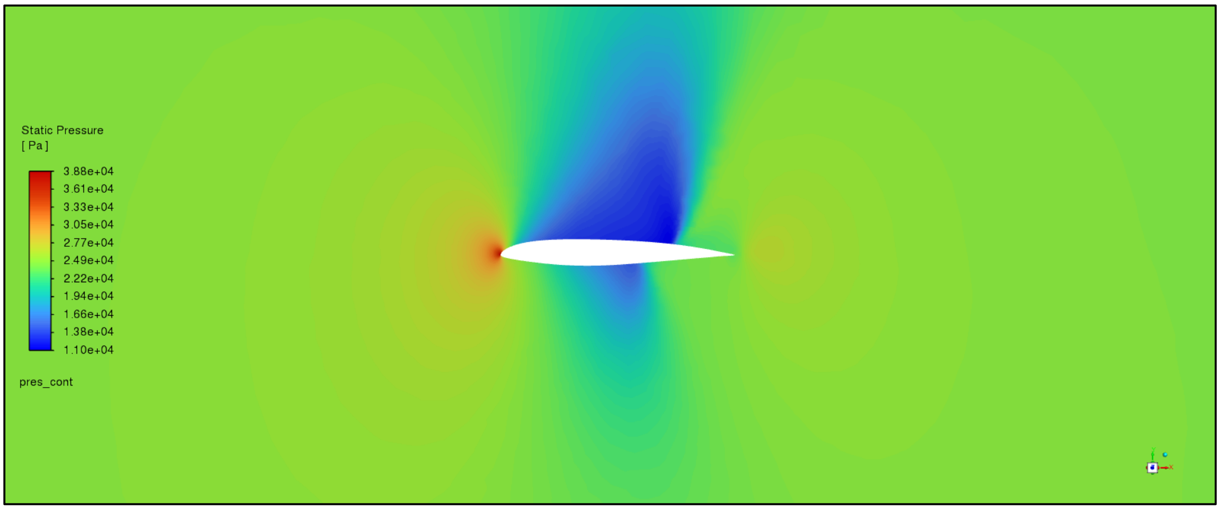

Figure 4-24 - Pressure contours of the initial BACXXX design

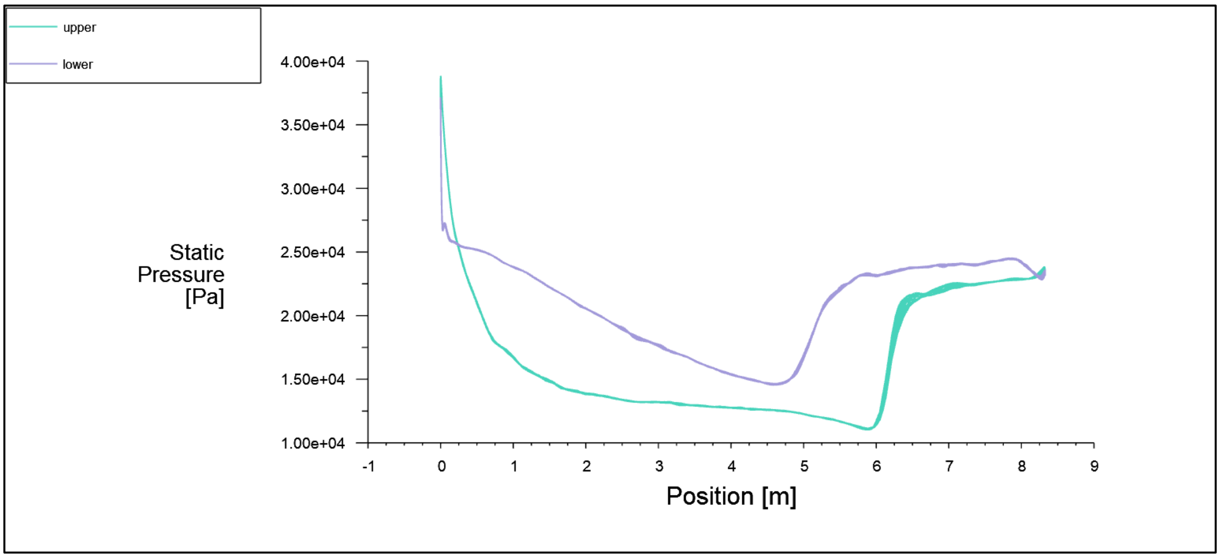

Figure 4-25 - Pressure plot of the initial BACXXX design

4.3.1 Optimisation

The improvements made from optimised mesh can be seen in Table 4-12. Overall, the lift-to-drag ratio was improved in both the constant and variable length trials. In the constant length trial, both the lift and drag coefficients increased. In the variable length trial, the drag coefficient decreased and the lift coefficient increased.

The coordinates of the optimised designs are visible in Figure 4-26 and Figure 4-27. The constant length design features a highly curved trailing edge.

The variable length design is very streamlined, and, as shown in Figure 4-29 and Figure 4-31, eliminates the transonic shock wave entirely, massively reducing the drag coefficient.

Table 4-12 - Optimised BACXXX drag and lift coefficients

BACXXX	C_d	C_l	Lift-to-drag Ratio
Original	0.058444341	0.367406172	6.286428467
Constant Length	0.03250261	0.60611258	18.64812026
Variable Length	0.00670604	0.3196861	47.6713679
Δ% Constant	-44.39%	64.97%	196.64%
Δ% Variable	-88.53%	-12.99%	658.32%

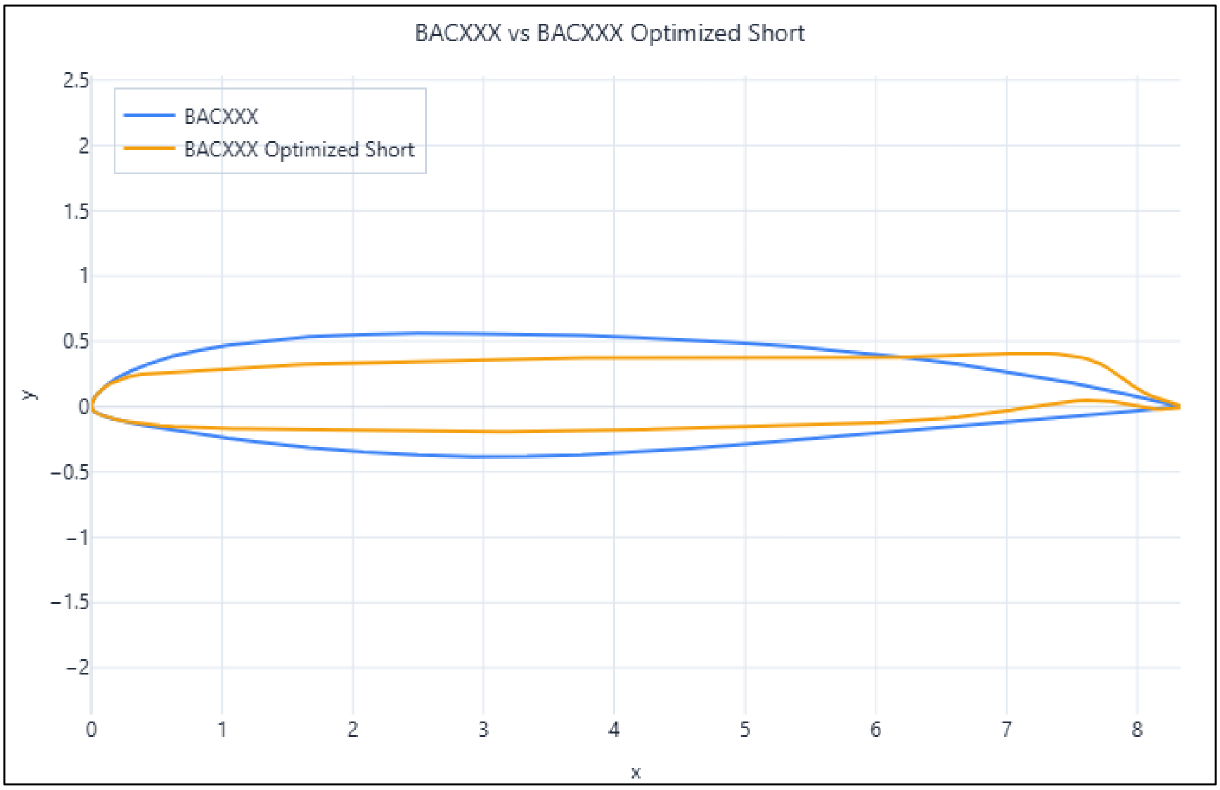

Figure 4-26 - Initial vs optimised BACXXX coordinates (constant length)

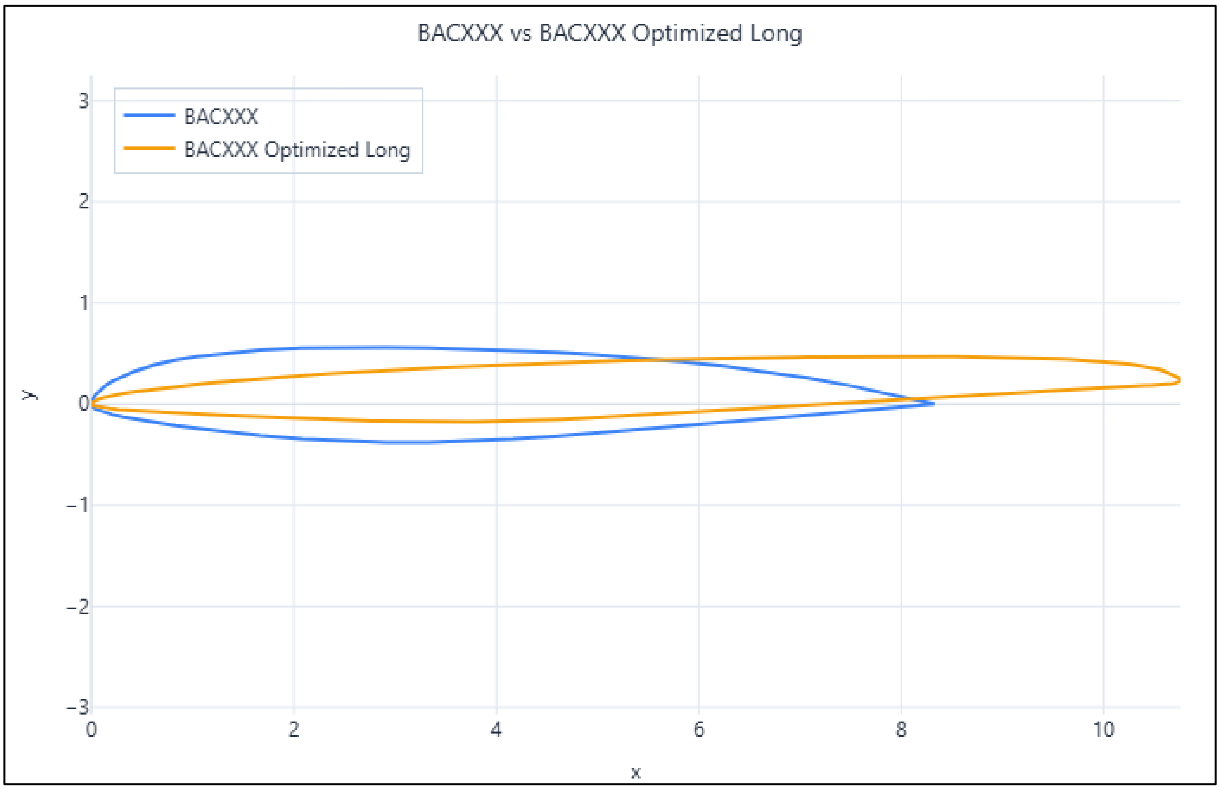

Figure 4-27 - Initial vs optimised BACXXX coordinates (variable length)

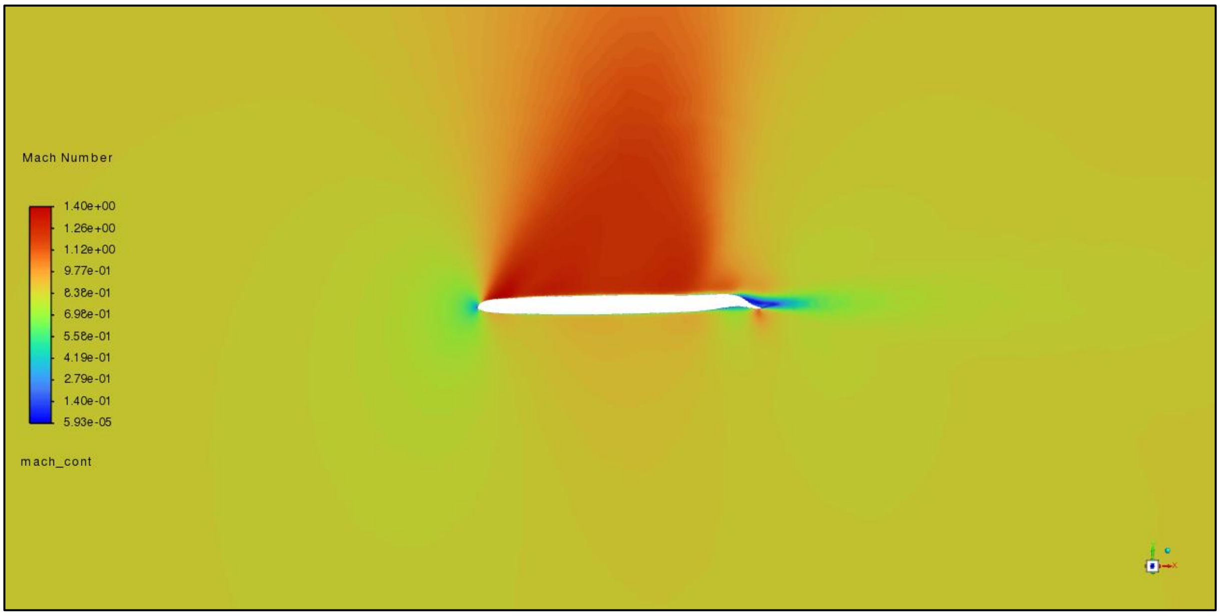

Figure 4-28 - Mach contours of the optimised BACXXX design (constant length)

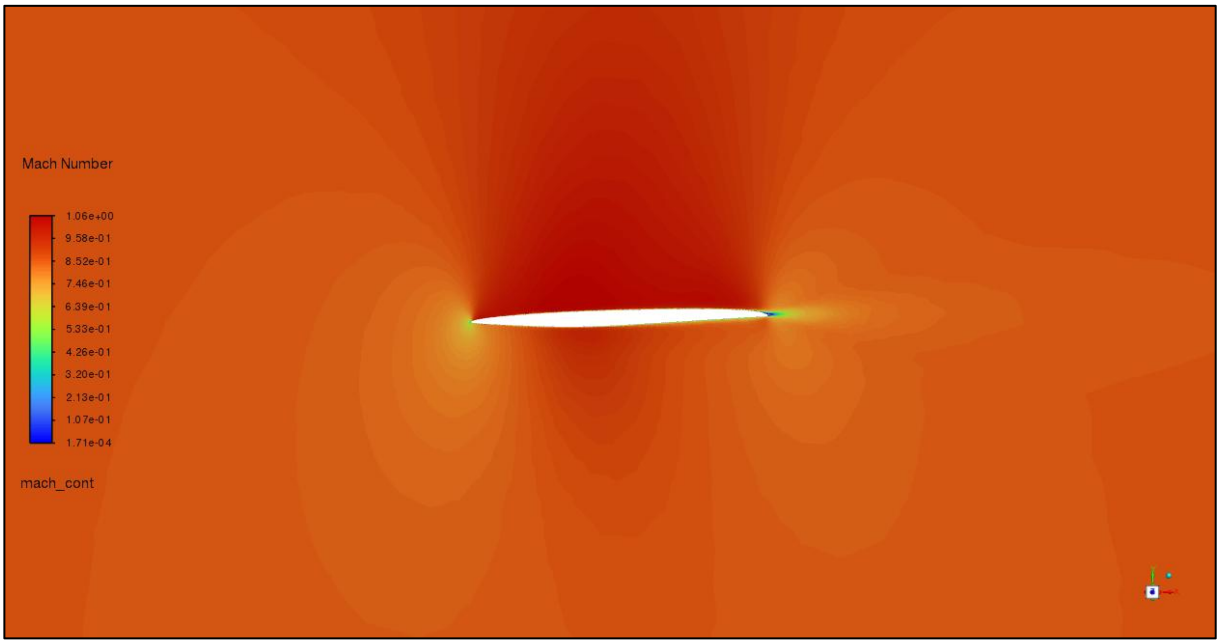

Figure 4-29 - Mach contours of the optimised BACXXX design (variable length)

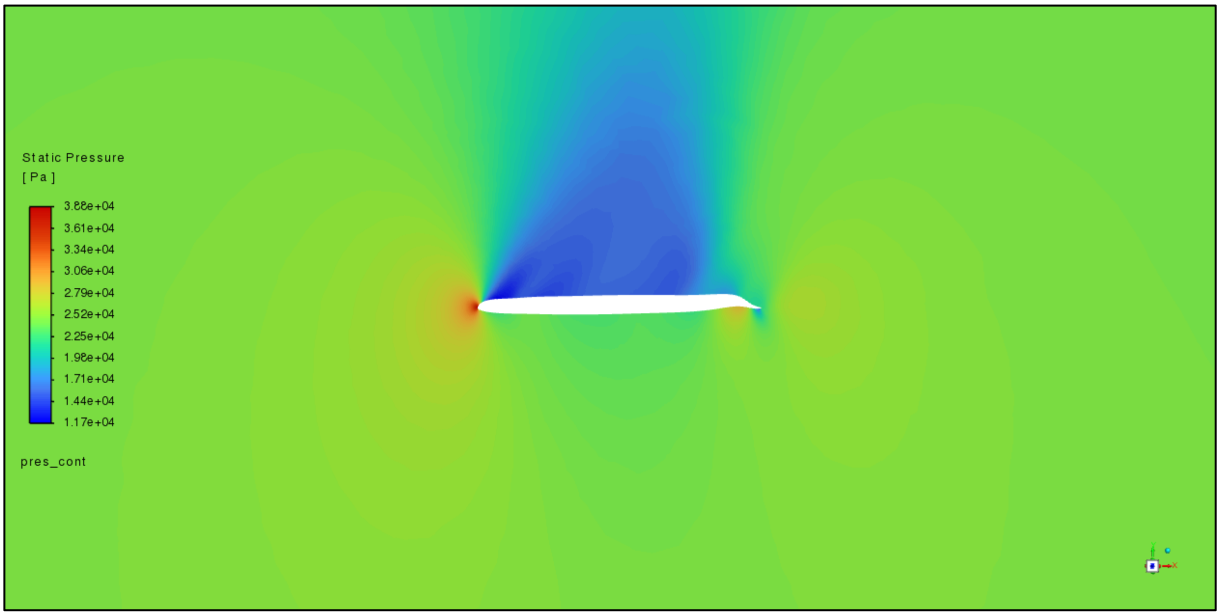

Figure 4-30 - Pressure contours of the optimised BACXXX design (constant length)

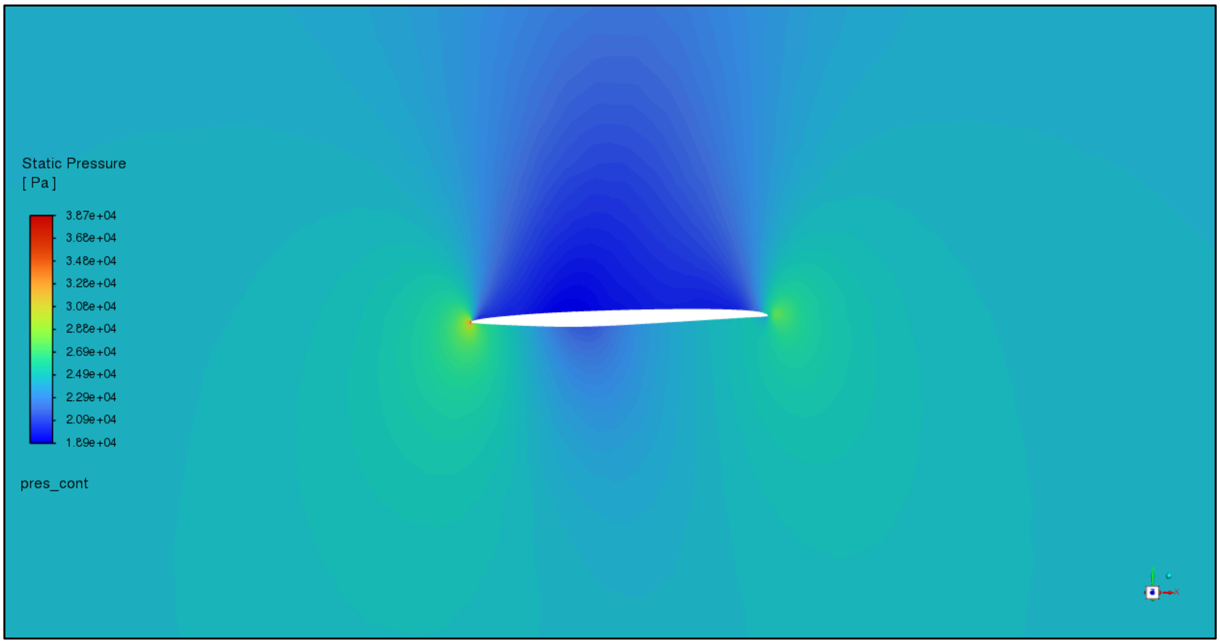

Figure 4-31 - Pressure contours of the optimised BACXXX design (variable length)

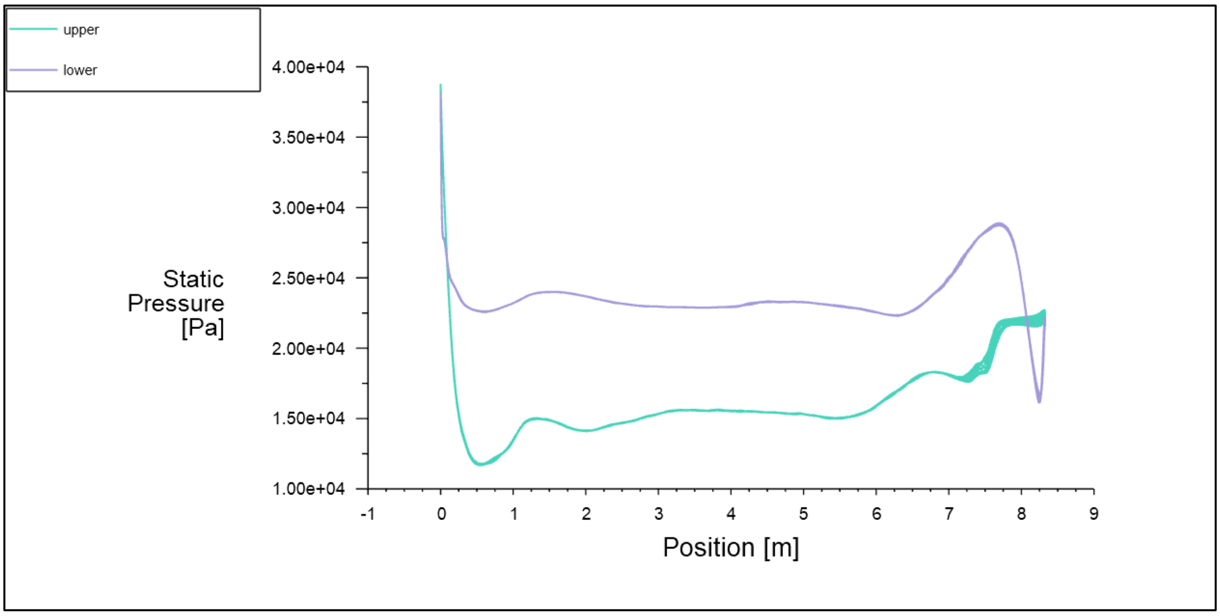

Figure 4-32 - Pressure plot of the optimised BACXXX design (constant length)

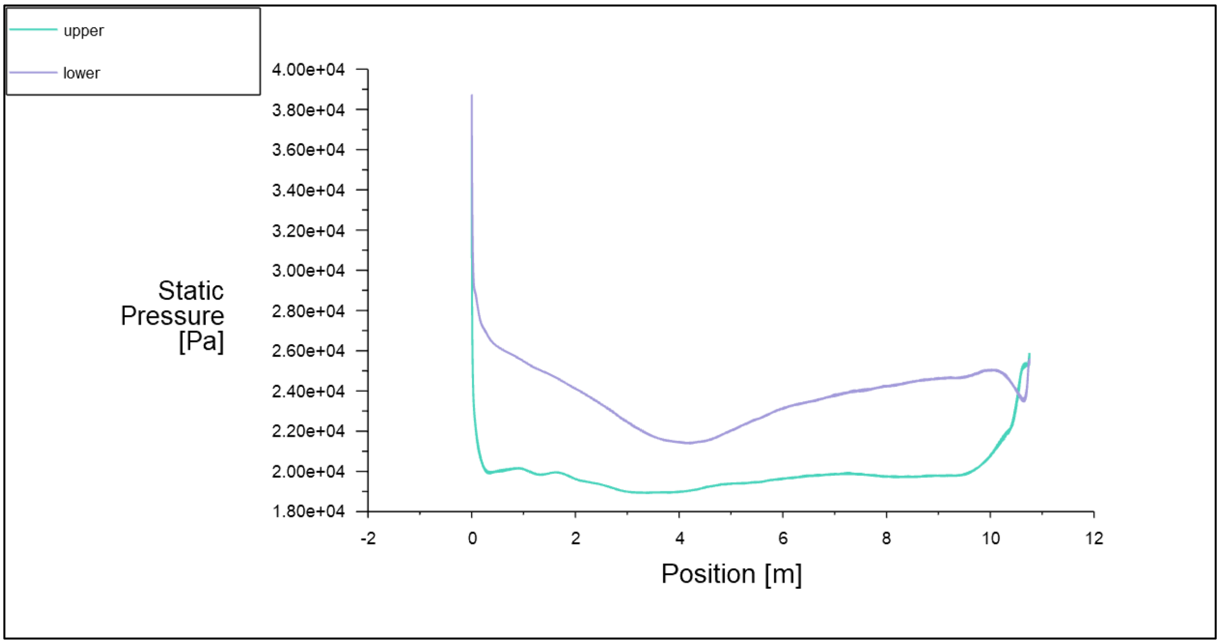

Figure 4-33 - Pressure plot of the optimised BACXXX design (variable length)

4.4 NASA SC(2)-0714 in Transonic Flow

As above, the quality of the meshes is shown in Table 4-13. The results are shown in Table 4-14. Table 4-15 compares the results to [38].

Table 4-13 - NASA SC(2)-0714 mesh quality

Variable	NASA_SC(2)_0714	NASA_SC(2)_0714_2D
Maximum Skewness	0.64	
Minimum Orthogonal Quality	0.111	0.600745
Face Count	52,488	
Cell Count	456,363	24,262
Time Taken (s)	218.16	1170.9
Mesh File Size (bytes)	42,668,564	649,988

Table 4-14 - Boundary conditions and results of the NASA SC(2)-0714 test and reference cases

Variable	Test		Reference	
Nickname	act_3D	act_2D	ref_3D	ref_2D
Dimensions	3D	2D	3D	2D
Altitude (m)	11,581.83	11,581.83	0.00	0.00
AoA (°)	2.00	2.00	0.00	0.00
Chord Length (m)	3.51	3.51	1.00	1.00
Mach Number	0.850	0.850	0.800	0.800
C_d	0.07842	0.07471	0.02001	0.01813
C_l	0.19462	0.08484	0.23339	0.20521
Lift-to-drag Ratio	2.48	1.14	11.66	11.32
Iterations Needed	2,665	483	315	2,000
y^+	2.00	2.00	1.20	1.20
Cell Count	456,363	24,262	456,363	24,262
Time Taken (s)	3,624.66	178.40	1,892.46	517.07
Time per iteration (s)	1.36	0.37	6.01	0.26
Reynolds Number	2.10E+07	2.10E+07	1.86E+07	1.86E+07

Table 4-15 - NASA SC(2)-0714 results compared to [38]

Variable	ref_3D	ref_2D	Sri	Δ%
C_d	0.02001	0.01813	0.0087	56.52%
C_l	0.23339	0.20521	0.068	70.86%
Lift-to-drag Ratio	11.66	11.32	7.82	32.99%

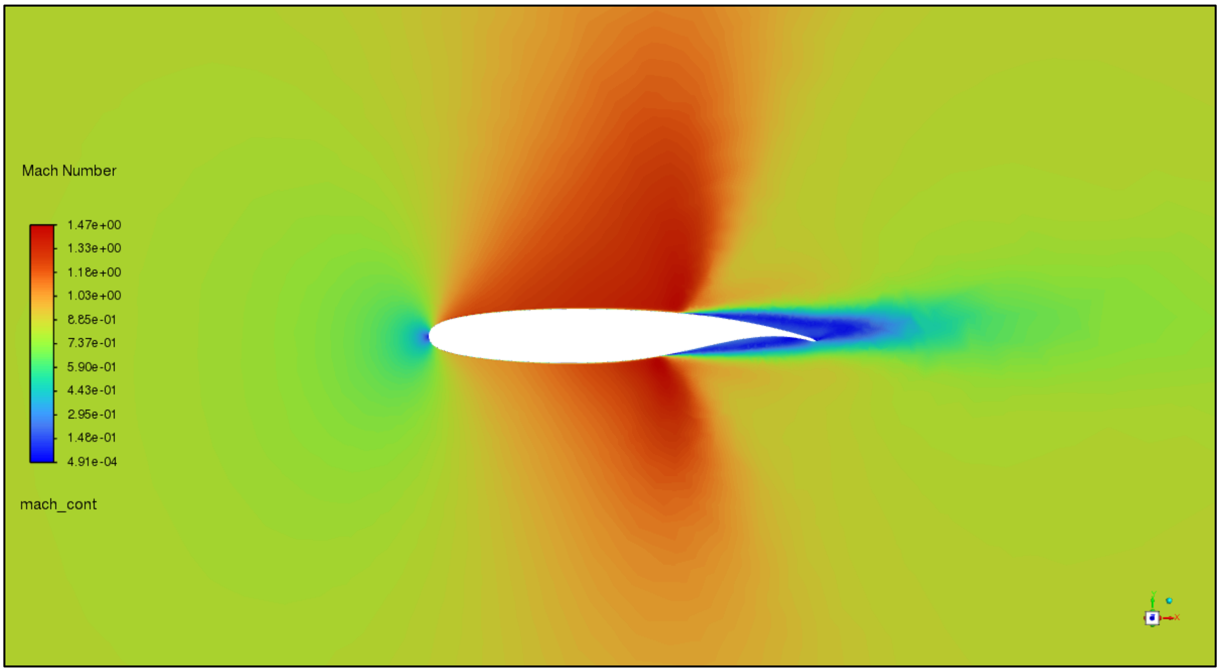

Figure 4-34 - Mach contours of the initial NASA SC(2)-0714 design

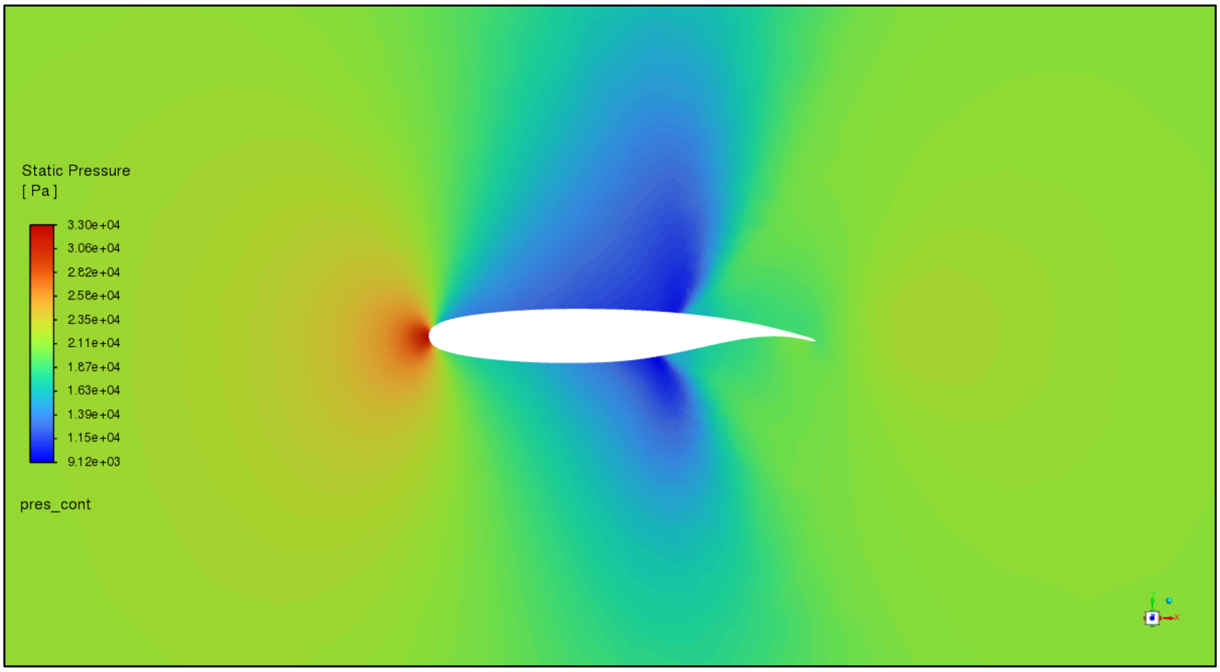

Figure 4-35 - Pressure contours of the initial NASA SC(2)-0714 design

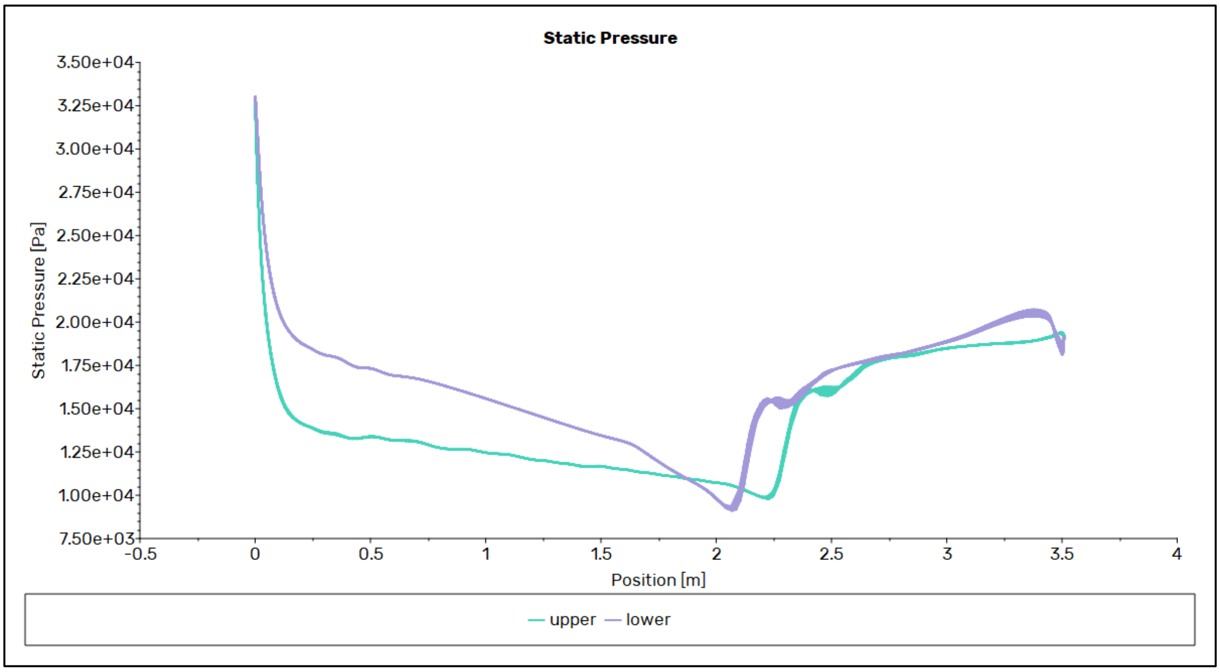

Figure 4-36 - Pressure plot of the initial NASA SC(2)-0714 design

4.4.1 Optimisation

The results from the optimised geometries are shown in Table 4-16. Both trials showed similar improvement, with a decrease in drag coefficient and increase in lift coefficient.

Figure 4-37 and Figure 4-38 show the optimised airfoil outlines. Figure 4-39 and Figure 4-40 show the optimised Mach contours, Figure 4-41 and Figure 4-42 show the optimised pressure contours, and Figure 4-43 and Figure 4-44 show the optimised pressure plots.

Table 4-16 - Optimised NASA SC(2)-0714 drag and lift coefficients

NASA SC(2)-0714	C_d	C_l	Lift-to-drag Ratio
Original	0.07842316	0.19462367	2.481711653
Constant Length	0.03486004	0.68061969	19.52435195
Variable Length	0.03481286	0.68170404	19.58196023
Δ% Constant	-55.55%	249.71%	686.73%
Δ% Variable	-55.61%	250.27%	689.05%

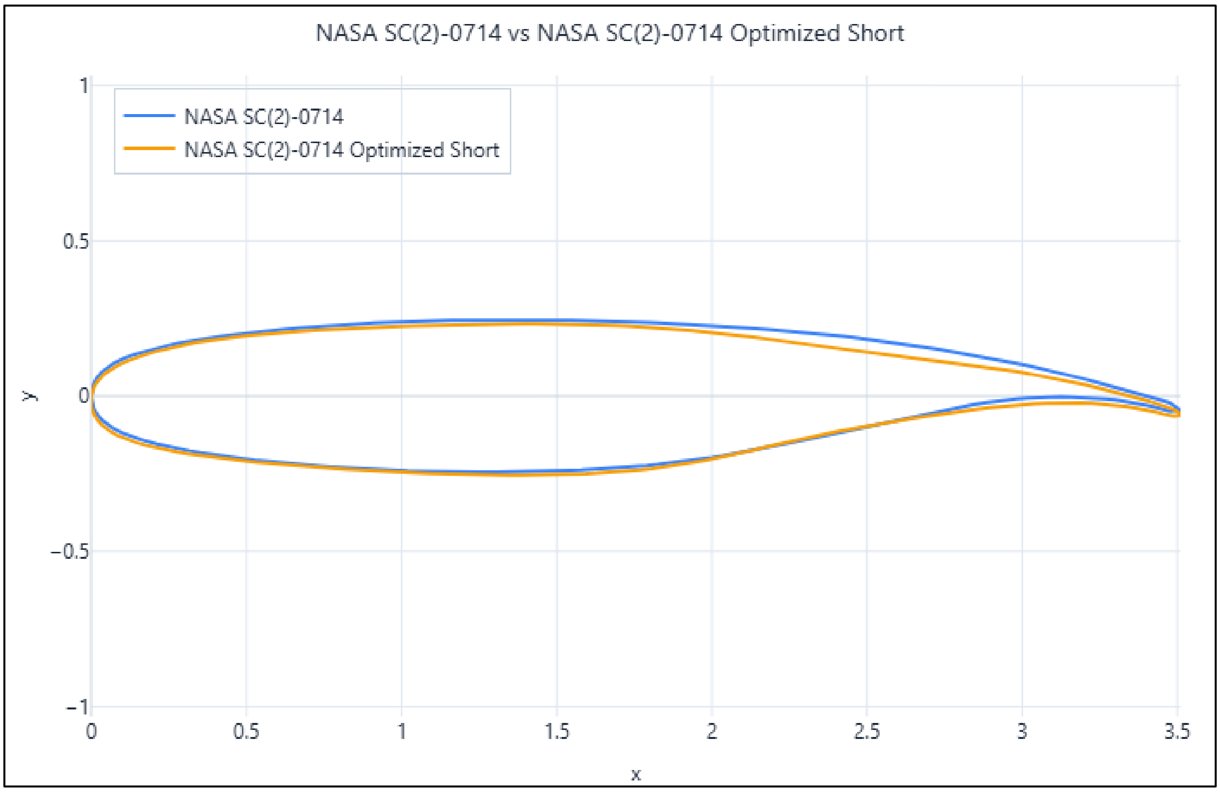

Figure 4-37 - Initial vs optimised NASA SC(2)-0714 coordinates (constant length)

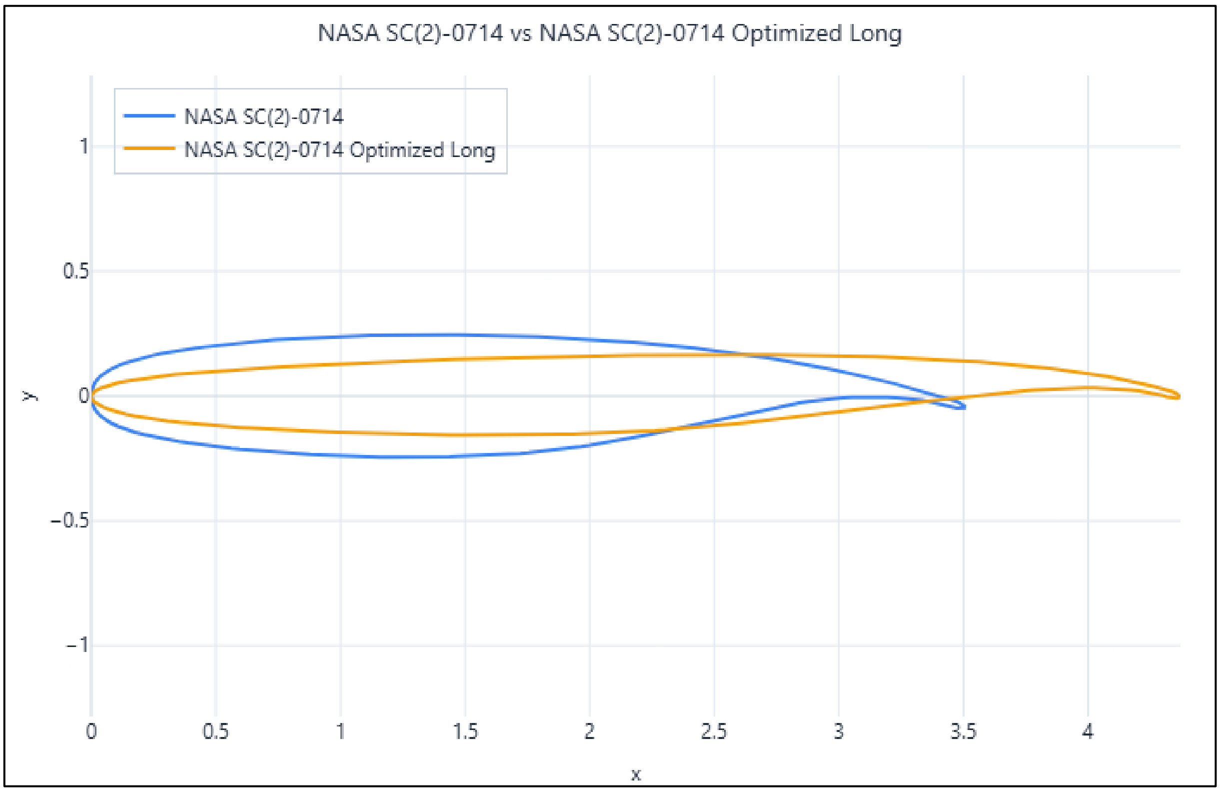

Figure 4-38 - Initial vs optimised NASA SC(2)-0714 coordinates (variable length)

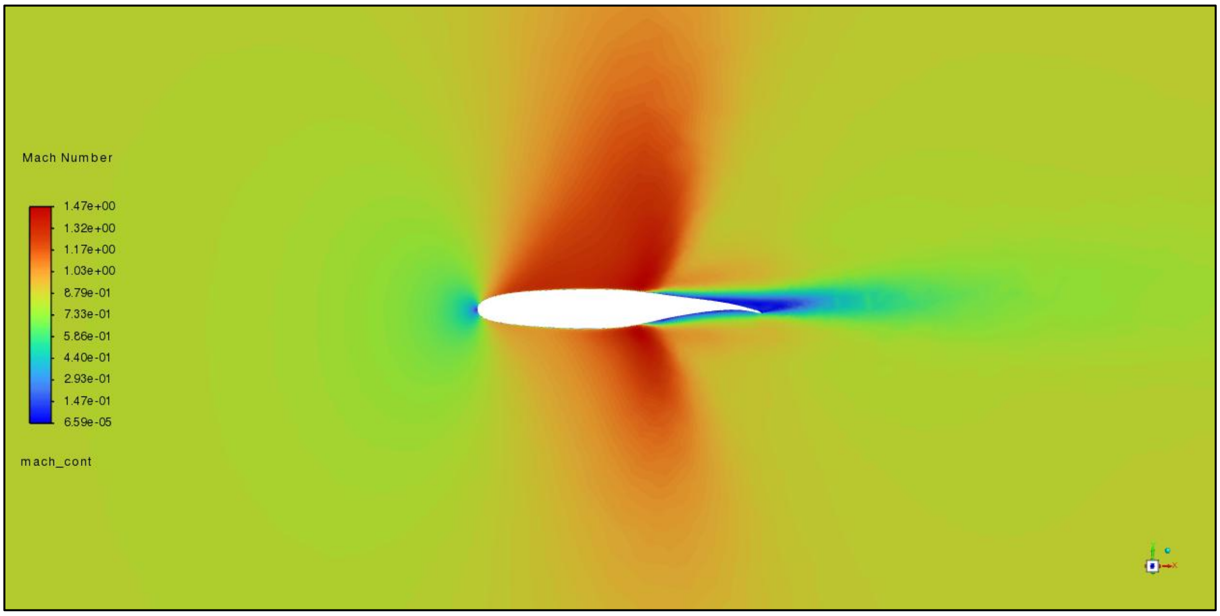

Figure 4-39 - Mach contours of the optimised NASA SC(2)-0714 design (constant length)

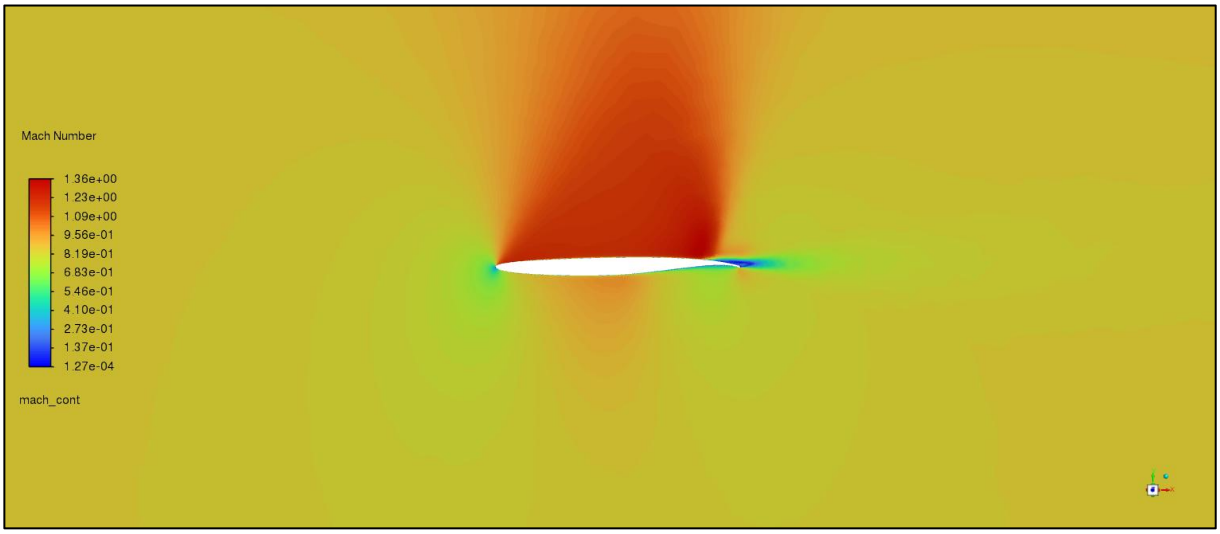

Figure 4-40 - Mach contours of the optimised NASA SC(2)-0714 design (variable length)

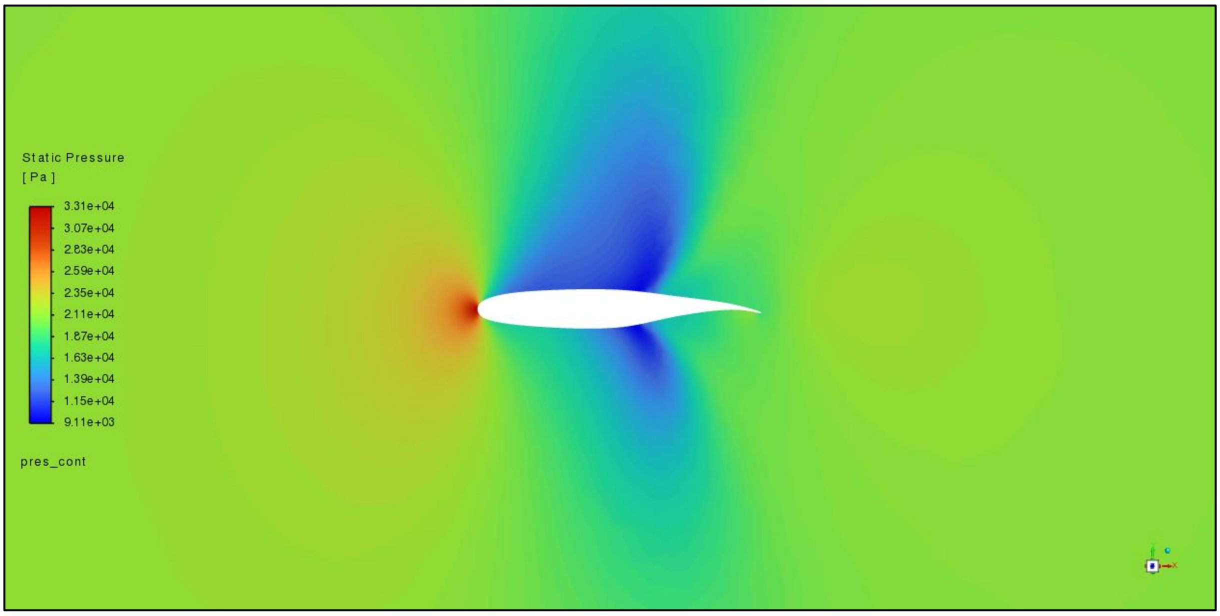

Figure 4-41 - Pressure contours of the optimised NASA SC(2)-0714 design (constant length)

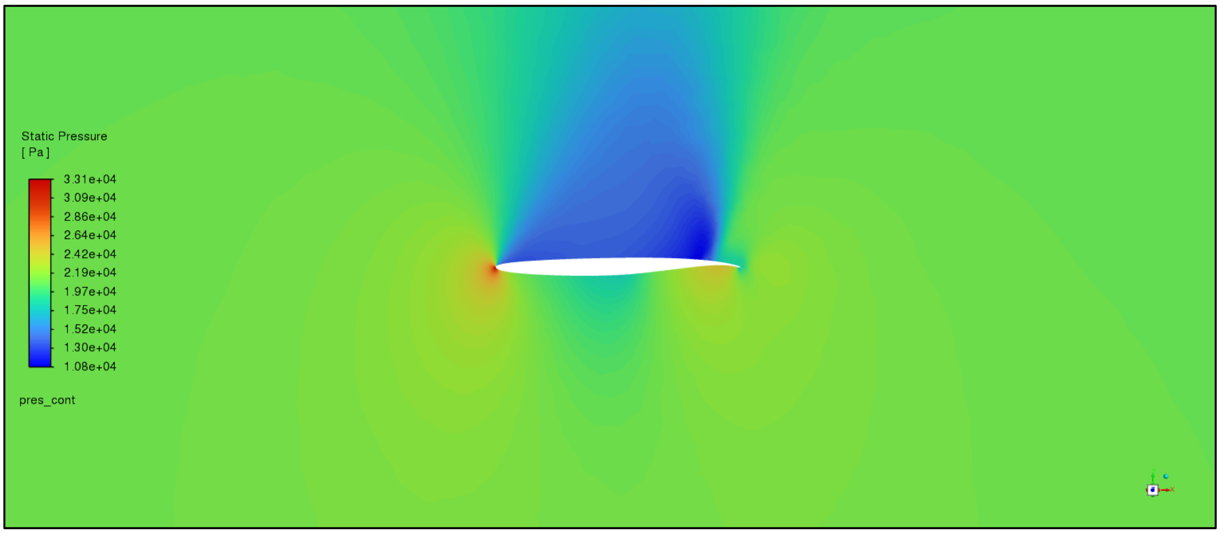

Figure 4-42 - Pressure contours of the optimised NASA SC(2)-0714 design (variable length)

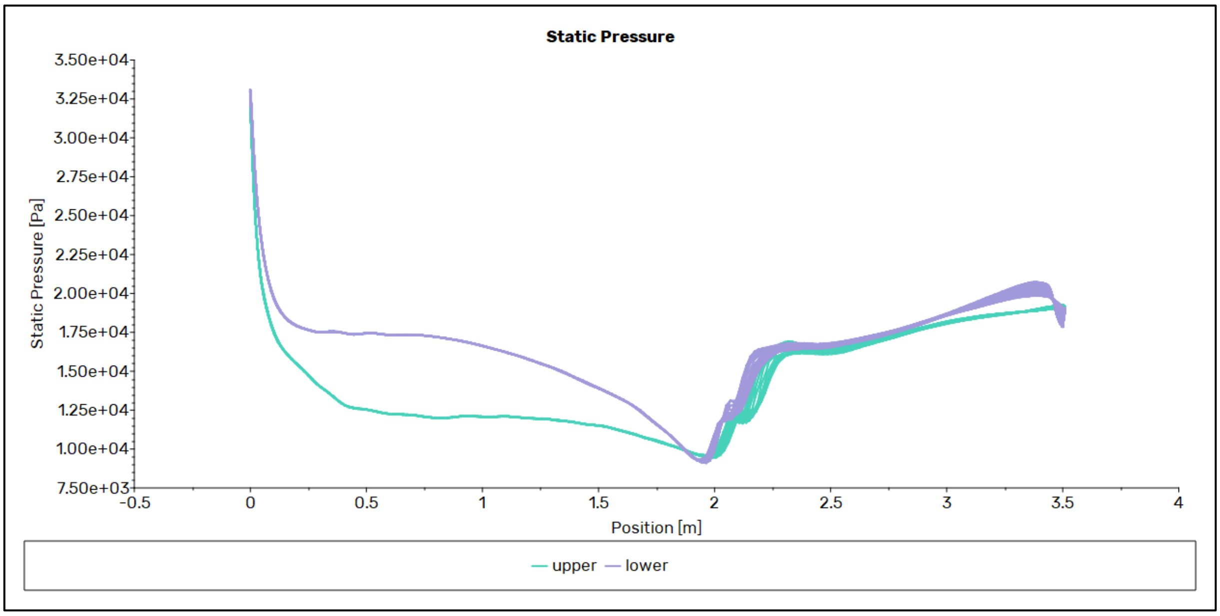

Figure 4-43 - Pressure plot of the optimised NASA SC(2)-0714 design (constant length)

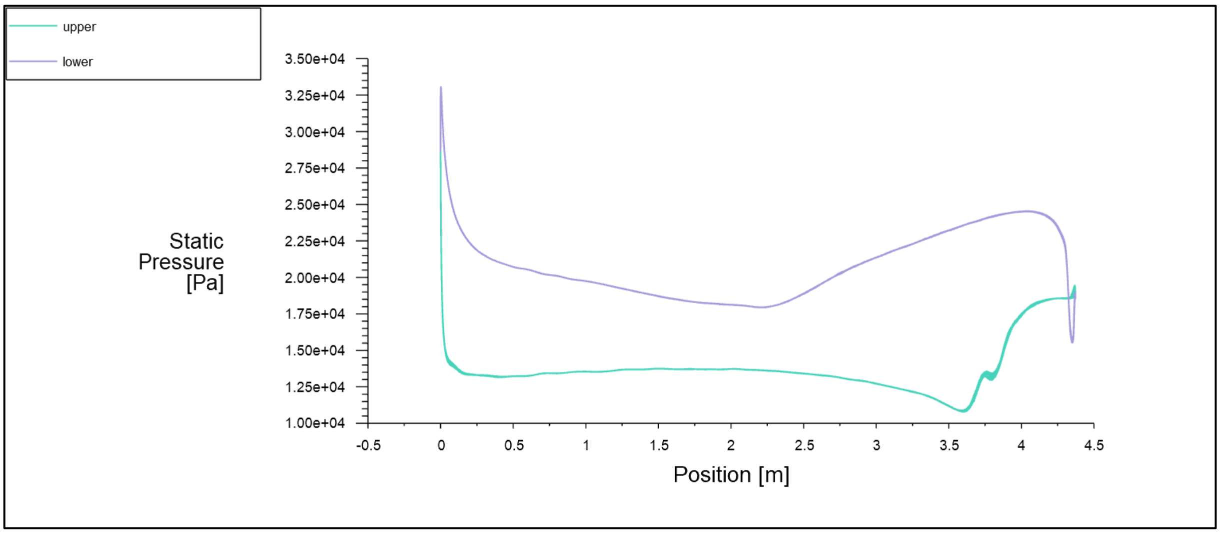

Figure 4-44 - Pressure plot of the optimised NASA SC(2)-0714 design (variable length)

4.5 NASA SC(2)-0714 in Supersonic Flow

The NASA SC(2)-0714 airfoil was also tested at supersonic speeds of $M = 1.8$. The trials used the same mesh as shown in Table 4-13. The results are shown in Table 4-17, and the comparison between recorded and theoretical values is shown in Table 4-18.

Table 4-17 - Boundary conditions and results of the NASA SC(2)-0714 test and reference cases at supersonic speeds

Variable	**Test**		**Reference**	
Nickname	act_3D	act_2D	ref_3D	ref_2D
Dimensions	3D	2D	3D	2D
Altitude (m)	13,715.33	13,715.33	0.00	0.00
AoA (°)	0.00	0.00	0.00	0.00
Chord Length (m)	3.51	3.51	1.00	1.00
Mach Number	1.800	1.800	1.200	1.200
C_d	0.12941	0.12695	0.08475	0.08345
C_l	0.02985	0.02744	0.03699	0.03466
Lift-to-drag Ratio	0.23	0.22	0.44	0.42
Iterations Needed	508	2,000	304	190
y^+	2.75	2.75	1.50	1.50
Cell Count	456,363	24,262	456,363	24,262
Time Taken (s)	940.81	517.07	2,268.71	389.48
Time per iteration (s)	1.85	0.26	7.46	2.05
Reynolds Number	3.42E+07	3.42E+07	2.80E+07	2.80E+07

Table 4-18 - NASA SC(2)-0714 results at supersonic speeds compared to [38]

Variable	**ref_3D**	**ref_2D**	**Sri**	**Δ%**
C_d	0.08475	0.08345	0.024	71.68%
C_l	0.03699	0.03466	0.013	64.85%
Lift-to-drag Ratio	0.44	0.42	0.54	-24.11%

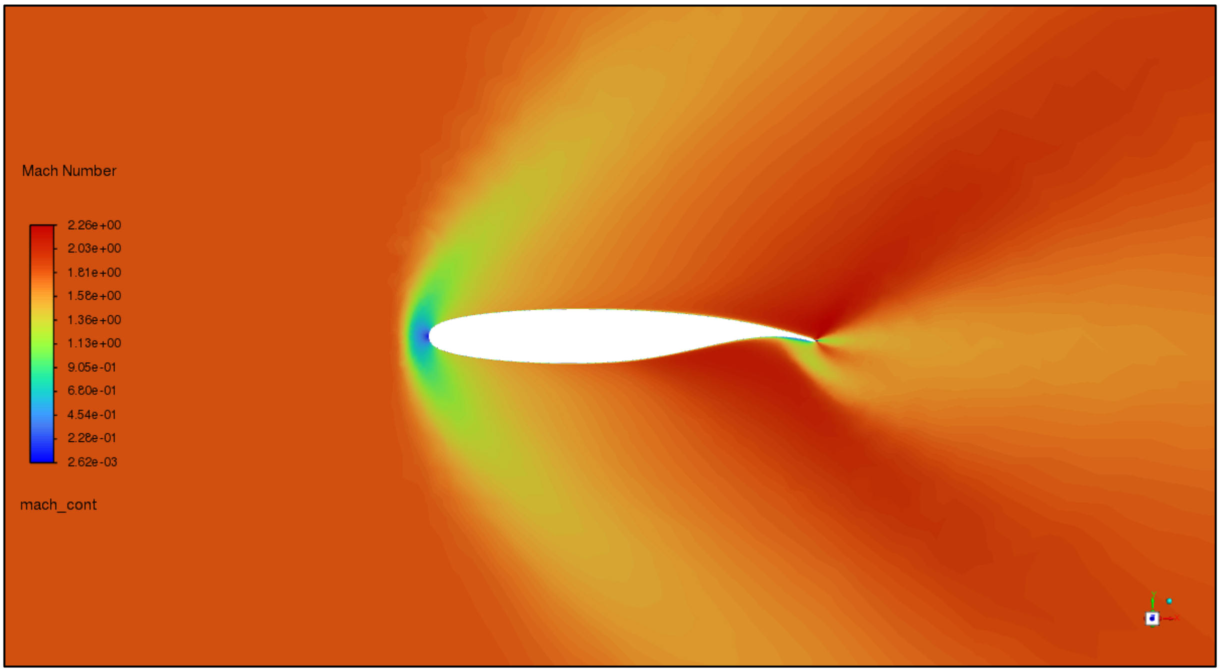

Figure 4-45 - Mach contours of the initial NASA SC(2)-0714 design at $M = 1.8$

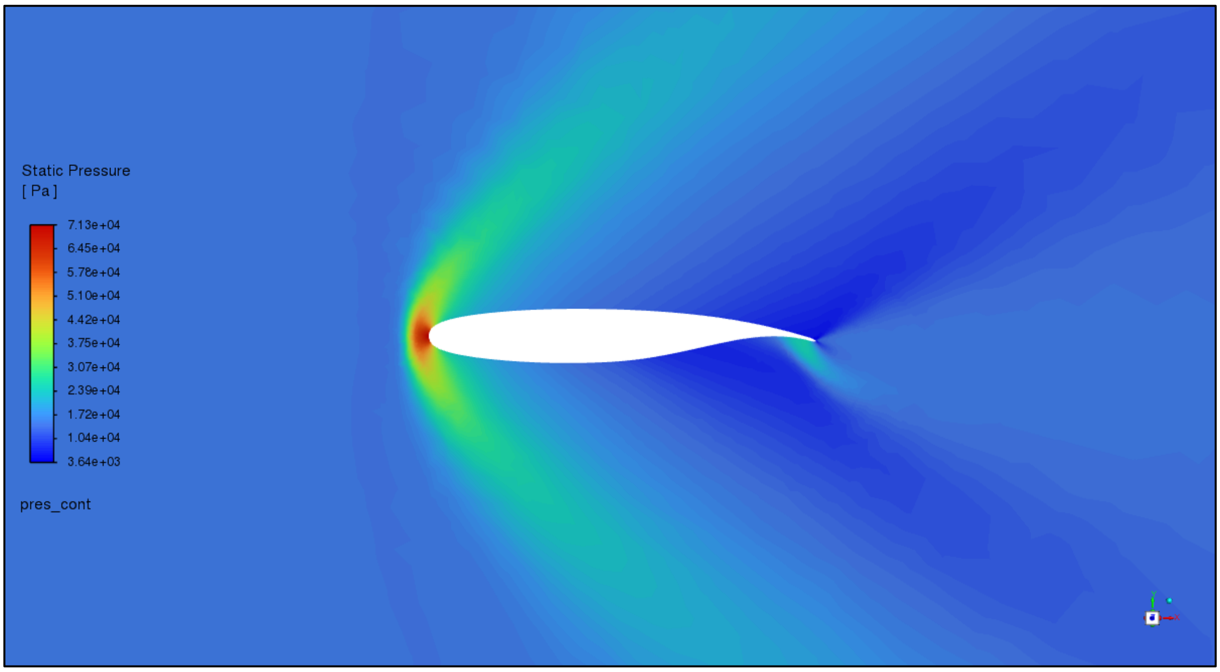

Figure 4-46 - Pressure contours of the initial NASA SC(2)-0714 design at $M = 1.8$

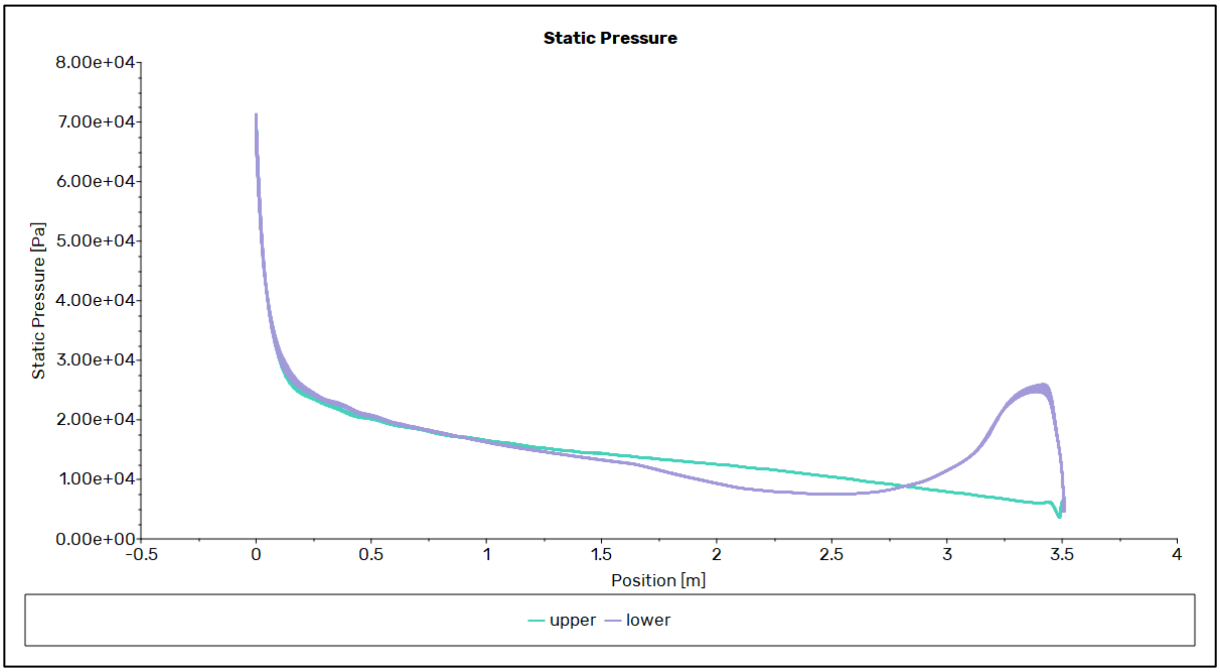

Figure 4-47 - Pressure plot of the initial NASA SC(2)-0714 design at $M = 1.8$

4.5.1 Optimisation

The results from the optimised geometries are shown in Table 4-19. Both trials showed improvement, with the variable length trial showing much greater improvement. The drag and lift coefficients were increased on the constant length trial, whereas for the variable length trial, the drag coefficient was decreased, and the lift coefficient was increased.

Figure 4-48 and Figure 4-49 show the optimised geometries. Figure 4-50 and Figure 4-51 show the Mach contours, Figure 4-52 and Figure 4-53 show the optimised pressure contours, and Figure 4-54 and Figure 4-55 show the optimised pressure plots.

Table 4-19 - Optimised NASA SC(2)-0714 (supersonic) drag and lift coefficients

NASA SC(2)-0714	C_d	C_l	Lift-to-drag Ratio
Original	0.12940847	0.02984689	0.230640931
Constant Length	0.176917	0.14721263	0.832099968
Variable Length	0.10656289	0.15569199	1.461033855
Δ% Constant	36.71%	393.23%	260.78%
Δ% Variable	-17.65%	421.64%	533.47%

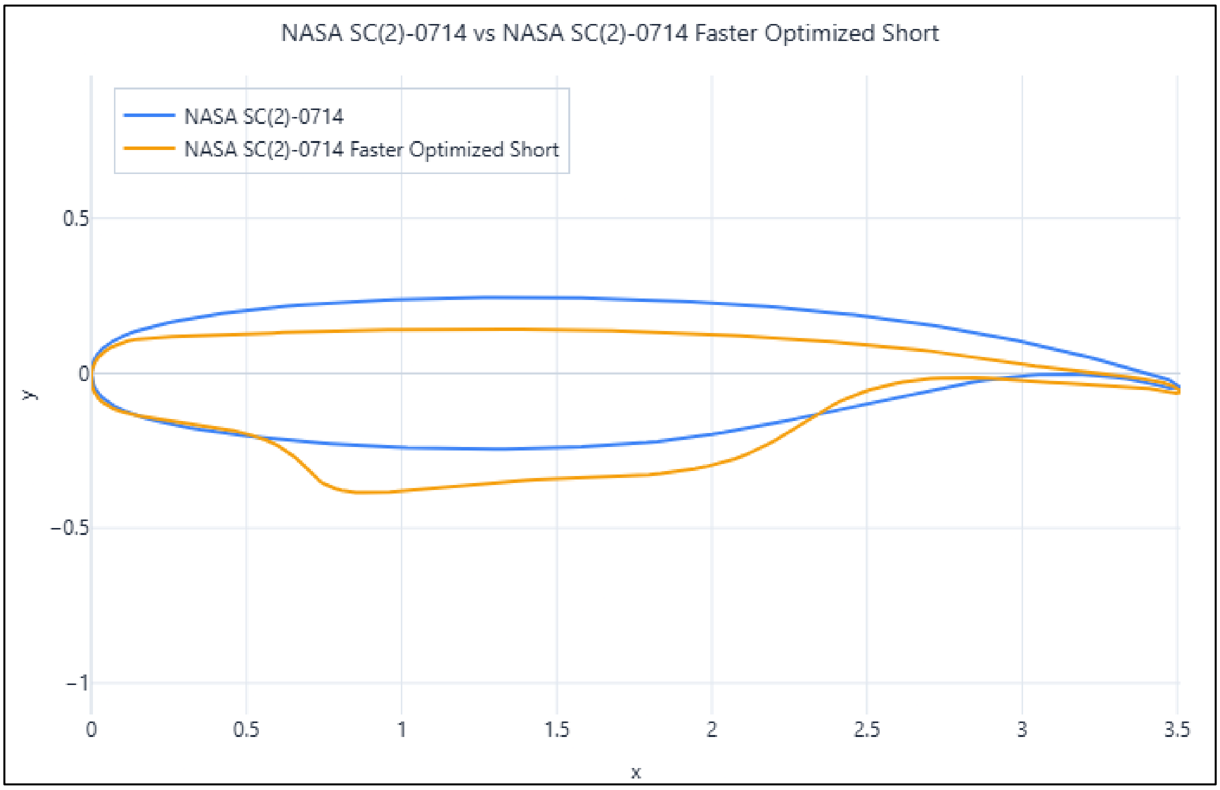

Figure 4-48 - Initial vs optimised NASA SC(2)-0714 coordinates (constant length) at $M = 1.8$

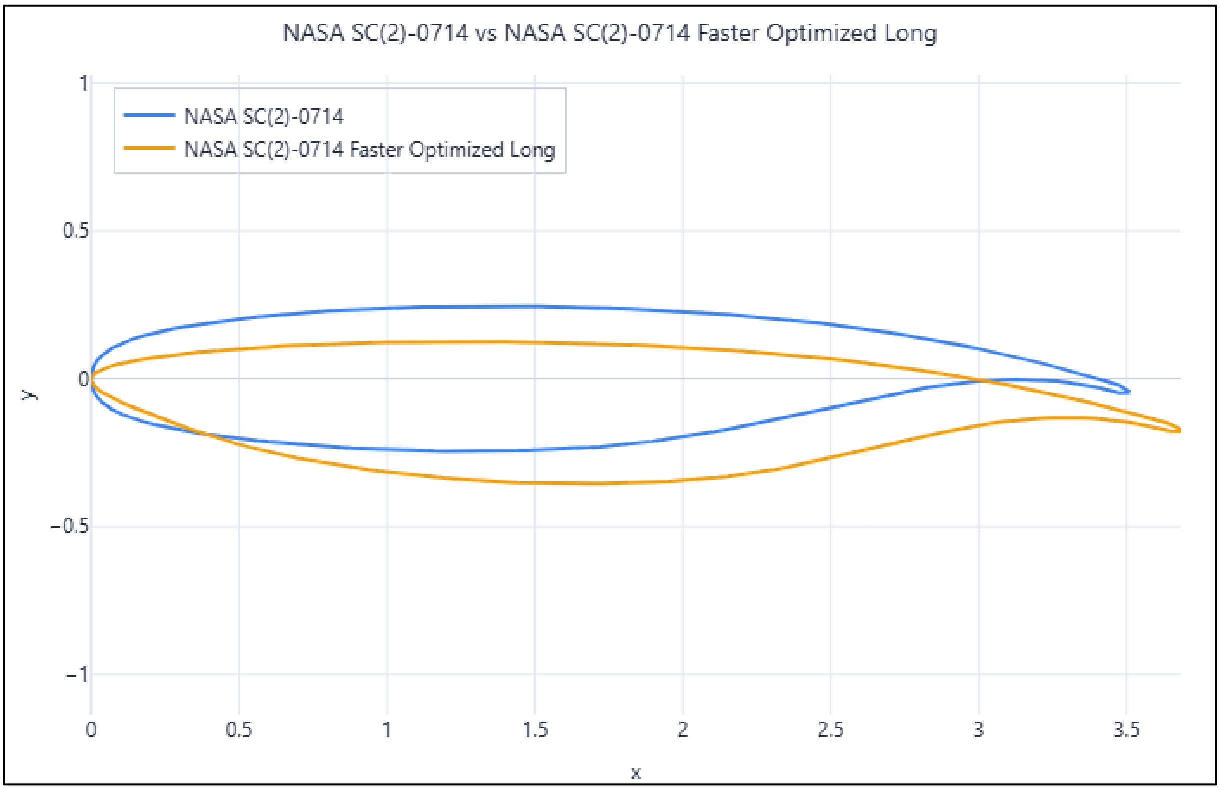

Figure 4-49 - Initial vs optimised NASA SC(2)-0714 coordinates (variable length) at $M = 1.8$

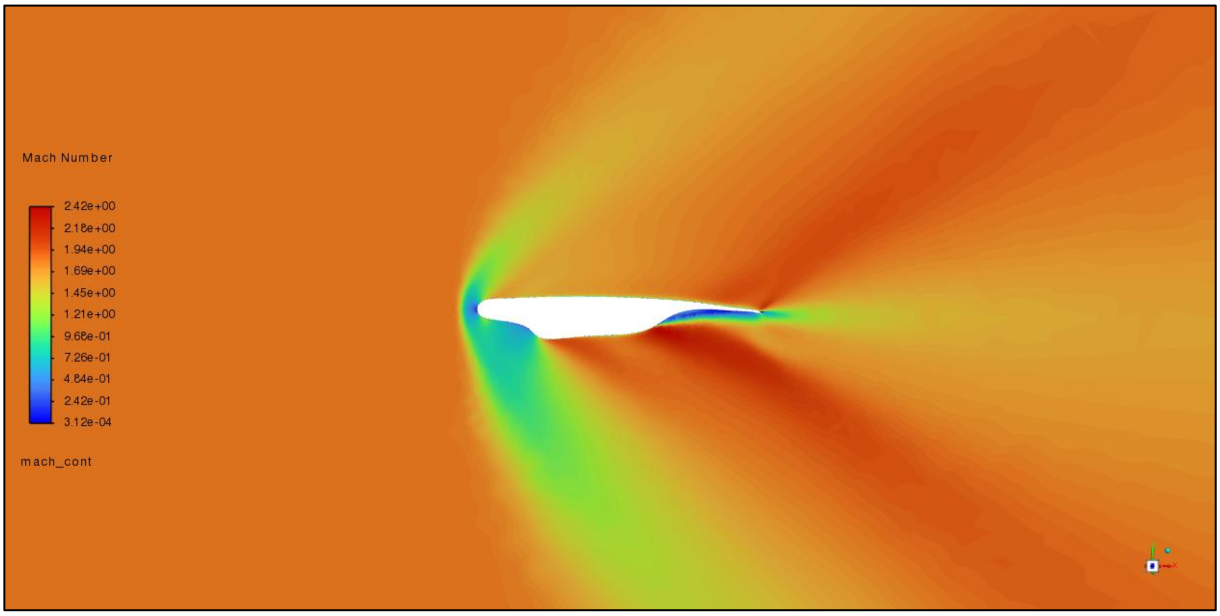

Figure 4-50 - Mach contours of the optimised NASA SC(2)-0714 design (constant length) at $M = 1.8$

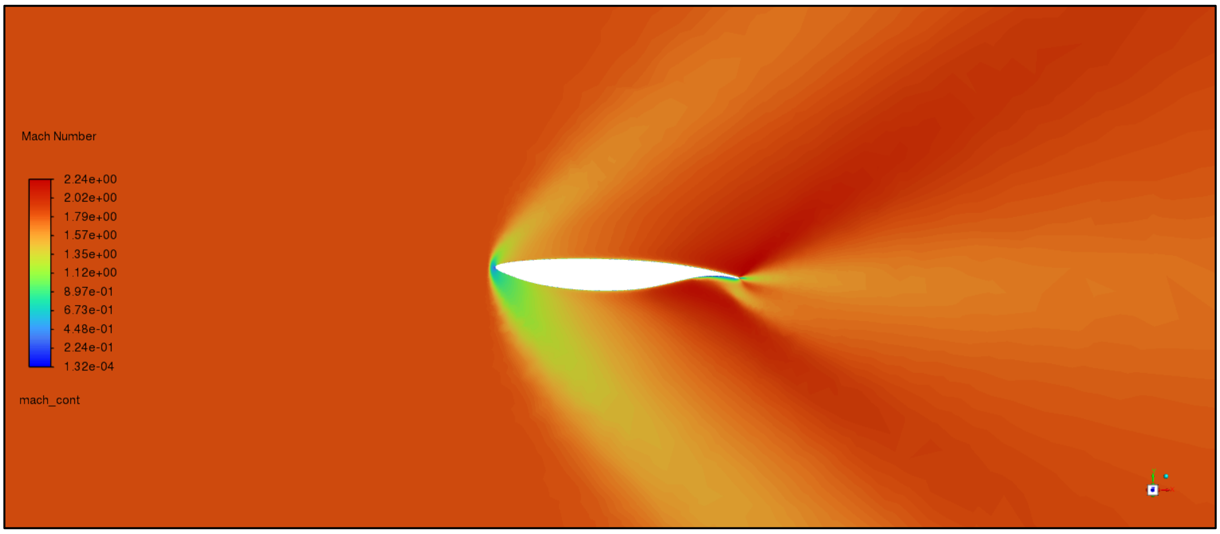

Figure 4-51 - Mach contours of the optimised NASA SC(2)-0714 design (variable length) at $M = 1.8$

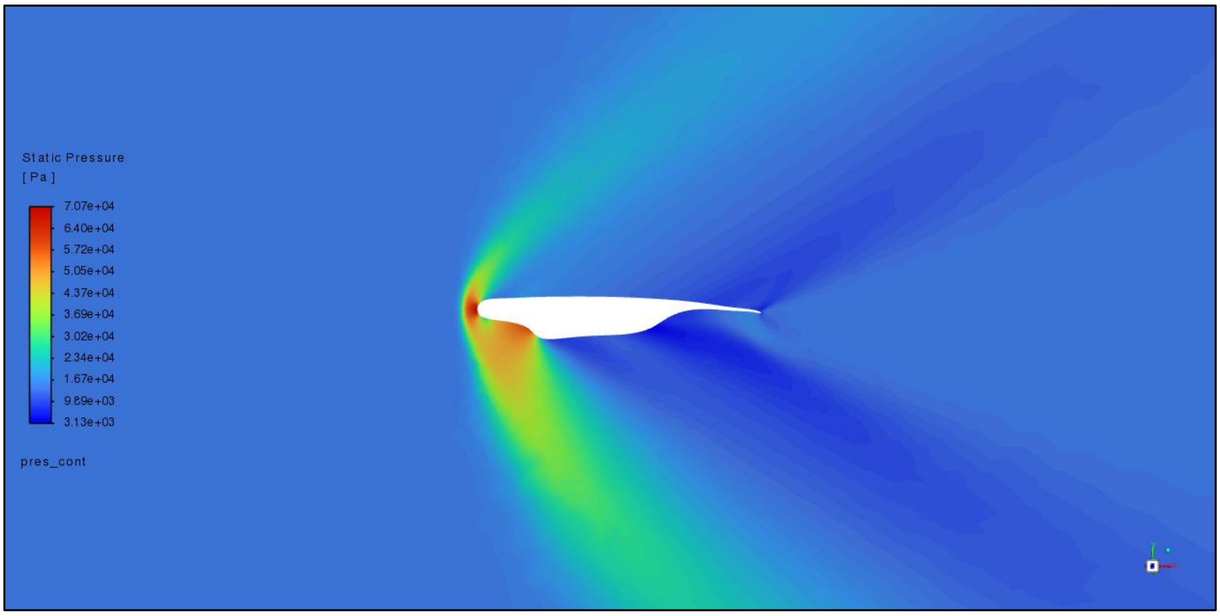

Figure 4-52 - Pressure contours of the optimised NASA SC(2)-0714 design (constant length) at $M = 1.8$

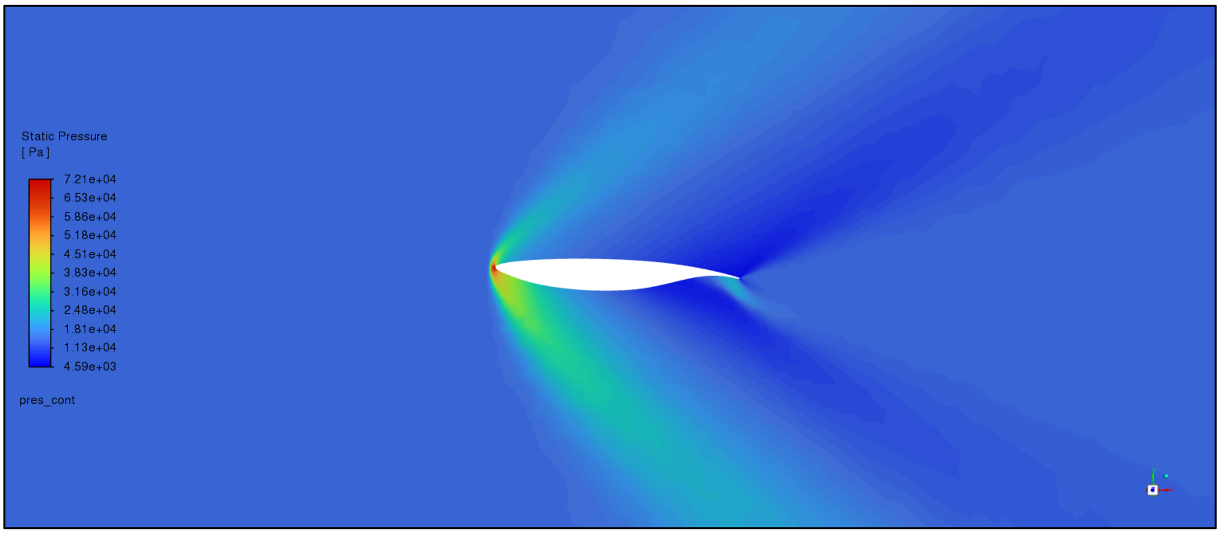

Figure 4-53 - Pressure contours of the optimised NASA SC(2)-0714 design (variable length) at $M = 1.8$

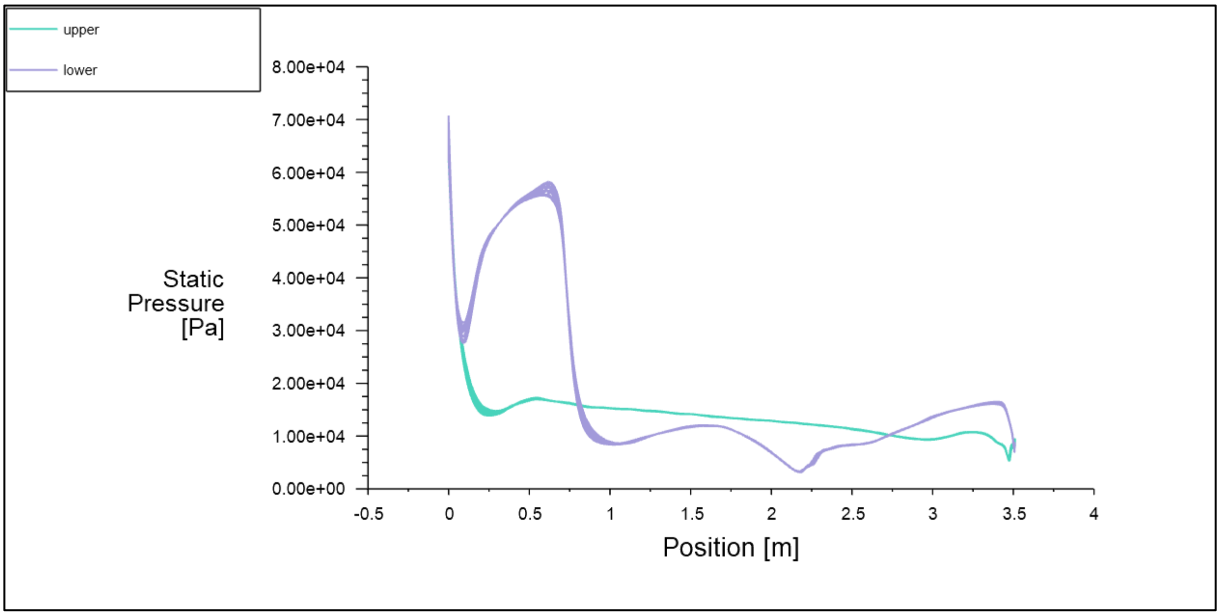

Figure 4-54 - Pressure plot of the optimised NASA SC(2)-0714 design (constant length) at $M = 1.8$

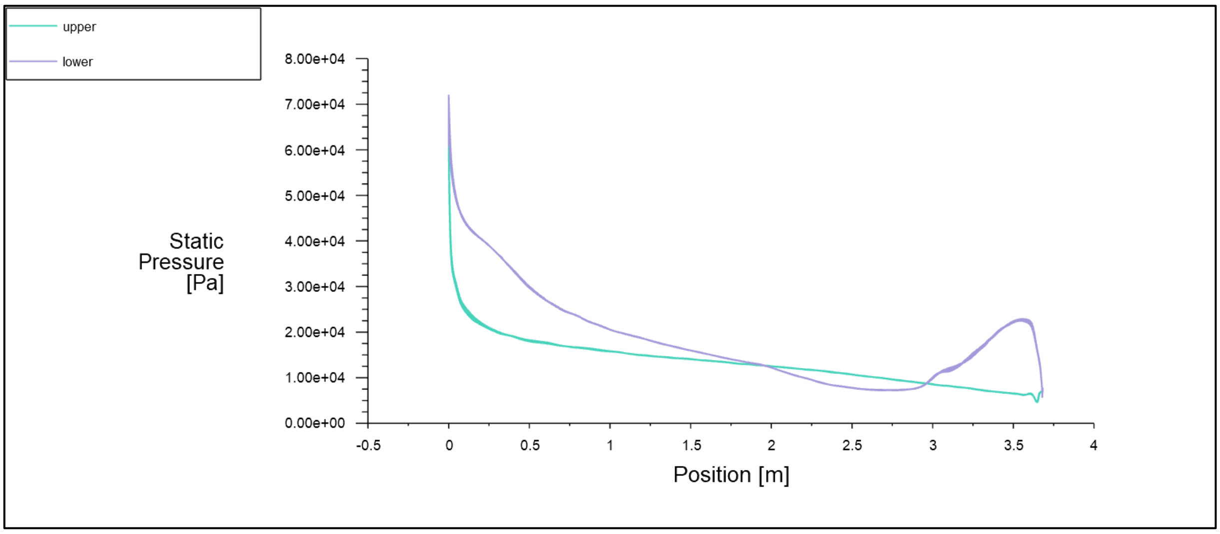

Figure 4-55 - Pressure plot of the optimised NASA SC(2)-0714 design (variable length) at $M = 1.8$

4.6 Additional Trials

4.6.1 y^+ Trial

Simulation cases with the same conditions but different meshes were analysed. The meshes were made with different estimated y^+ values. Three cases were tested at subsonic ($M = 0.197$), transonic ($M = 0.75$), and supersonic ($M = 1.8$) speeds. The drag and lift coefficients for the subsonic, transonic, and supersonic cases can be seen in Table 4-20, Table 4-23, and Table 4-26. The percentage difference between the value recorded at $y^+ = 1$ and other y^+ values is shown in Table 4-21, Table 4-24, and Table 4-27. Table 4-22, Table 4-25, and Table 4-28 show the number of cells or faces for each y^+ value.

Table 4-20 - NACA 2412 y^+ trial at M=0.197

y^+	C_d (3D)	C_d (2D)	C_l (3D)	C_l (2D)	Lift-to-drag Ratio (3D)	Lift-to-drag Ratio (2D)
1	0.009560	0.009382	0.215590	0.214329	22.551567	22.845311
2	0.009507	0.009274	0.217525	0.214396	22.881204	23.118564
3	0.009495	0.009266	0.217208	0.214074	22.875660	23.103361
4	0.009596	0.009385	0.224891	0.213336	23.436194	22.730911
5	0.009801	0.009561	0.215452	0.213284	21.982491	22.307946
10	0.010087	0.009851	0.223604	0.214604	22.166875	21.784159
20	0.009810	0.009537	0.226920	0.214864	23.130784	22.528897
30	0.009771	0.009522	0.228235	0.215292	23.358699	22.609758
50	0.010003	0.009713	0.228843	0.216398	22.877180	22.279935
100	0.010638	0.012233	0.233049	0.223952	21.907670	18.307130

Table 4-21 - Percentage error in drag and lift between the value recorded at $y^+ = 1$ at $M = 0.197$

y+	3D		2D	
	C_d Δ%	C_l Δ%	C_d Δ%	C_l Δ%
2	-0.56%	0.90%	-1.15%	0.03%
3	-0.68%	0.75%	-1.23%	-0.12%
4	0.38%	4.31%	0.04%	-0.46%
5	2.52%	-0.06%	1.91%	-0.49%
10	5.52%	3.72%	5.01%	0.13%
20	2.62%	5.26%	1.66%	0.25%
30	2.21%	5.87%	1.50%	0.45%
50	4.64%	6.15%	3.53%	0.97%
100	11.28%	8.10%	30.39%	4.49%

Table 4-22 - Mesh details at $M = 0.197$

y^+	Cell Count (3D)	Face Count (2D)
1	965,636	22,475
2	475,737	21,907
3	333,525	21,514
4	127,115	21,342
5	246,927	21,056
10	123,666	20,478
20	116,692	19,938
30	111,854	19,658
50	98,841	19,106
100	85,993	18,707

Table 4-23 - NACA 2412 y^+ trial at M=0.75

y^+	C_d (3D)	C_d (2D)	C_l (3D)	C_l (2D)	Lift-to-drag Ratio (3D)	Lift-to-drag Ratio (2D)
1	0.012472	0.014271	0.24487321	0.329778	19.634057	23.108079
2	0.013536	0.014672	0.25224511	0.333613	18.634524	22.737851
3	0.014212	0.014361	0.26979231	0.315834	18.983590	21.992106
4	0.014571	0.014188	0.27911849	0.314610	19.155267	22.173645
5	0.014636	0.014056	0.27987293	0.314031	19.122698	22.341791
10	0.015573	0.014461	0.31382543	0.317812	20.151971	21.976578
20	0.015693	0.014204	0.33103849	0.319999	21.095332	22.528077
30	0.016227	0.014085	0.35603723	0.323105	21.940714	22.940297
50	0.016173	0.014198	0.36104785	0.323980	22.323697	22.819290
100	0.016376	0.014524	0.36021148	0.326030	21.995995	22.447245

Table 4-24 - Percentage error in drag and lift between the value recorded at $y^+ = 1$ at $M = 0.75$

y+	3D		2D	
	C_d Δ%	C_l Δ%	C_d Δ%	C_l Δ%
2	8.54%	3.01%	2.81%	1.16%
3	13.95%	10.18%	0.63%	-4.23%
4	16.83%	13.98%	-0.58%	-4.60%
5	17.35%	14.29%	-1.51%	-4.77%
10	24.86%	28.16%	1.33%	-3.63%
20	25.82%	35.19%	-0.47%	-2.97%
30	30.11%	45.40%	-1.31%	-2.02%
50	29.68%	47.44%	-0.51%	-1.76%
100	31.31%	47.10%	1.77%	-1.14%

Table 4-25 - Mesh details at M=0.75

y^+	Cell Count (3D)	Face Count (2D)
1	1,263,487	23,781
2	620,168	23,159
3	406,305	22,772
4	649,713	22,549

5	524,010	22,298
10	300,616	21,668
20	201,099	21,120
30	105,879	20,780
50	99,523	20,344
100	96,591	19,780

Table 4-26 - NACA 2412 y^+ trial at M=1.8

y+	C_d (3D)	C_d (2D)	C_l (3D)	C_l (2D)	Lift-to-drag Ratio (3D)	Lift-to-drag Ratio (2D)
1	0.100612	0.101869	-0.022448	-0.021196	-0.223116	-0.208076
2	0.100671	0.102051	-0.02229	-0.022431	-0.221412	-0.219807
3	0.101411	0.101679	-0.022236	-0.021697	-0.21927	-0.213392
4	0.101793	0.101711	-0.022441	-0.020106	-0.220458	-0.19768
5	0.101894	0.10277	-0.023186	-0.022353	-0.227553	-0.217505
10	0.102302	0.101896	-0.022231	-0.019586	-0.21731	-0.192214
20	0.102311	0.102985	-0.022842	-0.02156	-0.223258	-0.20935
30	0.101802	0.102991	-0.021253	-0.020194	-0.208771	-0.196076
50	0.101957	0.102878	-0.021754	-0.020561	-0.213361	-0.199862
100	0.102223	0.102641	-0.022046	-0.021529	-0.215664	-0.209753

Table 4-27 - Percentage error in drag and lift between the value recorded at $y^+ = 1$ at $M = 1.8$

y^+	3D		2D	
	C_d Δ%	C_l Δ%	C_d Δ%	C_l Δ%
2	0.06%	-0.71%	0.18%	5.83%
3	0.79%	-0.94%	-0.19%	2.36%
4	1.17%	-0.03%	-0.16%	-5.14%
5	1.27%	3.29%	0.88%	5.46%
10	1.68%	-0.97%	0.03%	-7.60%
20	1.69%	1.75%	1.10%	1.71%
30	1.18%	-5.32%	1.10%	-4.73%
50	1.34%	-3.09%	0.99%	-3.00%
100	1.60%	-1.79%	0.76%	1.57%

Table 4-28 - Mesh details at $M = 1.8$

y^+	Cell Count (3D)	Face Count (2D)
1	3,099,916	23,889
2	1,611,302	23,979
3	1,088,472	23,326
4	862,436	22,987
5	683,365	22,827
10	375,219	22,364
20	240,149	21,672
30	129,576	21,330
50	119,513	20,968
100	112,675	20,310

4.6.2 Mesh Type Trial

Different volume fill types for 3D mesh were compared. The meshes were solved with the same boundary conditions in order to measure the difference between results.

The boundary conditions of the trial are shown in Table 4-29. Table 4-30 shows the recorded drag and lift coefficients, as well as cell count and time taken to solve. As seen, all fill types except Hexcore recorded similar results. Polyhedra was the mesh with the smallest cell count and fastest solve time.

Table 4-29 - The boundary conditions of the mesh type trial

Variable	Test
Airfoil	BACXXX
Dimensions	3D
Altitude (m)	4,114.80
AoA (°)	0.00
Chord Length (m)	1.49
Mach Number	0.197
Reynolds Number	4.65E+06

Table 4-30 - Drag and lift coefficients of different volume fill types of BACXXX mesh

Airfoil	Hexcore	Poly-Hexcore	Polyhedra	Tetrahedral
C_d	0.00765997	0.00860639	0.00860459	0.00886742
C_l	0.20502815	0.18451001	0.18152376	0.18742733
Cell Count	1,251,959	581,626	514,409	1,146,752
Time Taken (s)	4,999.78	1,649.76	1,179.98	2,827.01
Time per Iteration (s)	9.0905	4.1347	3.3522	6.9631

4.6.3 GUI Trial and Core Count Trial

The meshing and solution functions were tested with the Graphical User Interface (GUI) enabled and disabled. The trials were repeated using a different number of processing cores.

Figure 4-56 and Figure 4-57 show the results of this trial for meshing and solving, respectively. For meshing, the fastest solve was achieved using 2 cores. For solving, 4 cores were the fastest[9].

[9] All meshes used in the solve trial were generated using 4 cores. It is possible that the most efficient number of cores when solving is the number of cores used to generate the mesh.

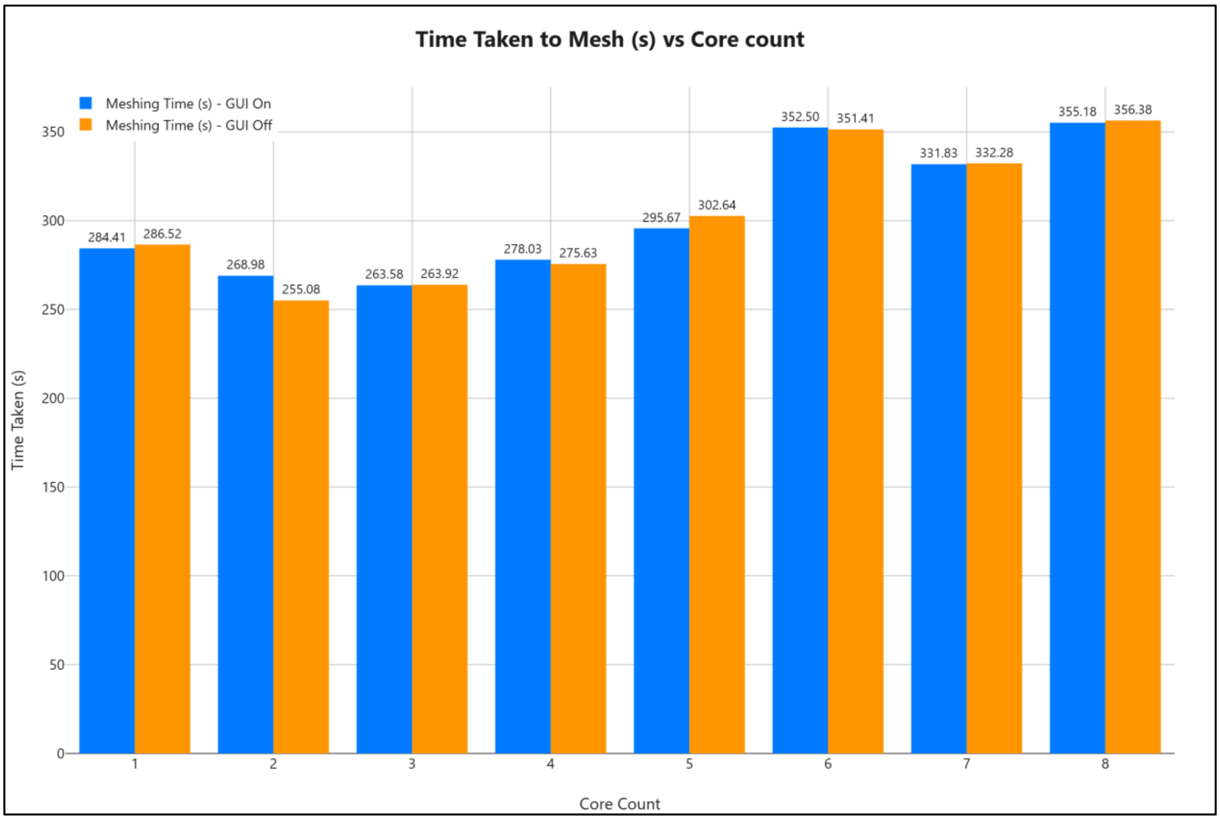

Figure 4-56 - Core count and GUI trial, meshing

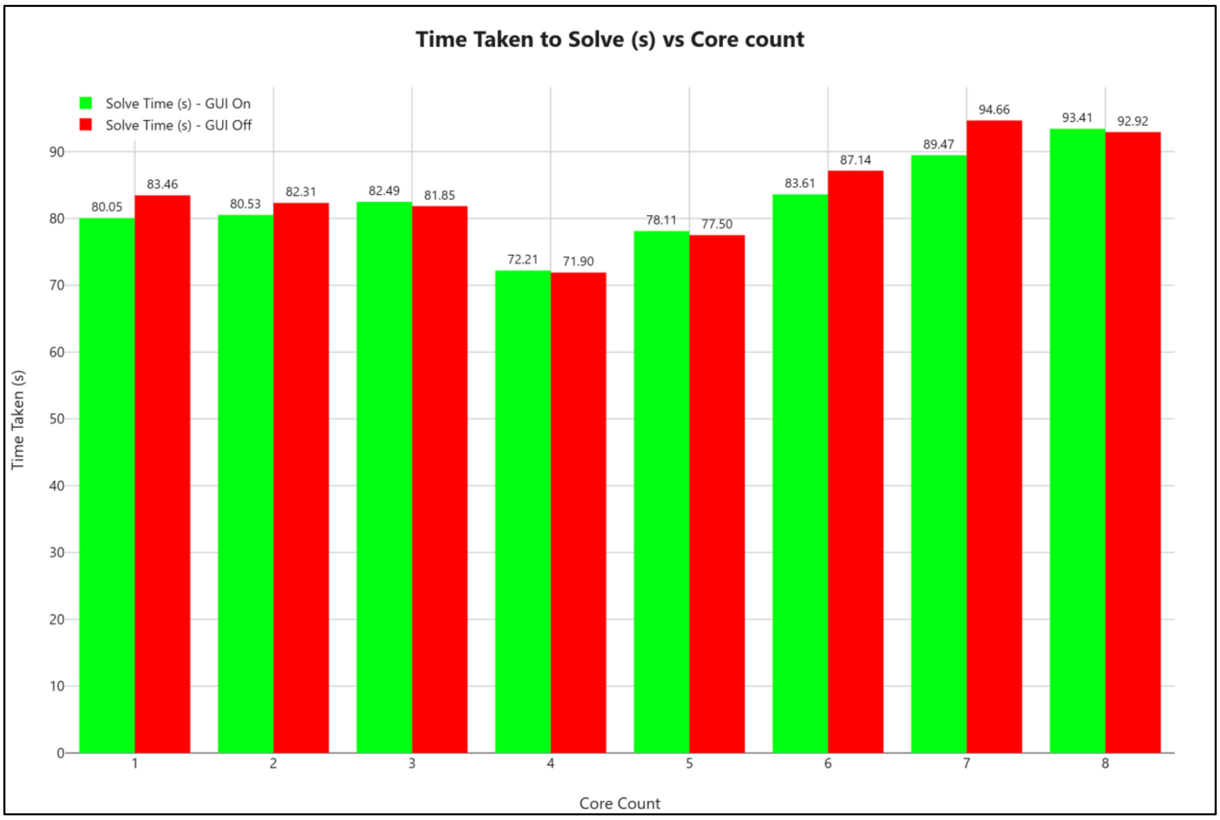

Figure 4-57 - Core count and GUI trial, solving

4.6.4 Time Scale Trial

A trial was done to compare the number of iterations required to solve based on the pseudo time scale value. The initial boundary conditions for this trial are shown in Table 4-31. The results are shown in Table 4-32. The number of iterations was limited to 500.

Table 4-31 - Boundary conditions used in the time scale trial

Variable	Value
Airfoil	NACA 0012
Dimensions	3D
Altitude (m)	2,675.16
AoA (°)	1.55
Chord Length (m)	1.00
Mach Number	0.72
Reynolds Number	1.30E+07

Table 4-32 - Iterations needed to converge for different pseudo time scales

Pseudo Time Scale	Iterations Needed to Solve
Off	146
0.001	500+
0.01	500+
0.1	500+
1	145
2	138
3.5	134
5	133
10	130
15	129
20	129
25	129

4.6.5 Turbulence Method Trial

The $k-\omega$ GEKO, $k-\omega$ SST, and Spalart-Allmaras turbulence methods were compared. The number of iterations required to solve is shown in Figure 4-58. In order to minimise the impact of external disturbances on the solve time, the same case was done 5 times. Figure 4-59 displays the time taken per iteration, calculated as the total time taken to solve divided by the number of iterations. As seen, Spalart-Allmaras was consistently the fastest turbulence model, both based on the time and based on the number of iterations. The slowest turbulence model, with the highest number of iterations required was $k-\omega$ GEKO.

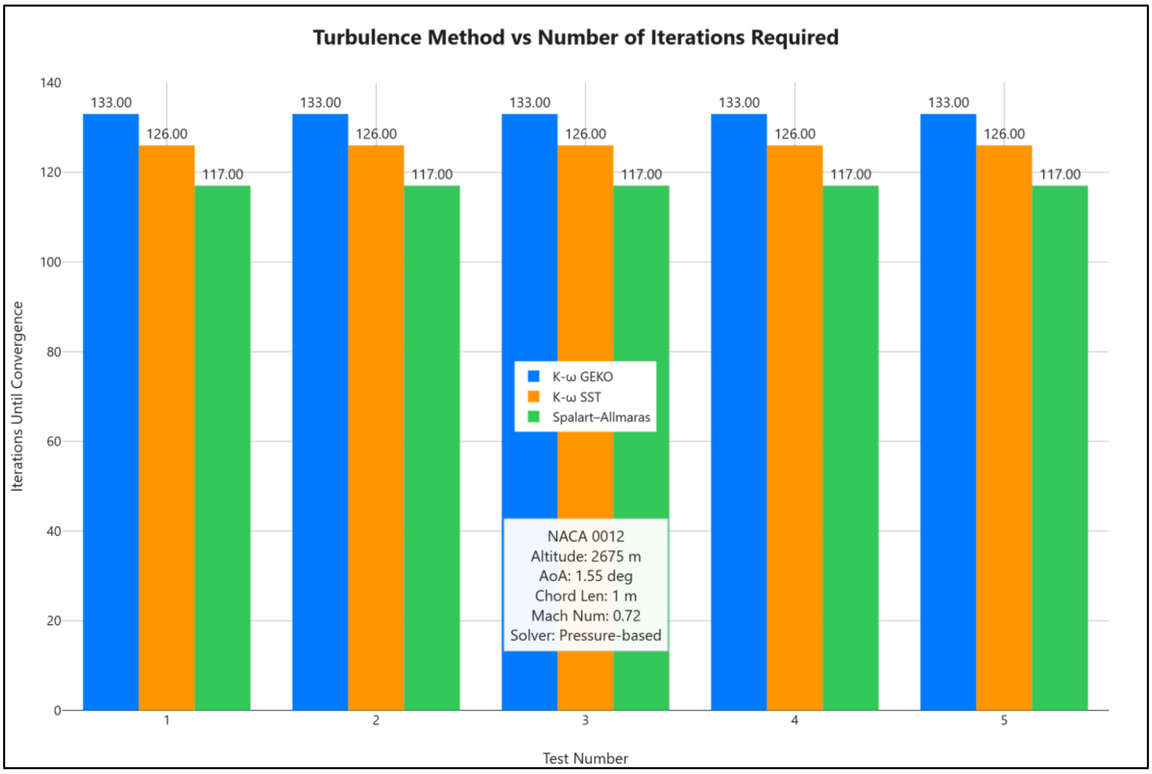

Figure 4-58 - Number of iterations required by turbulence model

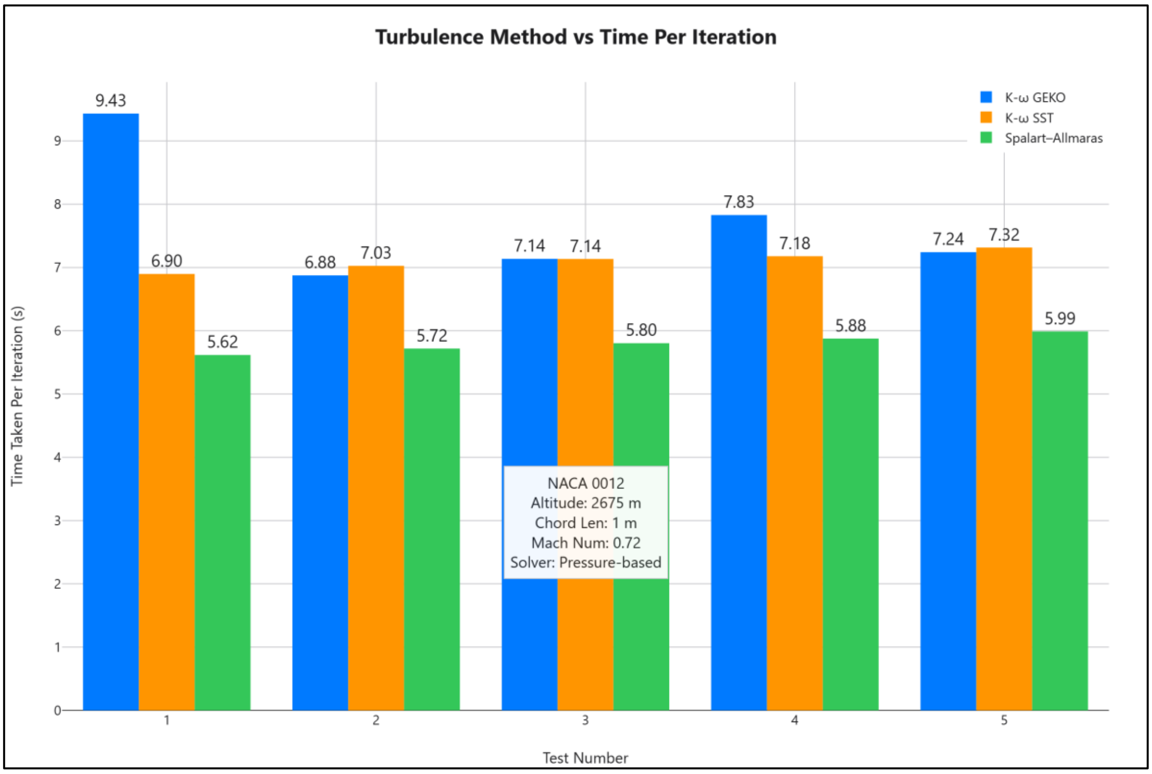

Figure 4-59 - Time (s) taken per iteration for different turbulence models

4.6.6 Flow Scheme Trial

Different types of flow schemes were tested and compared. The boundary conditions are shown in Table 4-33. The number of iterations required to solve is shown in Table 4-34. The number of iterations were limited to 2000.

Figure 4-60 shows the time per iteration. According to Table 4-34, the coupled solver consistently converged with the least iterations. Both PISO and SIMPLE solvers required over 2000 iterations to solve. All drag/lift coefficients were equal between solvers once converged. According to Figure 4-60, the coupled solver was the slowest solver per iteration in every run.

Table 4-33 - The boundary conditions of the flow scheme trial

Variable	Test
Airfoil	NASA SC(2)-0714
Dimensions	2D
Mach Number	0.3
Altitude (m)	11,581.83
C_d	0.0116065
C_l	0.66589255
Lift-to-drag Ratio	57.37
AoA (°)	2.00
Chord Length (m)	3.51
Reynolds Number	7.42E+06

Table 4-34 - Number of iterations needed to solve in the flow scheme trial

Trial	Flow Scheme	Iterations Needed
1	Coupled	187
1	PISO	2000+
1	SIMPLE	2000+
1	SIMPLEC	1056
2	Coupled	187
2	PISO	2000+
2	SIMPLE	2000+
2	SIMPLEC	1056
3	Coupled	187
3	PISO	2000+
3	SIMPLE	2000+
3	SIMPLEC	1056
4	Coupled	187
4	PISO	2000+
4	SIMPLE	2000+
4	SIMPLEC	1056

5	Coupled	187
5	PISO	2000+
5	SIMPLE	2000+
5	SIMPLEC	1056

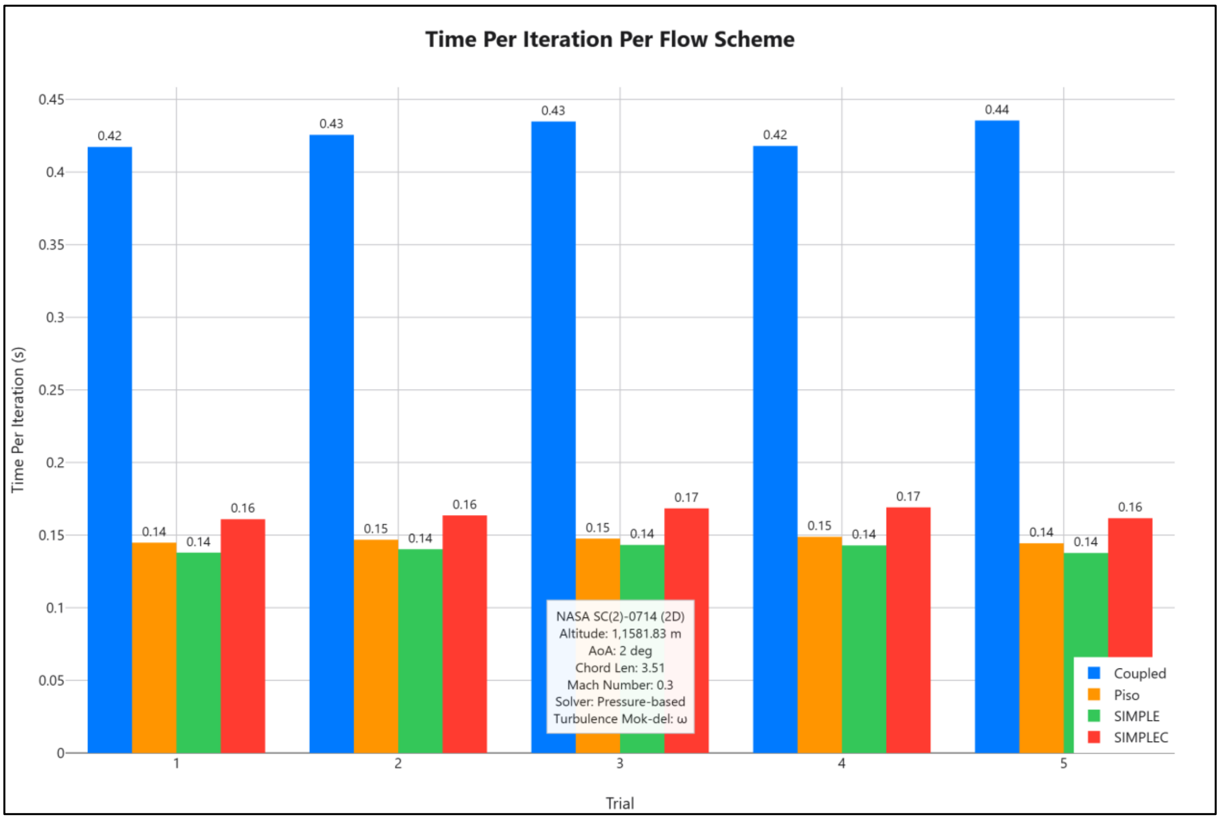

Figure 4-60 - Time per iteration for different flow schemes

4.6.7 Drag Divergence Trial

A drag divergence trial was also conducted on the NACA 0012 airfoil by varying the Mach number from $M = 0.258$ to $M = 1.2$ and keeping all other conditions the same. The boundary conditions of the study are shown in Table 4-35. The drag and lift coefficients are shown in Table 4-36.

Figure 4-61 and Figure 4-62 show the drag and lift coefficients, respectively, at varying Mach numbers. As shown, the drag starts increasing between $M = 0.7$ and $M = 0.75$. At M=0.75, the lift coefficient decreases suddenly.

Table 4-35 - Boundary conditions of the drag divergence trial

Variable	Value
Airfoil	NACA 0012

Dimensions	2D
Altitude (m)	0.00
AoA (°)	2.00
Chord Length (m)	1.00

Table 4-36 - The drag and lift coefficients of NACA 0012 at different Mach numbers

Mach Number	C_d	C_l	**Lift-to-drag Ratio**
0.258	0.00846501	0.22087909	26.093178
0.4	0.00806637	0.23145443	28.693758
0.6	0.00813805	0.26468709	32.524621
0.7	0.00890325	0.30445521	34.195952
0.75	0.01560326	0.33231323	21.297678
0.8	0.0234274	0.17980688	7.675069
0.85	0.05254719	0.00237474	0.045193
0.9	0.10963913	0.12726525	1.160765
0.95	0.12080992	0.19305553	1.598011
1	0.11631657	0.1829023	1.572453
1.1	0.10877814	0.16906634	1.554231
1.2	0.10588125	0.15374067	1.45201

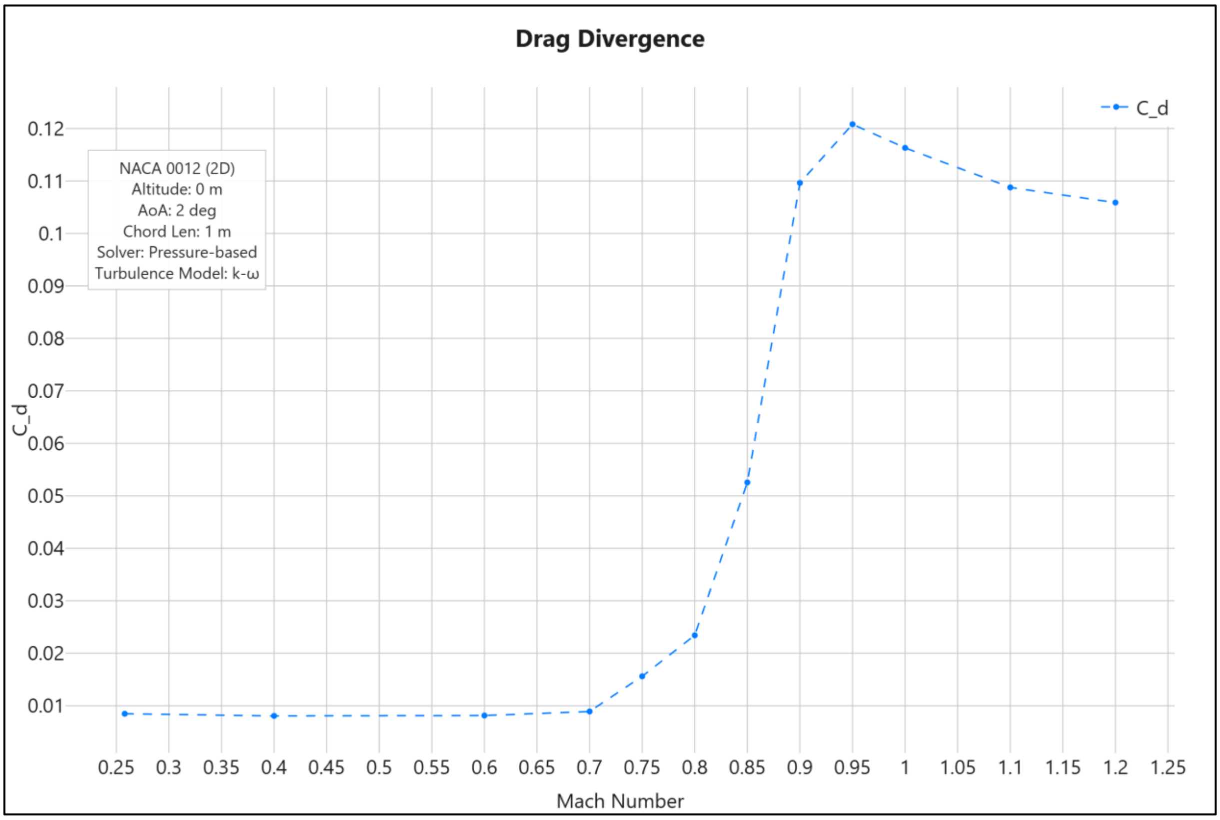

Figure 4-61 - Drag coefficient at different Mach numbers

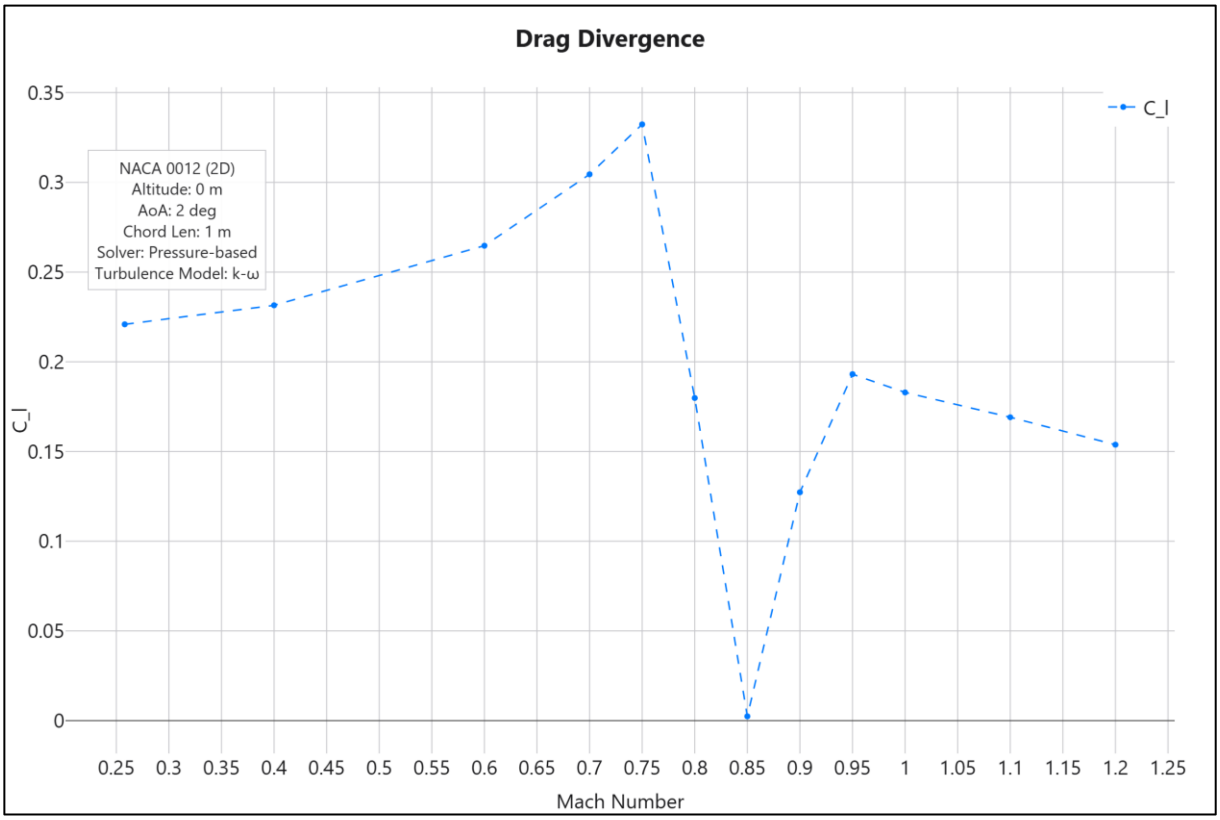

Figure 4-62 - Lift coefficient at different Mach numbers

4.6.8 AoA Trial

The drag and lift coefficients were measured at different angles of attack using the $k-\omega$ GEKO and Spalart-Allmaras models. The stall angles using both models were determined.

Initially, the angle of attack was changed in 1° increments until 10°, then in 2° until the final angle of 40°. The stall angle was determined to be between $14° - 16°$ for both $k-\omega$ and Spalart-Allmaras models. Hence, the increment was reduced to 0.1° in this range in order to find, more precisely, the stall angle.

Table 4-37 displays the boundary conditions of the trial. Table 4-38 shows the resulting drag and lift coefficients and the difference between the turbulence models. Figure 4-63 and Figure 4-64 show the drag and lift coefficients, respectively, of both turbulence models at different angles of attack.

Table 4-37 - The boundary conditions of the AoA trial

Variable	Value
Airfoil	NACA 0012
Dimensions	2D
Altitude (m)	0.00
Chord Length (m)	1.00
Reynolds Number	6.01E+06

Table 4-38 - The drag and lift coefficients at various angles of attack of the $k-\omega$ and Spalart-Allmaras turbulence models

AoA (°)	C_d $k-\omega$	C_d SA	C_d Δ%	C_l $k-\omega$	C_l SA	C_l Δ%
0	0.00810737	0.00824406	-1.69%	0.00064259	0.00060963	5.13%
1	0.00818804	0.00832175	-1.63%	0.11099709	0.11151154	-0.46%
2	0.00846501	0.00859522	-1.54%	0.22087909	0.22182622	-0.43%
3	0.00893344	0.0090599	-1.42%	0.33024017	0.33183126	-0.48%
4	0.00958203	0.00969744	-1.20%	0.43851895	0.44082062	-0.52%
5	0.01041448	0.0105122	-0.94%	0.54406491	0.547174	-0.57%
6	0.01144675	0.01151626	-0.61%	0.64593558	0.64994414	-0.62%
7	0.01269997	0.01273174	-0.25%	0.74370284	0.74915553	-0.73%
8	0.01420595	0.01417737	0.20%	0.83757313	0.84434014	-0.81%
9	0.0159913	0.0158787	0.70%	0.9255198	0.93429015	-0.95%
10	0.01808	0.01785186	1.26%	1.00738452	1.01876173	-1.13%
12	0.02344747	0.02277821	2.85%	1.14948386	1.16956523	-1.75%
14	0.03156503	0.02969766	5.92%	1.24346491	1.28100792	-3.02%
14.1	0.0320864	0.03013368	6.09%	1.2459921	1.28503087	-3.13%
14.2	0.08979203	0.03057715	65.95%	0.2536781	1.28886965	-408.07%
14.3	0.03320714	0.03103132	6.55%	1.25121502	1.29252183	-3.30%
14.4	0.10629744	0.03150105	70.37%	0.41899665	1.29595363	-209.30%
14.5	0.1494091	0.03198472	78.59%	0.2317241	1.29919182	-460.66%
14.6	0.12262277	0.03248487	73.51%	0.67827715	1.30214953	-91.98%
14.7	0.11735968	0.0329994	71.88%	0.50990654	1.30485983	-155.90%
14.8	0.16713967	0.03352315	79.94%	0.24008196	1.30711165	-444.44%
14.9	0.20653939	0.034638	83.23%	0.9275805	1.31100719	-41.34%
15	0.12732152	0.03523956	72.32%	0.64255398	1.31258347	-104.28%
15.1	0.10742093	0.035855	66.62%	0.90153676	1.31382741	-45.73%
15.2	0.37882315	0.03650574	90.36%	0.9629632	1.31480001	-36.54%
15.3	0.13446325	0.03719937	72.33%	0.60878193	1.31512257	-116.03%
15.4	0.14364782	0.03791721	73.60%	0.5296328	1.31566752	-148.41%
15.5	0.31028172	0.03869846	87.53%	0.76714909	1.3143921	-71.33%
15.6	0.17064734	0.03948439	76.86%	1.37631814	1.31331481	4.58%
15.7	0.14871247	0.04035235	72.87%	0.70833893	1.31206326	-85.23%
15.8	0.15584297	0.04125205	73.53%	0.28572334	1.31027643	-358.58%
15.9	0.17140548	0.04220709	75.38%	0.6966701	1.30783211	-87.73%
16	0.21024153	0.04322953	79.44%	0.80179782	1.30481557	-62.74%
16.1	0.17625195	0.04427685	74.88%	0.73046082	1.30064335	-78.06%
16.2	0.17744973	0.04543317	74.40%	0.64084331	1.29995086	-102.85%
16.3	0.20781865	0.04643967	77.65%	0.80808796	1.2843533	-58.94%

16.4	0.15452533	0.09094461	41.15%	0.35809978	0.80275451	-124.17%
16.5	0.14100753	0.10822862	23.25%	0.77897078	1.08162932	-38.85%
16.6	0.21038977	0.08926815	57.57%	0.66643582	0.6393998	4.06%
16.7	0.22043383	0.10756707	51.20%	0.88382495	1.00283826	-13.47%
16.8	0.24745572	0.08474225	65.75%	0.64236435	0.73437755	-14.32%
16.9	0.17579155	0.09542167	45.72%	0.55642551	1.04298971	-87.44%
18	0.18666397	0.11218014	39.90%	0.52853417	0.8488385	-60.60%
20	0.2353268	0.23324442	0.88%	0.62915728	0.74908943	-19.06%
22	0.27448547	0.29576367	-7.75%	0.68016261	0.75588119	-11.13%
24	0.32296823	0.35134564	-8.79%	0.70487013	0.78578998	-11.48%
26	0.36860693	0.40441822	-9.72%	0.73030648	0.81621828	-11.76%
28	0.41356695	0.45765526	-10.66%	0.75299779	0.84626609	-12.39%
30	0.45976281	0.50949119	-10.82%	0.77583147	0.86939222	-12.06%
32	0.50486874	0.56335438	-11.58%	0.79184263	0.89180406	-12.62%
34	0.55079253	0.6132526	-11.34%	0.80476594	0.90218029	-12.10%
36	0.59897567	0.66677908	-11.32%	0.81634615	0.91352623	-11.90%
38	0.64693629	0.71875177	-11.10%	0.82291129	0.9177102	-11.52%
40	0.69505548	0.76743393	-10.41%	0.8255462	0.91377338	-10.69%

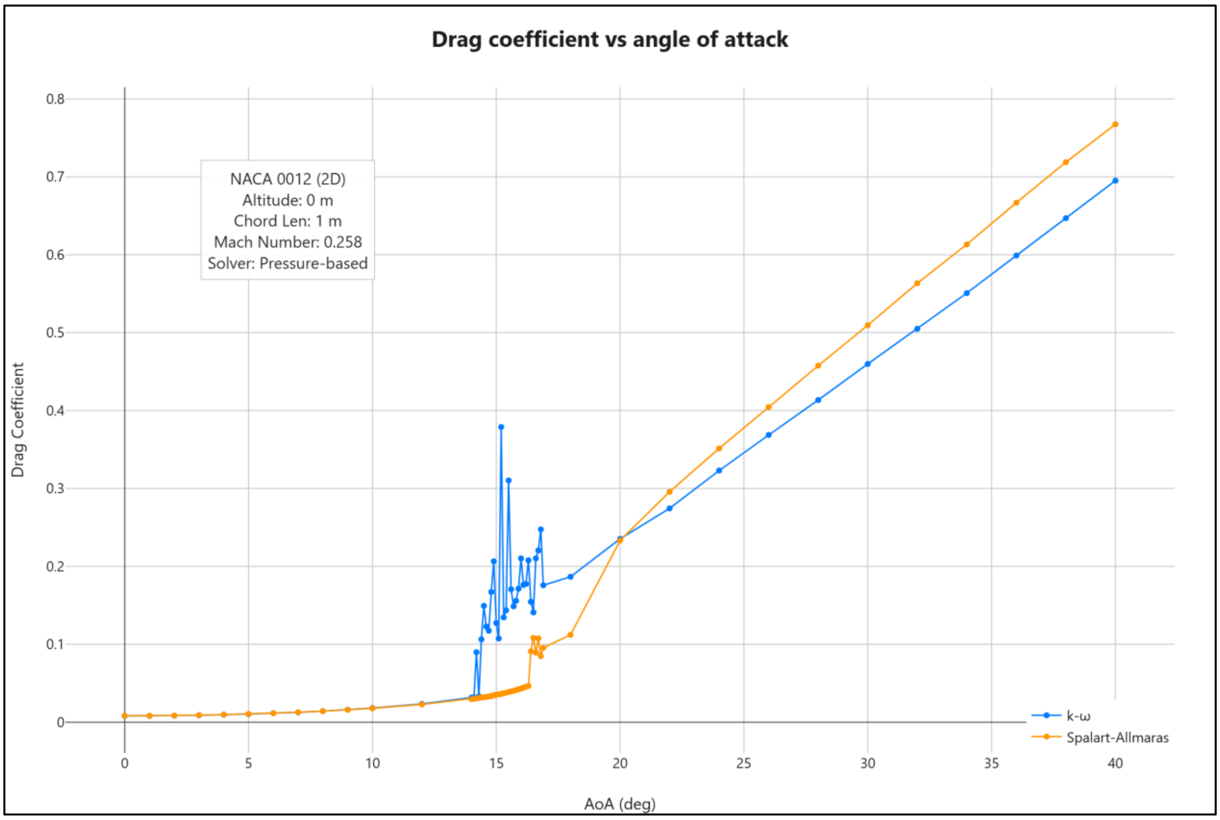

Figure 4-63 - Drag coefficient at different angles of attack

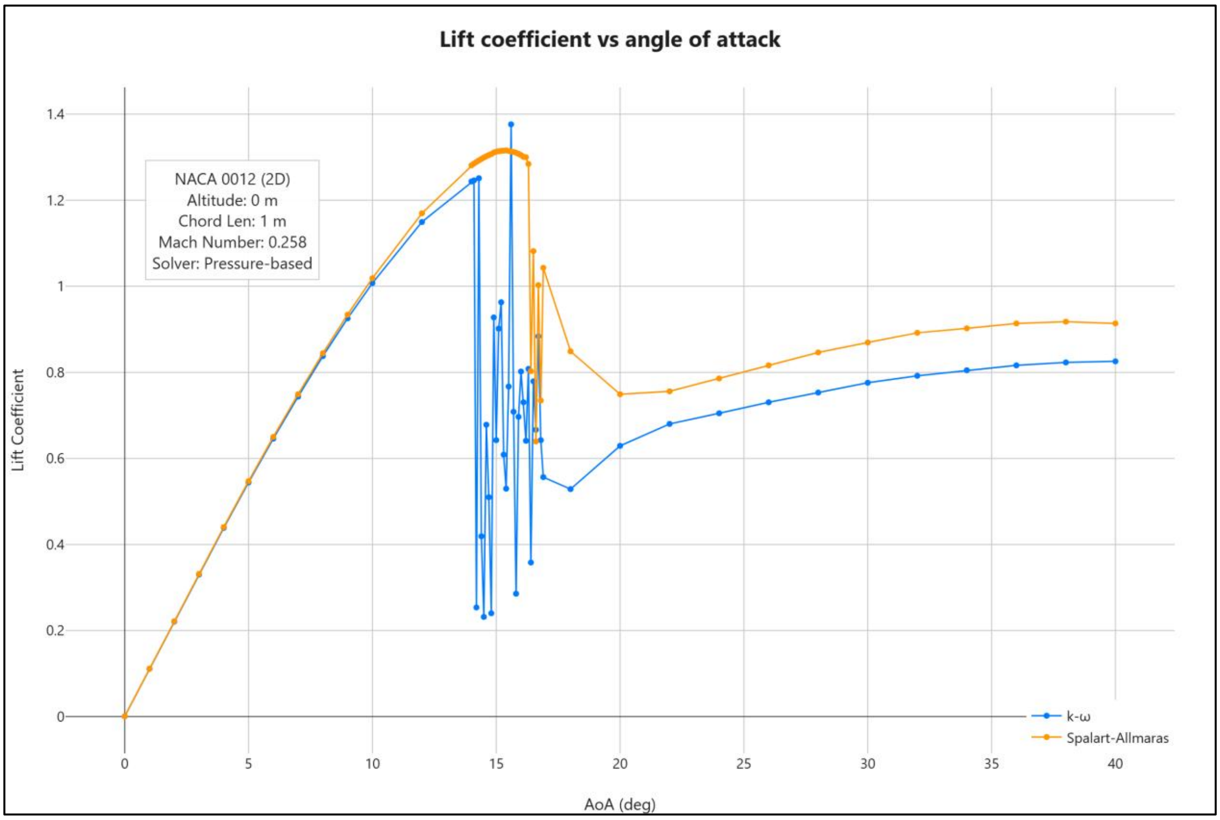

Figure 4-64 - Lift coefficient at different angles of attack

4.6.9 Downforce Trial

The adjoint solver programme was configured to achieve the minimum lift-to-drag ratio, instead of the maximum. This resulted in downforce, instead of lift. The boundary conditions and drag and lift coefficients can be seen in Table 4-39. The comparison between initial and final coordinates is shown in Figure 4-65. The Mach and pressure contours are shown in Figure 4-66 and Figure 4-67, respectively. Figure 4-68 shows the pressure plot of the optimised design.

Table 4-39 - Results from the downforce trial

Variable	**Result**
Dimensions	2D
Altitude (m)	2,675.17
AoA (°)	0.00
Chord Length (m)	1.00
Mach Number	0.716
C_d	0.01064
C_l	-0.45670
Lift-to-drag Ratio	-42.93
Reynolds Number	1.30E+07

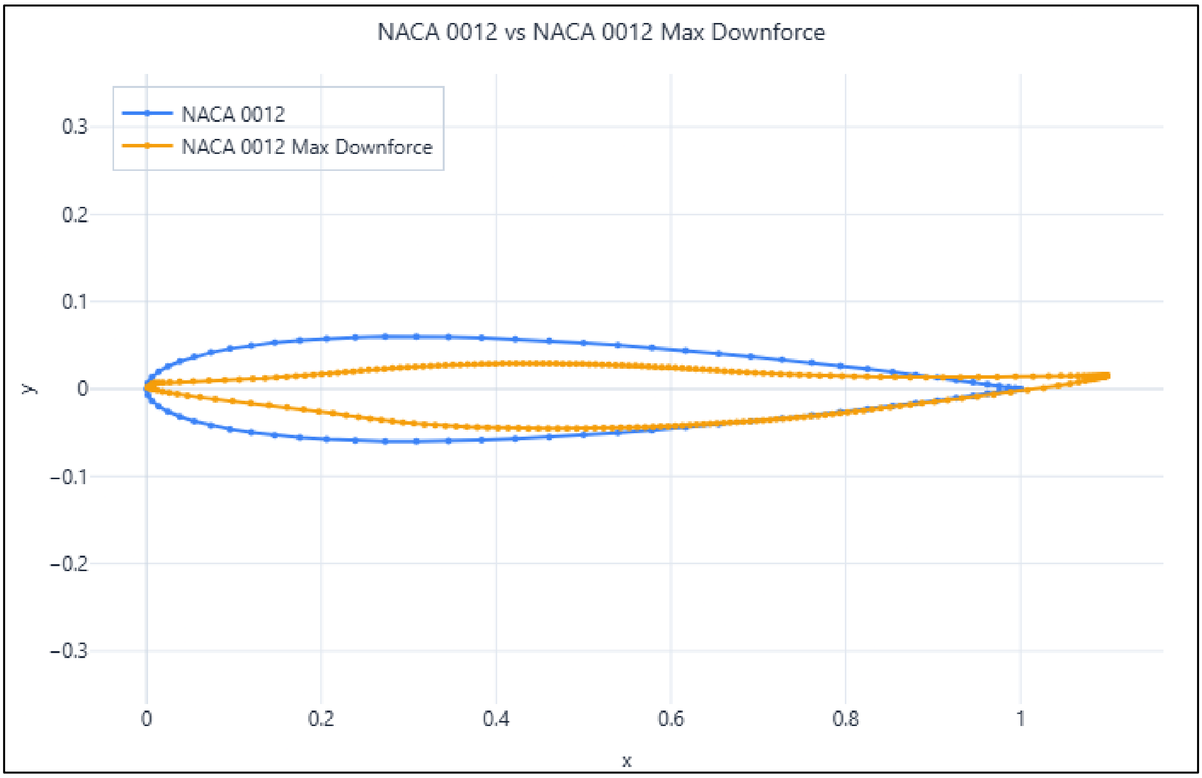

Figure 4-65 - Initial vs optimised NACA 0012 coordinates (maximum downforce)

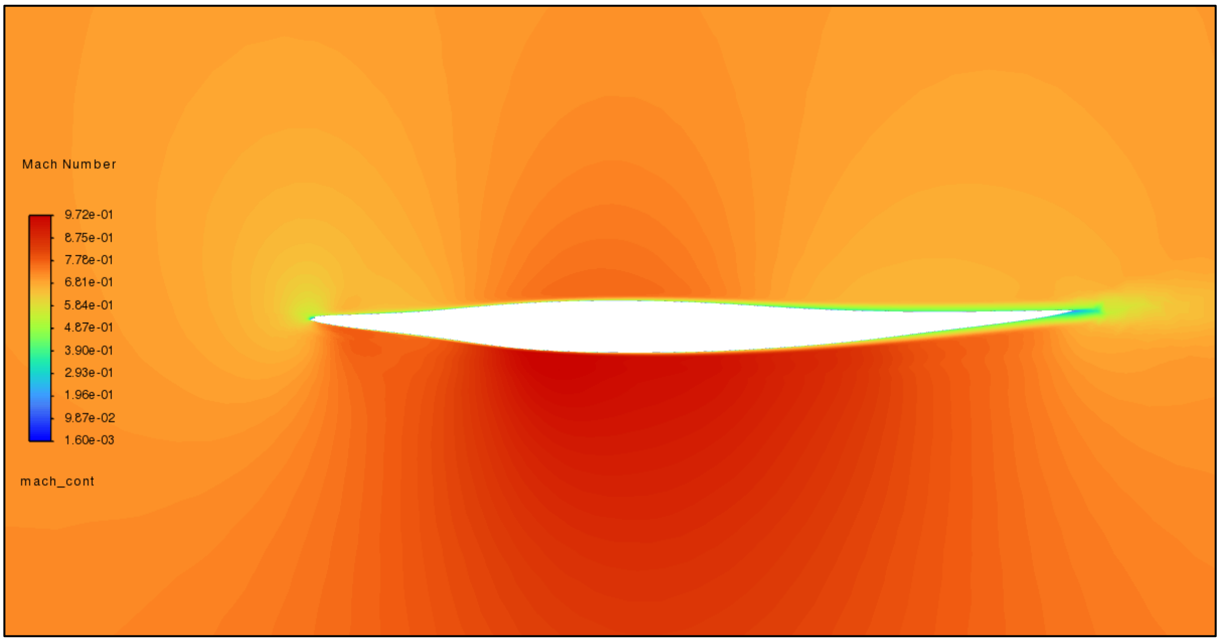

Figure 4-66 - Mach contours of the NACA 0012 optimised for downforce

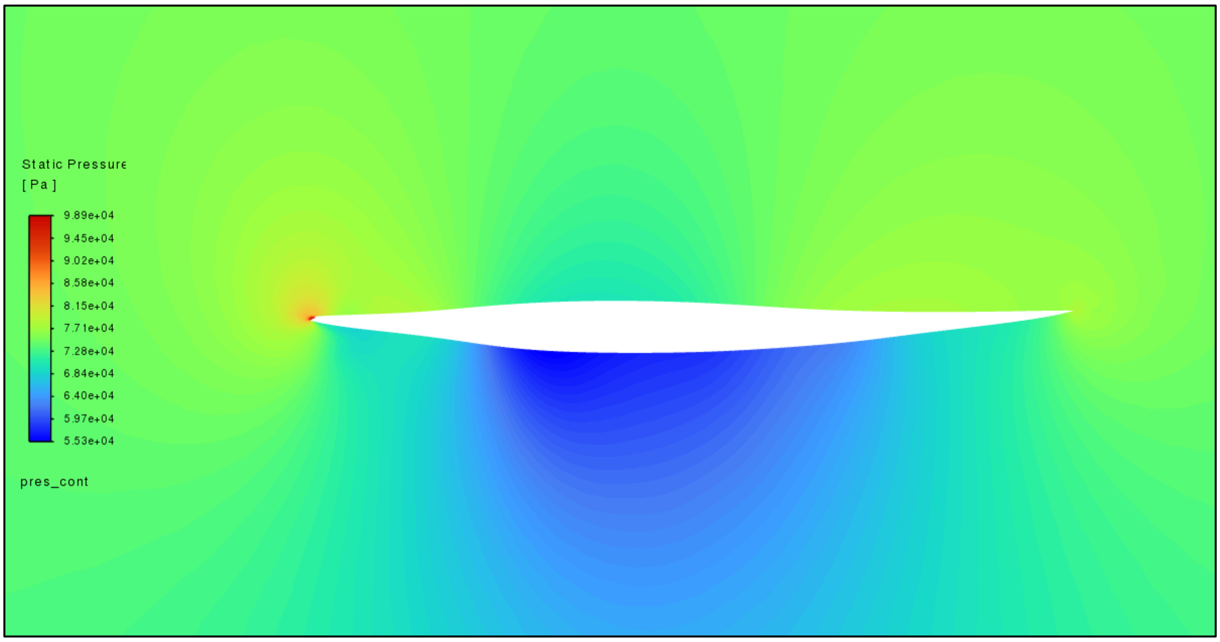

Figure 4-67 - Pressure contours of the NACA 0012 optimised for downforce

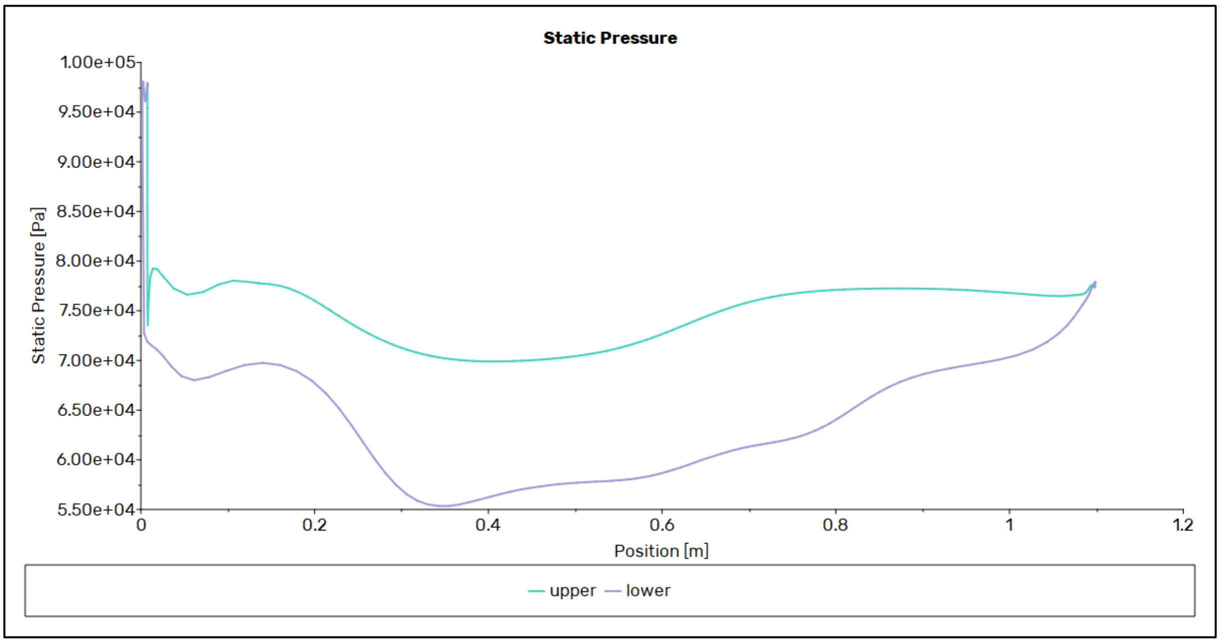

Figure 4-68 - Pressure plot of the NACA 0012 optimised for downforce

4.7 Fuel Savings

The NACA 2412 airfoil was modelled on boundary conditions based on the Cessna 172 aeroplane. Using Eqn. 14, it is possible to estimate the fuel savings.

As shown in Table 4-8, the lift-to-drag ratio of the NACA 2412 airfoil improved from 21.89 to 46.54 (for the constant length profile). Estimating a total of 100 flight hours per year per aircraft at a cruising speed of $M = 0.197$ ($63.8\ m/s$), at 25 flights per year and 4 hours per flight:

$$R = Flight\ Seconds\ (in\ seconds) * Speed(in\ m/s) = 918{,}720\ m \quad \text{Eqn. 15}$$

The TSFC was assumed to be $5 * 10^{-5}$ [45]. The fuel demand of the initial design is:

$$\frac{W_{start}}{W_{end}} = e^{\frac{14{,}400*5*10^{-5}}{(21.89)}} \Rightarrow W_{fuel} = 25.21\ kg \quad \text{Eqn. 16}$$

Vs the optimised design:

$$\frac{W_{start}}{W_{end}} = e^{\frac{14{,}400*5*10^{-5}}{(46.54)}} \Rightarrow W_{fuel} = 11.76\ kg \quad \text{Eqn. 17}$$

For a total of 13.46 kg of fuel saved per flight. [46] states that over 45,000 Cessna 172s were produced. Assuming 20,000 still in operation (a conservative estimate, based on the age of the model), the total fuel saving is 6,728.62 tonnes per year.

For the BACXXX airfoil, used on the Boeing 747-400, [47] states that 255 are in service. According to the ranges in [48], a total of 200 10-hour flights were estimated annually. With a dry mass of 182,480 kg [49], a TSFC of $1.6 * 10^{-5}$ [50], cruising at $M = 0.85$ ($252.1\ m/s$), the improvement was 11,785.47 kg of fuel per flight, for a total of 601,058.98 tonnes of fuel saved per year.

Assuming 3.15 kg of CO_2 emitted per kg of aviation fuel [51], the total carbon savings are equivalent to 21,195.15 tonnes per year for the Cessna 172 and 1,893,335.78 tonnes per year for the Boeing 747-400.

Table 4-40 - The fuel and carbon savings of improved profiles

Value	Cessna 172	Boeing 747-400
Initial L/D	21.89	6.29
Improved L/D	46.54	18.65
Initial Fuel Demand (kg)	25.21	17,509.83
Optimised Fuel Demand (kg)	11.76	5,724.36
Fuel Saved Per Year (tonnes)	6,728.62	601,058.98
CO_2 Saved Per Year (tonnes)	21,195.15	1,893,335.78

5 Discussion

The primary goal of this project was twofold: Firstly, to optimise the given airfoil profiles for a better lift-to-drag. Secondly, to determine and outline the optimal process, with minimal resource usage, for meshing, solving, and optimising airfoils. The latter was achieved by automating the meshing, solution, and optimisation steps using Ansys PyFluent and testing different configurations to compare solve time and accuracy.

5.1 interprets the meshing, solving, and optimisation results of each of the five cases tested. For each airfoil, the section includes verification of the mesh quality, by using orthogonal quality and skewness, and verification of the solution, by using a reference case to recreate flow conditions from previously published papers and comparing the recorded values.

5.2 interprets the results of all additional trials performed in 4.6 Additional Trials, such as the y^+ trial and downforce trial. Mainly, this section aims to clarify how some settings in Ansys Fluent can influence results, and which settings are appropriate for differing flow conditions.

5.3 discusses the advantages and disadvantages of using PyFluent in Python to automate this process. The setup, inputs, and outputs for each of the scripts are also analysed.

5.4 lists the recommended settings for meshing, solving, and optimising, for each flow condition (subsonic, transonic, supersonic), according to the results of 4.6 Additional Trials.

5.5 discusses the differences between 2D and 3D simulations.

5.6 discusses the advantages and disadvantages of using the adjoint solver for optimising airfoils, as well as the possibility of utilising the codebase created for this research for other, non-airfoil, applications.

5.1 Airfoil Solving and Optimisation

It is important to verify the CFD results before any conclusion can reliably be drawn. The methods for verifying the data are discussed in 3.2.10 Data Analysis and Results Verification. The y^+ value requirements are discussed in 5.2.1 y^+ Trial.

5.1.1 NACA 0012

The NACA 0012 mesh, as shown in Table 4-1, was of acceptable quality in both 2D and 3D. The reference case, which aims to validate the solver settings, had overall good correlation with Hasan [35], with the lift-to-drag ratio within 3% (see Table 4-3). With respect to Ladson [34], while the lift coefficient was within 1% of the value shown in the paper, the recorded drag coefficient was 36.68% higher than the reference, and thus the lift-to-drag ratio was off by -57.24%. Ladson [34] tested the airfoil in a wind tunnel, whereas Hasan [35] was a CFD study, which likely explains the difference between values.

The test case for the airfoil, simulating $M = 0.72$ at $h = 2675\ m$, and $\alpha = 1.55°$, measured a lift-to-drag ratio of 23.18 in 3D and 27.37 in 2D. Since the solver settings were verified using the verification trial, and since the drag and lift coefficients recorded were within a realistic range, the results of this trial are consistent with other CFD studies, although further direct experimental verification is required.

The Mach contours (see Figure 4-1) show the acceleration of the flow over the upper surface. Since the airflow over the top surface (shown in Figure 4-2), reaches a speed above Mach 1, the flow can be considered transonic, and the formation of a transonic shock wave is visible. Figure 4-3 confirms the location of the low-pressure zones above and below the airfoil. These flow characteristics agree with previous research discussed in 2.1.8 Transonic Flow.

5.1.1.1 Optimised design

The optimised results, shown in Table 4-4, show an increase in both the drag and the lift coefficient. Both the constant and variable length airfoils improved by $\sim$44%. The variable length profile did not change in length during optimisation, suggesting that the airfoil length is not critical to its lift-to-drag ratio at this flow. Both optimised profiles (Figure 4-4 and Figure 4-5) show a reduced chord thickness, slight camber, and a curled trailing edge. The thickest point is also shifted downstream, increasing the length of the curve on the upper surface, and generating more lift.

5.1.2 NACA 2412

The reference case of NACA 2412 also had great correlation with [35], with a 2.92% difference in total lift-to-drag ratio (22.67 recorded vs 23.33 for [35]). The test case showed a lower lift-to-drag ratio (21.89) when compared to the reference. This is expected considering the larger chord length, which increases drag due to a larger overall airfoil profile, and due to the lower Mach number and lower pressure, which decreases lift. Overall, both the verification and test cases match the expected values.

The Mach and Pressure contours (Figure 4-12 and Figure 4-13, respectively) clearly show the subsonic flow pattern with low pressure on the upper surface.

5.1.2.1 Optimised design

The optimised NACA 2412 profiles showed great improvement, with the constant length case improving the lift-to-drag ratio by 112.65%, and the variable length profile by 153.67%. Similarly to the NACA 0012, both the drag and lift coefficients were increased. The constant length profile showed a slightly reduced maximum thickness and a pronounced curl on the trailing edge.

Unlike the NACA 0012, the variable length profile did change in length, becoming shorter. It also features a sharper leading edge and is thinner. The trailing edge was moved below the chord line, in effect increasing the angle of attack (although the boundary conditions were not changed).

5.1.3 BACXXX

The reference case of the BACXXX airfoil has by far the largest difference between the theoretical and simulated values. The source of this discrepancy is likely [37]. The study can be called into question since both the drag and lift coefficients are unreasonably high for the test case. [37] stated a lift coefficient of $C_l = 4.5$ and drag coefficient of $C_d = 0.85$, which is far higher than any simple airfoil can achieve in normal flow conditions ([52] states that a conventional airfoil stalls at around $C_l = 1.5$).

However, the lift and drag coefficients, when divided by the chord length of $14.34\ m$, give values of $C_d = 0.85/14.34 = 0.05927$ and $C_l = 4.5/14.34 = 0.31381$. These coefficients are more realistic, and, in fact, close to the simulated values of $C_d = 0.05838$ (a difference of -1.54%) and $C_l = 0.36604$ (a difference of 14.34%). It is possible that [37] has a non-dimensionalisation error, where the surface area was not scaled when scaling the mesh, which, in Ansys Fluent, inflates the value of the coefficients by a multiple of that amount (in this case, 14.34), explaining the discrepancy.

Both the reference and test cases for BACXXX showed a larger difference between the 2D and 3D drag and lift coefficients than previous airfoils, suggesting that compared to subsonic flows, spanwise flow effects play a larger role in transonic conditions.

The test case produced a similar lift-to-drag ratio to the reference case. This was expected since the test and reference cases are similar. The Mach and pressure contours (shown in Figure 4-23 and Figure 4-24, respectively) show a clear transonic shock wave, matching previous research discussed in 2.1.8 Transonic Flow.

5.1.3.1 Optimised design

Both optimised BACXXX profiles show striking changes. The constant length model increases both the drag and lift coefficients, resulting in a 196.64% increase in the lift-to-drag ratio. It also reduces the transonic shock wave (see Figure 4-28) compared to the initial profile. This is achieved by minimising the overall thickness and curvature of both the upper and lower surfaces of the airfoil, helping keep the flow attached. Furthermore, the optimised profile introduces a highly complex, curved trailing edge which seemingly resembles the adjoint solver's attempt at creating a Gurney flap [53].

The variable length trial for the BACXXX airfoil was the most impactful result of the study. The result was a thin profile with a sharp leading edge, akin to a supersonic airfoil. Although the sharp leading edge is expected to experience sensitivity and instability at off-design conditions, the advantage of the shape is the complete elimination of a transonic shock wave (see Figure 4-29), thus reducing the drag coefficient by 88.53%.

5.1.4 NASA SC(2)-0714 in Transonic Flow

The NASA SC(2)-0714 test case could not recreate the results of [38]. However, the results of the study are questionable; [38] states a $C_d = 0.0087$ at $M = 0.8$; however, [54] found the drag divergence to be $M = 0.726$ with a drag coefficient at that point of $C_d = 0.0097$. Considering that the test case of $M = 0.8$ is far past this point, the drag is expected to be higher.

The test case of NASA SC(2)-0714 shows a sharp rise in drag and a sharp drop in lift, as expected considering the test case is past the drag divergence number. The Mach and pressure contours (Figure 4-34 and Figure 4-35, respectively) show another clear example of a transonic shock wave. The difference between 2D and 3D results is similar to the BACXXX trial, confirming that spanwise effects are more important in transonic flows than in any other flow condition.

5.1.4.1 Optimised design

Both optimised designs reduced the drag coefficient by $\sim 55\%$ and improved the lift coefficient by $\sim 250\%$. The overall improvement was similar between the constant and variable length profiles, suggesting that the length was not critical in improving the performance of the airfoil in these flow conditions.

The constant length trial produced a remarkably similar profile to the original. The only notable differences are the slightly reduced thickness, and a large reduction in the curvature of the upper surface at $\sim 70\%$ of the chord.

The variable length trial, similarly to the variable length trial for the BACXXX airfoil, produced a longer, thinner profile with a sharp leading edge. This also produced a massively reduced transonic shock wave but is likely to be an unstable design.

5.1.5 NASA SC(2)-0714 in Supersonic Flow

Analogous to the previous NASA SC(2)-0714 trial, the difference between the reference case and [38] is large. However, the validity of [38] is still questionable; the drag coefficient, $C_d = 0.024$, is quite low for supersonic flight.

The contours of the test case (Figure 4-45 and Figure 4-46) show a well-defined shock wave, matching the theory discussed in 2.1.9 Supersonic Flow. With a very small difference between 2D and 3D results, and a low number of iterations to converge for the 3D solve, it suggests that well-defined shock waves are as easy to simulate as subsonic airfoils with laminar flow.

5.1.5.1 Optimised design

Both the constant and variable length cases produced a large improvement in the lift-to-drag ratio. The constant length trial increased the drag and lift coefficients, for a total lift-to-drag ratio increase of 260.78%, producing a unique profile (see Figure 4-48). The profile features a “pocket” on the lower surface at $\sim 20\%$ of the chord, creating a high-pressure zone (shown in Figure 4-52) and generating lift.

The variable length trial, in comparison, produces a more orthodox airfoil profile. The optimised airfoil features a sharper trailing edge, reduced thickness, and reduced curvature on the upper surface. The trailing edge is also moved below the leading edge, in effect giving the profile an increased angle of attack. It is visible in Figure 4-51 that the shock wave forms much closer to the trailing edge, when compared to the original profile.

Even after optimisation, the lift-to-drag ratio is still low for both profiles (0.83 and 1.46). This is expected for supersonic test cases, since the shock wave is a large source of drag. Even if the absolute performance remains below other test cases, the increase in lift-to-drag ratio compared to baseline is notable.

5.2 Additional Trials

5.2.1 y^+ Trial

Traditionally, a CFD study would begin by analysing mesh convergence, where mesh size is varied and the difference between outputs is observed. However, in this study, a y^+ convergence trial was completed instead. The reason for this is due to the peculiarities of Ansys Fluent Meshing. To achieve low skewness and high orthogonal quality, the minimum size of the mesh must scale with the first layer height. The maximum size of the mesh depends on the growth rate, a factor which describes how fast the cell size can grow (i.e., for a growth rate of 1.2, a layer of cells will be 1.2 times larger than the previous layer). However, a high growth rate also degrades the mesh quality. Therefore, both the minimum and the maximum cell sizes are limited based on the first layer height, which itself is based on the desired y^+ value. Hence, varying y^+ changes the resolution of the entire mesh.

The trial was done in 2D and 3D at subsonic, transonic, and supersonic flows. All trials used the NACA 2412, with a chord length of $l = 1.0\ m$, altitude of $4114.8\ m$, and an angle of attack of $\alpha = 0°$.

For subsonic and supersonic flows, as shown in Table 4-21 and Table 4-27, the results were within $\pm 5\%$ for any y^+ value below 50. In fact, the supersonic trial was within $\pm 2\%$ even at $y^+ = 100$. The transonic trial (Table 4-24) showed increasing error in both drag and lift coefficients for 3D, while the 2D solves were largely unaffected.

Another observation can be made from Table 4-22, Table 4-25, and Table 4-28; in 2D simulations, the face count (and thus solving speed) is affected very little by changes in y^+.

Furthermore, all solves with high y^+ values struggled to converge in 3D, with some requiring high-order term relaxation settings to converge, suggesting that lowering the first layer height can help with non-converging solves.

For 2D solves, using the smallest y^+ value possible is recommended, since the solve speed is largely unaffected by first layer height. For 3D solves, the trade-off between solve time and accuracy must be considered: smaller y^+ values are more accurate but take longer to solve. The exception for both 2D and 3D are transonic flow conditions; at that speed, the smallest possible y^+ value should be used.

5.2.2 Mesh Type Trial

The mesh type trial considered only the four volume fill options available for 3D models in Ansys Fluent Meshing: Hexcore, Poly-Hexcore, Polyhedra, and Tetrahedral. All meshes were tested in the same boundary conditions.

As shown in Table 4-30, apart from Hexcore, all volume fill methods recorded the same drag and lift coefficients and Mach/pressure contours. Polyhedra had the smallest cell count and therefore was fastest to solve, suggesting that Polyhedra should be used when meshing 3D domains.

5.2.3 GUI and Core Count Trial

Running Ansys Fluent with a disabled GUI (headless mode) was one of the main objectives of automating the process using PyFluent. The meshing and solving speed was tested with the GUI on and off using the same settings.

Figure 4-56 and Figure 4-57 show the results of the study. As seen, there was no significant difference in time when disabling the GUI. The biggest difference between solves was due to the change in core count. Increasing the core count did not result in decreased solve times. In fact, the optimal setup uses 2 cores for meshing and 4 for solving.

This is attributable to the equipment: Running any background tasks while solving could require several free cores. Thus, with a high core count, the demands of the system are interfering with the solve and inflating the required time. Another explanation of the phenomenon is Amdahl's Law [55], which states that an increase in parallelisation is only effective for larger tasks, since the compute required to divide the task outweighs the benefit of utilising additional cores. However, Amdahl's Law only predicts a plateau in processing speeds, instead of an increase.

Overall, the trial is inconclusive, since the comparison was limited to only a single mesh and solution case. Furthermore, as all studies were done only once, the timing could have been affected by external factors, such as temperature variations or running background processes. For a more reliable conclusion, the trials should be done with different mesh types, sizes, dimensions, number of iterations, and with more repetitions of each case.

5.2.4 Time Scale Trial

Different time scales were tested when using the global time step setting, to establish the difference in number of iterations required. Table 4-32 shows the results of the trial[10]. All tests were conducted in transonic flow just before Mach divergence. A smaller number of iterations was required at higher pseudo time scales with diminishing returns. Notably, the large time scale values can cause divergence. This requires restarting the solve, thus negating any time saved. Considering that the overall improvement is marginal, but the risk of divergence is high, disabling the global time step by default is recommended. In select cases where the solver fails to converge entirely without a pseudo time scale, a smaller scale is recommended. If the solve is repeated multiple times, the time scale can be increased marginally until divergence is observed.

5.2.5 Turbulence Method Trial

Different turbulence methods were tested to establish the optimal. In general, there was great agreement in drag and lift coefficients between the methods.[11] The most notable difference between them was number of iterations and the time taken per iteration. The results are shown in Figure 4-59.

[10] The number of iterations for this trial was limited to 500. Therefore, all results with 500+ are essentially undefined, as they never converged.

[11] For the boundary conditions shown in the text box of Figure 4-58. As seen in 5.2.8 AoA Trial, the models start diverging at high angles of attack.

The fastest solver both in number of iterations and time taken was Spalart-Allmaras, a one-equation model. There was little difference between $k - \omega\, SST$ and $k - \omega$ GEKO. The results are expected considering that the $k - \omega$ models use two equations, and thus more computing power.

[23] states that the Spalart-Allmaras model can handle attached flow, or mild flow separation. Since it is the fastest solver, using it is recommended for slower, attached flows. If the flow condition is unclear, the $k - \omega$ GEKO model should be used as the baseline.

5.2.6 Flow Scheme Trial

The Coupled, PISO, SIMPLE, and SIMPLEC flow schemes were tested to establish the difference. Table 4-34 shows the number of iterations required to converge, and Figure 4-60 shows the time per iteration. The Coupled solver converged in the least number of iterations but was slowest per iteration. The PISO and SIMPLE solvers did not fully converge in 2000 iterations (the maximum number used in this study). Hence, using the Coupled solver by default is recommended to ensure the fastest convergence.

5.2.7 Drag Divergence Trial

The NACA 0012 airfoil was tested at various Mach numbers to establish the drag divergence speed and to demonstrate the ability of the process to solve for different Mach numbers. Figure 4-61 shows that the drag divergence speed is between 0.7 and 0.75, which agrees with [56]. Figure 4-62 shows that the lift coefficient experiences a sudden drop at $M = 0.75$, which [57] describes as shock stall, the sudden loss of lift experienced when reaching the critical Mach number. The lift coefficient reaches 0 at $M = 0.85$ and only partially recovers.

5.2.8 AoA Trial

The NACA 0012 airfoil was tested at various angles of attack, to establish the stall angle and demonstrate the code's ability to solve for different angles of attack. The $k - \omega$ GEKO and Spalart-Allmaras models were used and compared.

Figure 4-64 shows the lift coefficient for both solvers. As seen, there is close agreement between the turbulence methods until $\alpha = 14°$, when the $k - \omega$ model predicts stall. From this point, the solver struggles to converge, and large variations in the drag and lift coefficients are visible. The Spalart-Allmaras model predicts an attached flow until $\alpha = 16°$.

Overall, the results confirm expectations: The models had great agreement pre-stall ($\pm 6\%$ before $14°$). The Spalart-Allmaras model, designed for attached flows or mild separation [23], predicted a more favourable condition.

5.2.9 Downforce Trial

To demonstrate the usability of the code in other applications, the process described in 3.2.6 Adjoint Solver Setup was repeated, but the solver was set to reduce the lift-to-drag ratio instead of increasing it. The resulting profile, presented in Figure 4-65, generated downforce.

The output resembles an upside-down transonic airfoil, with a sharp leading and trailing edge. The pressure contours in Figure 4-67 show the low-pressure area generated at the curve. This trial demonstrates the versatility of the Ansys adjoint solver.

The fact that re-configuring the solver from optimising lift to optimising downforce instead required a single change in setup demonstrates the versatility of the automation scripts.

5.3 PyFluent Automation Process

It could be argued that the usage of PyFluent was unnecessary. The use of automation is justified by the scale of the simulations. Hundreds of simulations were done. If done manually, the results would require a significant number of man-hours. Furthermore, running each case automatically removes the possibility of human error.

The benefits of using PyFluent are as follows:

- ***Repeatability:*** *All trials could be recreated with the exact same solver settings and boundary conditions, ensuring consistency between solves when comparing results at different parametric conditions.*
- ***Advanced Parametrisation:*** *Ansys has limited parameter settings. It is possible to parametrise certain boundary conditions like Mach number, but some values, like mesh size settings or solver settings (such as the viscosity method) cannot be parametrised.*
- ***Elimination of Human Error:*** *Manual setup of hundreds of individual solves will almost certainly introduce human error. Since the setup process was automated, human error was only present while setting up the input file. However, since all inputs used are saved as a separate file, the inputs for all solves are verifiable.*

The approach, however, comes with limitations:

- ***Development Time:*** *Writing the scripts required a significant time investment. Furthermore, adding or editing steps during setup required modifying the code and debugging required re-running upstream steps, inflating the total development time.*
- ***Limited Documentation:*** *Ansys PyFluent documentation is outdated and incomplete. There are very few examples of the full code available. Several steps included a large amount of trial-and-error. Hence, adding new features took much longer than expected.*
- ***Limited Geometry/Domain Shapes:*** *All domains used in the experiment must be similar in shape to Figure 3-2, with a split upper and lower surface. Thus, if this script is to be adopted into an existing workflow, all domains must either be remade entirely or retroactively modified to suit the code.*

Overall, in this project, the benefits of automation at this scale outweighed the costs. The time sunk into developing the scripts was justified by their simplicity and reliability.

5.4 Recommended Settings

The following is a collection of the recommended settings, based on the trials presented above:

- *5.2.1 y^+ Trial recommends $y^+ < 1$ for 2D solves, since the simulation time is largely unaffected by the first layer height. For 3D solves, smaller y^+ values are more accurate at the cost of processing time. For all transonic solves, $y^+ < 1$ for maximum accuracy.*
- *5.2.2 Mesh Type Trial indicates that Polyhedra is the optimal volume fill method for 3D solves.*
- *5.2.3 GUI and Core Count Trial reveals that GUI makes no difference. The best core count for this setup was 2 cores for meshing and 4 cores for solving, although results were inconclusive and therefore no definitive recommendations can be made.*
- *As discussed in 5.2.4 Time Scale Trial, the optimal setup has a high time step scale, but only cases with stable flow. By default, the global time step feature should be disabled.*
- *It is suggested in 5.2.5 Turbulence Method Trial to use the $k - \omega$ GEKO model if the flow conditions are unclear. If it is known that the flow is attached and stable, Spalart-Allmaras is optimal.*
- *5.2.6 Flow Scheme Trial compared the flow schemes and found that the Coupled scheme is the most reliable.*

5.5 2D vs 3D

In general, it is evident that across all trials, 2D results exhibited a higher lift-to-drag ratio when compared to 3D trials. In fact, Fertahi [58] explored the difference between 2D and 3D simulations and concluded that since 2D simulations are unable to characterise the spanwise flow effects, they exhibit more favourable flow conditions (in the case of airfoils, better flow conditions result in a higher lift-to-drag ratio). The difference between 2D and 3D is most evident during transonic flows. For maximum accuracy, using 3D domains is recommended. However, for adjoint-based optimisation, 2D should be used, as 3D demands large amounts of compute. Instead, the output of adjoint should be verified via a final 3D solve.

5.6 Adjoint Optimisation

The Ansys Adjoint solver optimisation was overall effective in increasing the lift-to-drag ratio of all airfoils in all cases tested. However, the outputs are not perfect. As noted by [33] and mentioned above, single point optimisation produces gains only for the selected operating condition, with no regard for off-design performance. Furthermore, the extremely thin profiles (see Figure 4-27 or Figure 4-40) are likely unsuitable for real-world scenarios, considering that wings are structural components and need to hold fuel. Lastly, the output coordinates require manual smoothing and sorting.

It must be noted that improving the lift-to-drag ratio by increasing both the drag and lift coefficients can be achieved by simply increasing the airfoil's angle of attack instead. Thus, in cases where both the lift and drag coefficients were higher (like NACA 0012 and NACA 2412), the airfoil is closer to stall.

Despite these limitations, the adjoint solver demonstrated its usability as a design exploration tool. While it cannot fully automate optimisation in the current form, it can be utilised to identify the correct direction for optimisation.

5.6.1 Non-airfoil Optimisation

As demonstrated in 4.6.9 Downforce Trial, the solver can be used, unmodified, for any profile with a distinct upper and lower surface, for maximising or minimising the lift coefficient, drag coefficient, or lift-to-drag ratio. As mentioned above, the profiles produced are fragile and sensitive to slight changes in angle of attack. This makes the solver potentially applicable to applications with non-structural wings and consistent operating conditions, such as aerodynamic vehicles on cars (front or rear wings).

6 Conclusions & Future Work

The Boeing BACXXX airfoil is the best-case scenario for optimisation. The trial eliminated transonic shock entirely, which, if retrofitted on all current Boeing 747-400s, would yield an estimated 1.8 million tonnes of CO_2 savings annually. Across the four selected airfoils and five selected cases, all yielded an improvement in the lift-to-drag ratio.

The project had two goals: To find the optimised profiles for each airfoil, and to establish the ideal optimisation process, saving both man-hours and computing resources. Overall, it succeeded in both counts, although limitations were present and are suggested as future work.

All airfoils, for both constant and variable length cases had an improved lift-to-drag ratio. Both NACA 0012 profiles (constant and variable length) improved by over 44%. The NACA 2412 constant length profile improved by 112.65%, and the variable length version had a 153.67% higher lift-to-drag ratio. The BACXXX profile improved by 196.64% and 658.32% for the constant and variable cases, respectively. The variable length case eliminated transonic shock entirely, by reducing thickness and developing a sharp leading edge. NASA SC(2)-0714 in transonic flow was improved by 686.73% and 689.05% (large percentage improvement because the initial lift-to-drag ratio was so low), and in supersonic flow, the improvement was 260.78% for the constant length case and 533.47% for the variable length case.

The PyFluent-based automation scripts were developed and released as a contribution in Appendix E: Sample Code.

Overall, although full end-to-end automation of optimisation was not achieved in this project, the solver proved itself to be an accessible and valuable tool for optimisation during the early stages of airfoil design.

6.1 Limitations and Future Work

The project did have limitations:

- ***Single-point optimisation:*** *The most relevant and significant limitation of this study was the single-point optimisation. A natural course of action, therefore, is to extend the study to cover Ansys Adjoint multi-point optimisation, including boundary conditions simulating the ascent and descent stage ([33] used 6 separate boundary conditions). This way, the produced geometries are not specialised and more accurately reflect real-world conditions.*
- ***Validation:*** *The CFD results provided have been validated before optimisation. The resulting improved profiles have not been validated against experimental data since they are unique and no research exists. Thus, wind tunnel testing of the geometries is recommended in order to provide experimental validation.*
- ***Pressure-based solver****: As mentioned in 2.4.4 Pressure-Based vs Density-Based Solvers, [30] states that while the pressure-based solver is adequate for compressible flows, the density-based solver could produce more accurate results. Thus, a study should be done comparing the solvers to establish the accuracy (or inaccuracy) of the pressure-based solver at high-speed flows.*
- ***Core count trial:*** *The core count trial was inconclusive due to limited cases and number of trials. A more rigorous study, with multiple cases, mesh types, iterations, and dimensions is recommended.*
- ***Full PyFluent automation:*** *While the Python code automated most of the manual inputs required, downloading the coordinates, preparing them, and creating the domain in SolidWorks were all manual steps. For a complete process, these steps could also be automated.*
- ***Applicability:*** *The current code (available in Appendix E: Sample Code) is only suitable for airfoil optimisation, and could be modified to suit other applications as future work.*

- ***Hyperparameter-based meshing:*** *Utilising Optuna [59], a hyperparameter optimisation framework, for meshing, was briefly attempted but not pursued further due to time and resource constraints. Revisiting the hyperparameter framework is a potential future project.*

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8 Appendices

Appendix A: Computer Specifications

Table 8-1 - Specifications of the HP ENVY x360 used for this project

Component	Specification
Manufacturer	HP
Model	HP ENVY x360 Convertible 15-es0xxx
Processors	11th Gen Intel(R) Core(TM) i7-1165G7 @ 2.80GHz
Total Physical Memory	16,936,132,608 bytes
Total Physical Memory (GB)	15.77 GB
OS Name	Microsoft Windows 11 Home Single Language
OS Version	10.0.26200[12]
BIOS Version	HPQOEM - 2
BIOS Release Date	30/11/2021 00:00
Max Clock Speed	4.70 GHz
Number of Cores	4
Number of Logical Processors	8
L2 Cache Size	5,120 bytes
L3 Cache Size	12,288 bytes
GPU 1 Name	Intel(R) Iris(R) Xe Graphics
GPU 1 Adapter RAM	1,073,741,824 bytes
GPU 1 Adapter RAM (GB)	1.00 GB
GPU 1 Driver Version	30.0.101.1340
GPU 2 Name	NVIDIA GeForce MX450
GPU 2 Adapter RAM	2,147,483,648 bytes
GPU 2 Adapter RAM (GB)	2.00 GB
GPU 2 Driver Version	32.0.15.8183
Disk Model	NVMe SK hynix BC711 HFM512GD3JX013N
Disk Size	512,105,932,800 bytes
Disk Size (GB)	476.94 GB
Disk Media Type	Fixed hard disk media
RAM 1 Manufacturer	SK Hynix
RAM 1 Speed	3,200 MHz
RAM 1 Capacity	8,589,934,592 bytes
RAM 1 Capacity (GB)	8.00 GB
RAM 1 Part Number	HMA81GS6CJR8N-XN
RAM 2 Manufacturer	SK Hynix
RAM 2 Speed	3,200 MHz
RAM 2 Capacity	8,589,934,592 bytes
RAM 2 Capacity (GB)	8.00 GB
RAM 2 Part Number	HMA81GS6CJR8N-XN
PowerShell Version	7.5.4

[12] While the machine was updated to the latest version of Windows 11 throughout the project, all results shown in this report were obtained on the same version. No results acquired before updating are used in this project.

Appendix B: Airfoil Coordinates

a. NACA 0012

Table 8-2 - Coordinates of NACA 0012 [39]

NACA 0012

x	y	x	y
1	0	0.001541	-0.006877
0.998459	0.000224	0.006156	-0.013503
0.993844	0.000891	0.013815	-0.019854
0.986185	0.00199	0.024472	-0.025893
0.975528	0.003501	0.03806	-0.03158
0.96194	0.005399	0.054497	-0.036867
0.945503	0.007651	0.07368	-0.041705
0.92632	0.010221	0.095492	-0.046049
0.904508	0.013071	0.119797	-0.049854
0.880203	0.016158	0.146447	-0.053083
0.853553	0.019438	0.175276	-0.055708
0.824724	0.022869	0.206107	-0.057712
0.793893	0.026405	0.238751	-0.059088
0.761249	0.03	0.273005	-0.059841
0.726995	0.03361	0.308658	-0.059988
0.691342	0.037188	0.345492	-0.059557
0.654508	0.040686	0.383277	-0.058582
0.616723	0.044055	0.421783	-0.057108
0.578217	0.047242	0.46077	-0.055184
0.53923	0.050196	0.5	-0.052862
0.5	0.052862	0.53923	-0.050196
0.46077	0.055184	0.578217	-0.047242
0.421783	0.057108	0.616723	-0.044055
0.383277	0.058582	0.654508	-0.040686
0.345492	0.059557	0.691342	-0.037188
0.308658	0.059988	0.726995	-0.03361
0.273005	0.059841	0.761249	-0.03
0.238751	0.059088	0.793893	-0.026405
0.206107	0.057712	0.824724	-0.022869
0.175276	0.055708	0.853553	-0.019438
0.146447	0.053083	0.880203	-0.016158
0.119797	0.049854	0.904508	-0.013071
0.095492	0.046049	0.92632	-0.010221
0.07368	0.041705	0.945503	-0.007651
0.054497	0.036867	0.96194	-0.005399
0.03806	0.03158	0.975528	-0.003501
0.024472	0.025893	0.986185	-0.00199
0.013815	0.019854	0.993844	-0.000891
0.006156	0.013503	0.998459	-0.000224
0.001541	0.006877	1	0
0	0		

b. NACA 2412

Table 8-3 - Coordinates of NACA 2412 [39]

NACA 2412

x	y	x	y
1	0	0.002223	-0.006689
0.998474	0.000326	0.007479	-0.012828
0.993903	0.001298	0.015723	-0.018404
0.986314	0.002896	0.026892	-0.023408
0.975752	0.005092	0.040906	-0.027826
0.962276	0.007845	0.057669	-0.031651
0.945966	0.011105	0.077071	-0.034878
0.926917	0.014814	0.098987	-0.037507
0.90524	0.01891	0.123281	-0.039546
0.881064	0.023324	0.149805	-0.041012
0.854532	0.027985	0.178401	-0.041933
0.825802	0.032822	0.208902	-0.042343
0.795047	0.03776	0.241131	-0.04229
0.762452	0.042726	0.274904	-0.041827
0.728216	0.047648	0.310028	-0.041016
0.692545	0.052453	0.346303	-0.039923
0.655659	0.057071	0.383522	-0.038617
0.617783	0.061432	0.421645	-0.037134
0.579153	0.065468	0.460398	-0.035388
0.540006	0.069113	0.499413	-0.033414
0.500587	0.072303	0.538453	-0.031267
0.461143	0.074977	0.577282	-0.028997
0.421921	0.077082	0.615662	-0.026651
0.383032	0.078547	0.653358	-0.024269
0.34468	0.07918	0.690139	-0.021884
0.307289	0.07893	0.725775	-0.019529
0.271106	0.077795	0.760046	-0.017226
0.236371	0.07579	0.792738	-0.014999
0.203313	0.072945	0.823646	-0.012865
0.172151	0.069308	0.852575	-0.010842
0.143089	0.06494	0.879342	-0.008946
0.116313	0.059917	0.903777	-0.007191
0.091996	0.054325	0.925723	-0.005594
0.070289	0.048257	0.94504	-0.004169
0.051324	0.041808	0.961603	-0.002931
0.035214	0.035076	0.975305	-0.001896
0.022051	0.028152	0.986056	-0.001076
0.011907	0.02112	0.993785	-0.000481
0.004833	0.014049	0.998444	-0.000121
0.00086	0.006997	1	0
0	0		

c. BACXXX

Table 8-4 - Coordinates of BACXXX [39]

BACXXX

x	y
1	0
0.95	0.0116
0.9	0.0218
0.85	0.0307
0.8	0.0384
0.75	0.045
0.7	0.0503
0.65	0.0548
0.6	0.0585
0.55	0.0613
0.5	0.0636
0.45	0.0654
0.4	0.0667
0.35	0.0673
0.3	0.0674
0.25	0.0665
0.2	0.0643
0.15	0.0599
0.125	0.0566
0.1	0.0521
0.075	0.0465
0.05	0.0384
0.0375	0.0334
0.025	0.0271
0.02	0.0242
0.015	0.0207
0.005	0.0112
0.0025	0.0078
0.001	0.005
0.0005	0.0037
0	0

x	y
0.0005	-0.0018
0.001	-0.0027
0.0025	-0.0043
0.005	-0.0058
0.015	-0.0098
0.02	-0.0112
0.025	-0.0125
0.0375	-0.0152
0.05	-0.0175
0.075	-0.0216
0.1	-0.0254
0.125	-0.0288
0.15	-0.032
0.2	-0.0375
0.25	-0.0417
0.3	-0.0445
0.35	-0.0458
0.4	-0.0457
0.45	-0.0443
0.5	-0.0417
0.55	-0.0383
0.6	-0.0344
0.65	-0.0303
0.7	-0.026
0.75	-0.0218
0.8	-0.0174
0.85	-0.0132
0.9	-0.009
0.95	-0.0047
1	0

d. NASA SC(2)-0714

Table 8-5 - Coordinates of NASA SC(2)-0714 [39]

$NASA\ SC(2)-0714$					
x	y	x	y	x	y
1	-0.013	0.31	0.0689	0.34	-0.0696
0.99	-0.0063	0.3	0.0686	0.35	-0.0697
0.98	-0.0032	0.29	0.0683	0.36	-0.0697
0.97	-0.0002	0.28	0.0679	0.37	-0.0697
0.96	0.0027	0.27	0.0675	0.38	-0.0696
0.95	0.0056	0.26	0.067	0.39	-0.0695
0.94	0.0084	0.25	0.0665	0.4	-0.0693
0.93	0.0111	0.24	0.066	0.41	-0.0691
0.92	0.0137	0.23	0.0654	0.42	-0.0688
0.91	0.0162	0.22	0.0648	0.43	-0.0685
0.9	0.0187	0.21	0.0641	0.44	-0.0681
0.89	0.0211	0.2	0.0633	0.45	-0.0677
0.88	0.0234	0.19	0.0625	0.46	-0.0672
0.87	0.0256	0.18	0.0616	0.47	-0.0667
0.86	0.0278	0.17	0.0607	0.48	-0.0661
0.85	0.0299	0.16	0.0597	0.49	-0.0654
0.84	0.0319	0.15	0.0586	0.5	-0.0646
0.83	0.0339	0.14	0.0574	0.51	-0.0637
0.82	0.0358	0.13	0.0562	0.52	-0.0627
0.81	0.0376	0.12	0.0549	0.53	-0.0616
0.8	0.0394	0.11	0.0535	0.54	-0.0604
0.79	0.0411	0.1	0.0519	0.55	-0.0591
0.78	0.0427	0.09	0.0502	0.56	-0.0577
0.77	0.0443	0.08	0.0484	0.57	-0.0562
0.76	0.0458	0.07	0.0463	0.58	-0.0546
0.75	0.0473	0.06	0.044	0.59	-0.0529
0.74	0.0487	0.05	0.0414	0.6	-0.0511
0.73	0.05	0.04	0.0383	0.61	-0.0492
0.72	0.0513	0.03	0.0346	0.62	-0.0473
0.71	0.0525	0.02	0.0296	0.63	-0.0453
0.7	0.0537	0.01	0.0224	0.64	-0.0433
0.69	0.0548	0.005	0.01658	0.65	-0.0412
0.68	0.0559	0.002	0.01077	0.66	-0.0391
0.67	0.0569	0	0	0.67	-0.037
0.66	0.0579	0.002	-0.01077	0.68	-0.0348
0.65	0.0588	0.005	-0.01658	0.69	-0.0326
0.64	0.0597	0.01	-0.0224	0.7	-0.0304
0.63	0.0605	0.02	-0.0296	0.71	-0.0282
0.62	0.0613	0.03	-0.0345	0.72	-0.026
0.61	0.0621	0.04	-0.0382	0.73	-0.0238
0.6	0.0628	0.05	-0.0413	0.74	-0.0216

0.59	0.0635
0.58	0.0641
0.57	0.0647
0.56	0.0653
0.55	0.0658
0.54	0.0663
0.53	0.0668
0.52	0.0672
0.51	0.0676
0.5	0.068
0.49	0.0683
0.48	0.0686
0.47	0.0689
0.46	0.0691
0.45	0.0693
0.44	0.0695
0.43	0.0696
0.42	0.0697
0.41	0.0698
0.4	0.0699
0.39	0.0699
0.38	0.0699
0.37	0.0699
0.36	0.0698
0.35	0.0697
0.34	0.0696
0.33	0.0694
0.32	0.0692

0.06	-0.0439
0.07	-0.0462
0.08	-0.0483
0.09	-0.0501
0.1	-0.0518
0.11	-0.0534
0.12	-0.0549
0.13	-0.0562
0.14	-0.0574
0.15	-0.0586
0.16	-0.0597
0.17	-0.0607
0.18	-0.0616
0.19	-0.0625
0.2	-0.0633
0.21	-0.0641
0.22	-0.0648
0.23	-0.0655
0.24	-0.0661
0.25	-0.0667
0.26	-0.0672
0.27	-0.0677
0.28	-0.0681
0.29	-0.0685
0.3	-0.0688
0.31	-0.0691
0.32	-0.0693
0.33	-0.0695

0.75	-0.0194
0.76	-0.0173
0.77	-0.0152
0.78	-0.0132
0.79	-0.0113
0.8	-0.0095
0.81	-0.0079
0.82	-0.0064
0.83	-0.005
0.84	-0.0038
0.85	-0.0028
0.86	-0.002
0.87	-0.0014
0.88	-0.001
0.89	-0.0008
0.9	-0.0009
0.91	-0.0012
0.92	-0.0017
0.93	-0.0025
0.94	-0.0036
0.95	-0.005
0.96	-0.0067
0.97	-0.0087
0.98	-0.011
0.99	-0.0136
1	-0.013

Appendix C: Optimised Coordinates

a. NACA 0012 Short

Table 8-6 - List of the optimised coordinates for NACA 0012 (constant length)

NACA 0012 Short

x	y	x	y	x	y
0.00004	0.00000	0.70069	0.04597	0.66883	-0.03182
0.00020	0.00217	0.71046	0.04501	0.65903	-0.03279
0.00066	0.00436	0.72023	0.04402	0.64924	-0.03375
0.00144	0.00657	0.72999	0.04301	0.63944	-0.03467
0.00257	0.00881	0.73973	0.04198	0.62962	-0.03558
0.00412	0.01109	0.74942	0.04093	0.61976	-0.03646
0.00620	0.01350	0.75904	0.03987	0.60990	-0.03732
0.00897	0.01609	0.76865	0.03879	0.60005	-0.03815
0.01274	0.01891	0.77827	0.03768	0.59023	-0.03896
0.01796	0.02204	0.78787	0.03657	0.58044	-0.03973
0.02501	0.02539	0.79744	0.03543	0.57064	-0.04048
0.03412	0.02880	0.80696	0.03429	0.56081	-0.04121
0.04471	0.03193	0.81646	0.03312	0.55097	-0.04191
0.05566	0.03458	0.82598	0.03193	0.54113	-0.04258
0.06657	0.03680	0.83551	0.03070	0.53132	-0.04322
0.07735	0.03870	0.84510	0.02942	0.52153	-0.04383
0.08796	0.04036	0.85475	0.02809	0.51177	-0.04441
0.09839	0.04186	0.86447	0.02668	0.50204	-0.04497
0.10864	0.04322	0.87423	0.02518	0.49233	-0.04549
0.11877	0.04449	0.88402	0.02358	0.48263	-0.04598
0.12883	0.04569	0.89390	0.02186	0.47295	-0.04644
0.13881	0.04684	0.90390	0.01998	0.46329	-0.04688
0.14872	0.04793	0.91398	0.01795	0.45364	-0.04728
0.15859	0.04899	0.92410	0.01578	0.44402	-0.04766
0.16846	0.05001	0.93427	0.01348	0.43441	-0.04800
0.17834	0.05101	0.94446	0.01109	0.42480	-0.04831
0.18822	0.05198	0.95457	0.00870	0.41521	-0.04860
0.19812	0.05293	0.96379	0.00659	0.40563	-0.04885
0.20804	0.05385	0.97142	0.00493	0.39607	-0.04907
0.21797	0.05475	0.97764	0.00367	0.38652	-0.04926
0.22792	0.05561	0.98265	0.00273	0.37700	-0.04942
0.23786	0.05643	0.98670	0.00204	0.36749	-0.04955
0.24779	0.05722	0.98997	0.00153	0.35800	-0.04965
0.25770	0.05796	0.99261	0.00111	0.34853	-0.04972
0.26757	0.05865	0.99474	0.00075	0.33906	-0.04975
0.27742	0.05929	0.99646	0.00047	0.32961	-0.04976
0.28724	0.05987	0.99787	0.00030	0.32017	-0.04973
0.29704	0.06040	0.99901	0.00018	0.31072	-0.04967
0.30681	0.06088	1.00000	0.00001	0.30125	-0.04958

0.31656	0.06129
0.32628	0.06166
0.33598	0.06196
0.34565	0.06221
0.35532	0.06241
0.36498	0.06256
0.37463	0.06266
0.38429	0.06271
0.39395	0.06271
0.40361	0.06267
0.41329	0.06259
0.42298	0.06246
0.43269	0.06230
0.44242	0.06209
0.45216	0.06185
0.46194	0.06157
0.47175	0.06126
0.48160	0.06090
0.49148	0.06052
0.50138	0.06010
0.51131	0.05965
0.52126	0.05917
0.53125	0.05866
0.54127	0.05812
0.55132	0.05754
0.56138	0.05694
0.57144	0.05632
0.58149	0.05567
0.59150	0.05499
0.60149	0.05429
0.61151	0.05357
0.62154	0.05282
0.63156	0.05204
0.64156	0.05124
0.65150	0.05042
0.66139	0.04958
0.67126	0.04871
0.68109	0.04782
0.69090	0.04691

0.99901	-0.00016
0.99787	-0.00028
0.99647	-0.00046
0.99476	-0.00072
0.99263	-0.00106
0.99000	-0.00140
0.98674	-0.00178
0.98271	-0.00221
0.97770	-0.00268
0.97147	-0.00314
0.96379	-0.00354
0.95446	-0.00386
0.94414	-0.00408
0.93364	-0.00428
0.92306	-0.00453
0.91244	-0.00489
0.90179	-0.00539
0.89116	-0.00606
0.88062	-0.00687
0.87013	-0.00783
0.85966	-0.00891
0.84922	-0.01008
0.83888	-0.01133
0.82862	-0.01262
0.81843	-0.01393
0.80829	-0.01526
0.79819	-0.01659
0.78812	-0.01792
0.77805	-0.01922
0.76798	-0.02051
0.75794	-0.02178
0.74795	-0.02301
0.73801	-0.02421
0.72808	-0.02538
0.71813	-0.02653
0.70820	-0.02764
0.69830	-0.02873
0.68845	-0.02979
0.67863	-0.03082

0.29178	-0.04946
0.28229	-0.04929
0.27279	-0.04910
0.26329	-0.04887
0.25378	-0.04860
0.24427	-0.04829
0.23477	-0.04794
0.22528	-0.04756
0.21581	-0.04713
0.20636	-0.04666
0.19696	-0.04616
0.18758	-0.04561
0.17821	-0.04503
0.16881	-0.04441
0.15939	-0.04375
0.14990	-0.04305
0.14029	-0.04232
0.13053	-0.04154
0.12059	-0.04072
0.11048	-0.03984
0.10011	-0.03889
0.08942	-0.03783
0.07846	-0.03661
0.06726	-0.03517
0.05586	-0.03338
0.04451	-0.03111
0.03371	-0.02827
0.02460	-0.02504
0.01766	-0.02181
0.01252	-0.01874
0.00880	-0.01593
0.00608	-0.01338
0.00405	-0.01100
0.00251	-0.00871
0.00139	-0.00647
0.00064	-0.00430
0.00019	-0.00215
0.00004	0.00000

b. NACA 0012 Long

Table 8-7 - List of the optimised coordinates for NACA 0012 (variable length)

NACA 0012 Long					
x	y	x	y	x	y
0.00004	0.00000	0.70069	0.04597	0.66883	-0.03182
0.00020	0.00217	0.71046	0.04501	0.65903	-0.03279
0.00066	0.00436	0.72023	0.04402	0.64924	-0.03375
0.00144	0.00657	0.72999	0.04301	0.63944	-0.03467
0.00257	0.00881	0.73973	0.04198	0.62962	-0.03558
0.00412	0.01109	0.74942	0.04093	0.61976	-0.03646
0.00620	0.01350	0.75904	0.03987	0.60990	-0.03732
0.00897	0.01609	0.76865	0.03879	0.60005	-0.03815
0.01274	0.01891	0.77827	0.03768	0.59023	-0.03896
0.01796	0.02204	0.78787	0.03657	0.58044	-0.03973
0.02501	0.02539	0.79744	0.03543	0.57064	-0.04048
0.03412	0.02880	0.80696	0.03429	0.56081	-0.04121
0.04471	0.03193	0.81646	0.03312	0.55097	-0.04191
0.05566	0.03458	0.82598	0.03193	0.54113	-0.04258
0.06657	0.03680	0.83551	0.03070	0.53132	-0.04322
0.07735	0.03870	0.84510	0.02942	0.52153	-0.04383
0.08796	0.04036	0.85475	0.02809	0.51177	-0.04441
0.09839	0.04186	0.86447	0.02668	0.50204	-0.04497
0.10864	0.04322	0.87423	0.02518	0.49233	-0.04549
0.11877	0.04449	0.88402	0.02358	0.48263	-0.04598
0.12883	0.04569	0.89390	0.02186	0.47295	-0.04644
0.13881	0.04684	0.90390	0.01998	0.46329	-0.04688
0.14872	0.04793	0.91398	0.01795	0.45364	-0.04728
0.15859	0.04899	0.92410	0.01578	0.44402	-0.04766
0.16846	0.05001	0.93427	0.01348	0.43441	-0.04800
0.17834	0.05101	0.94446	0.01109	0.42480	-0.04831
0.18822	0.05198	0.95457	0.00870	0.41521	-0.04860
0.19812	0.05293	0.96379	0.00659	0.40563	-0.04885
0.20804	0.05385	0.97142	0.00493	0.39607	-0.04907
0.21797	0.05475	0.97764	0.00367	0.38652	-0.04926
0.22792	0.05561	0.98265	0.00273	0.37700	-0.04942
0.23786	0.05643	0.98670	0.00204	0.36749	-0.04955
0.24779	0.05722	0.98997	0.00153	0.35800	-0.04965
0.25770	0.05796	0.99261	0.00111	0.34853	-0.04972
0.26757	0.05865	0.99474	0.00075	0.33906	-0.04975
0.27742	0.05929	0.99646	0.00047	0.32961	-0.04976
0.28724	0.05987	0.99787	0.00030	0.32017	-0.04973
0.29704	0.06040	0.99901	0.00018	0.31072	-0.04967
0.30681	0.06088	1.00000	0.00001	0.30125	-0.04958
0.31656	0.06129	0.99901	-0.00016	0.29178	-0.04946
0.32628	0.06166	0.99787	-0.00028	0.28229	-0.04929

0.33598	0.06196
0.34565	0.06221
0.35532	0.06241
0.36498	0.06256
0.37463	0.06266
0.38429	0.06271
0.39395	0.06271
0.40361	0.06267
0.41329	0.06259
0.42298	0.06246
0.43269	0.06230
0.44242	0.06209
0.45216	0.06185
0.46194	0.06157
0.47175	0.06126
0.48160	0.06090
0.49148	0.06052
0.50138	0.06010
0.51131	0.05965
0.52126	0.05917
0.53125	0.05866
0.54127	0.05812
0.55132	0.05754
0.56138	0.05694
0.57144	0.05632
0.58149	0.05567
0.59150	0.05499
0.60149	0.05429
0.61151	0.05357
0.62154	0.05282
0.63156	0.05204
0.64156	0.05124
0.65150	0.05042
0.66139	0.04958
0.67126	0.04871
0.68109	0.04782
0.69090	0.04691

0.99647	-0.00046
0.99476	-0.00072
0.99263	-0.00106
0.99000	-0.00140
0.98674	-0.00178
0.98271	-0.00221
0.97770	-0.00268
0.97147	-0.00314
0.96379	-0.00354
0.95446	-0.00386
0.94414	-0.00408
0.93364	-0.00428
0.92306	-0.00453
0.91244	-0.00489
0.90179	-0.00539
0.89116	-0.00606
0.88062	-0.00687
0.87013	-0.00783
0.85966	-0.00891
0.84922	-0.01008
0.83888	-0.01133
0.82862	-0.01262
0.81843	-0.01393
0.80829	-0.01526
0.79819	-0.01659
0.78812	-0.01792
0.77805	-0.01922
0.76798	-0.02051
0.75794	-0.02178
0.74795	-0.02301
0.73801	-0.02421
0.72808	-0.02538
0.71813	-0.02653
0.70820	-0.02764
0.69830	-0.02873
0.68845	-0.02979
0.67863	-0.03082

0.27279	-0.04910
0.26329	-0.04887
0.25378	-0.04860
0.24427	-0.04829
0.23477	-0.04794
0.22528	-0.04756
0.21581	-0.04713
0.20636	-0.04666
0.19696	-0.04616
0.18758	-0.04561
0.17821	-0.04503
0.16881	-0.04441
0.15939	-0.04375
0.14990	-0.04305
0.14029	-0.04232
0.13053	-0.04154
0.12059	-0.04072
0.11048	-0.03984
0.10011	-0.03889
0.08942	-0.03783
0.07846	-0.03661
0.06726	-0.03517
0.05586	-0.03338
0.04451	-0.03111
0.03371	-0.02827
0.02460	-0.02504
0.01766	-0.02181
0.01252	-0.01874
0.00880	-0.01593
0.00608	-0.01338
0.00405	-0.01100
0.00251	-0.00871
0.00139	-0.00647
0.00064	-0.00430
0.00019	-0.00215
0.00004	0.00000

c. NACA 2412 Short

Table 8-8 - List of the optimised coordinates for NACA 2412 (constant length)

NACA 2412 Short

x	y	x	y	x	y
0.00000	0.00000	0.66928	-0.00750	0.67349	0.06871
0.00006	-0.00218	0.67830	-0.00649	0.66405	0.06949
0.00043	-0.00430	0.68736	-0.00545	0.65450	0.07024
0.00108	-0.00637	0.69647	-0.00437	0.64487	0.07098
0.00204	-0.00845	0.70565	-0.00327	0.63517	0.07169
0.00339	-0.01058	0.71489	-0.00213	0.62547	0.07239
0.00514	-0.01270	0.72420	-0.00095	0.61575	0.07306
0.00740	-0.01486	0.73358	0.00026	0.60598	0.07372
0.01039	-0.01716	0.74304	0.00150	0.59616	0.07436
0.01442	-0.01961	0.75259	0.00278	0.58632	0.07498
0.01990	-0.02219	0.76230	0.00407	0.57649	0.07557
0.02718	-0.02477	0.77222	0.00538	0.56667	0.07613
0.03640	-0.02714	0.78236	0.00668	0.55686	0.07666
0.04695	-0.02906	0.79275	0.00793	0.54703	0.07717
0.05792	-0.03047	0.80342	0.00912	0.53719	0.07764
0.06890	-0.03149	0.81440	0.01017	0.52734	0.07807
0.07973	-0.03225	0.82574	0.01106	0.51748	0.07847
0.09031	-0.03282	0.83747	0.01171	0.50759	0.07883
0.10062	-0.03328	0.84966	0.01207	0.49768	0.07916
0.11069	-0.03365	0.86236	0.01209	0.48773	0.07944
0.12057	-0.03397	0.87557	0.01173	0.47778	0.07969
0.13029	-0.03424	0.88924	0.01093	0.46784	0.07990
0.13990	-0.03446	0.90318	0.00969	0.45791	0.08007
0.14943	-0.03465	0.91712	0.00799	0.44801	0.08020
0.15891	-0.03481	0.93071	0.00590	0.43812	0.08030
0.16836	-0.03493	0.94359	0.00355	0.42824	0.08038
0.17783	-0.03501	0.95538	0.00123	0.41836	0.08043
0.18732	-0.03506	0.96524	-0.00062	0.40844	0.08044
0.19683	-0.03507	0.97294	-0.00181	0.39848	0.08042
0.20636	-0.03505	0.97904	-0.00245	0.38850	0.08035
0.21592	-0.03500	0.98392	-0.00274	0.37847	0.08025
0.22549	-0.03491	0.98786	-0.00282	0.36842	0.08012
0.23509	-0.03480	0.99105	-0.00280	0.35832	0.07996
0.24471	-0.03466	0.99364	-0.00269	0.34818	0.07976
0.25435	-0.03449	0.99574	-0.00253	0.33799	0.07954
0.26401	-0.03430	0.99745	-0.00239	0.32773	0.07929
0.27368	-0.03408	0.99885	-0.00230	0.31741	0.07902
0.28335	-0.03384	0.99998	-0.00225	0.30703	0.07871
0.29302	-0.03358	1.00000	-0.00191	0.29657	0.07837
0.30267	-0.03330	0.99886	-0.00173	0.28606	0.07799
0.31232	-0.03301	0.99745	-0.00145	0.27548	0.07758

0.32196	-0.03269
0.33161	-0.03236
0.34125	-0.03201
0.35089	-0.03164
0.36053	-0.03126
0.37016	-0.03087
0.37979	-0.03046
0.38942	-0.03003
0.39904	-0.02960
0.40865	-0.02913
0.41826	-0.02865
0.42786	-0.02814
0.43746	-0.02760
0.44706	-0.02703
0.45666	-0.02643
0.46626	-0.02581
0.47586	-0.02517
0.48543	-0.02450
0.49496	-0.02381
0.50446	-0.02310
0.51391	-0.02237
0.52329	-0.02162
0.53262	-0.02085
0.54192	-0.02007
0.55120	-0.01927
0.56047	-0.01844
0.56972	-0.01760
0.57894	-0.01675
0.58811	-0.01588
0.59725	-0.01500
0.60633	-0.01411
0.61537	-0.01321
0.62437	-0.01229
0.63336	-0.01137
0.64234	-0.01042
0.65132	-0.00947
0.66029	-0.00849

0.99573	-0.00105
0.99359	-0.00051
0.99093	0.00015
0.98761	0.00103
0.98348	0.00225
0.97827	0.00400
0.97161	0.00642
0.96308	0.00970
0.95218	0.01382
0.93944	0.01829
0.92602	0.02256
0.91249	0.02650
0.89935	0.03012
0.88691	0.03341
0.87536	0.03639
0.86467	0.03907
0.85468	0.04150
0.84524	0.04371
0.83621	0.04573
0.82749	0.04758
0.81904	0.04929
0.81077	0.05089
0.80258	0.05240
0.79448	0.05383
0.78643	0.05519
0.77838	0.05649
0.77028	0.05775
0.76209	0.05896
0.75377	0.06013
0.74532	0.06125
0.73679	0.06233
0.72812	0.06336
0.71928	0.06435
0.71031	0.06530
0.70124	0.06621
0.69209	0.06707
0.68284	0.06791

0.26484	0.07713
0.25409	0.07662
0.24324	0.07605
0.23231	0.07541
0.22130	0.07469
0.21027	0.07387
0.19926	0.07295
0.18829	0.07191
0.17738	0.07073
0.16652	0.06942
0.15574	0.06794
0.14503	0.06629
0.13445	0.06446
0.12409	0.06245
0.11393	0.06025
0.10394	0.05785
0.09419	0.05525
0.08466	0.05243
0.07530	0.04938
0.06606	0.04608
0.05685	0.04251
0.04764	0.03865
0.03841	0.03447
0.02940	0.02998
0.02153	0.02555
0.01546	0.02157
0.01088	0.01801
0.00740	0.01476
0.00485	0.01183
0.00301	0.00922
0.00172	0.00682
0.00081	0.00449
0.00025	0.00222
0.00000	0.00000

d. NACA 2412 Long

Table 8-9 - List of the optimised coordinates for NACA 2412 (variable length)

NACA 2412 Long

x	y
0.00000	0.00000
0.00016	-0.00122
0.00067	-0.00239
0.00150	-0.00352
0.00263	-0.00461
0.00406	-0.00567
0.00577	-0.00670
0.00775	-0.00769
0.01006	-0.00869
0.01281	-0.00971
0.01595	-0.01071
0.01963	-0.01174
0.02411	-0.01284
0.02968	-0.01402
0.03667	-0.01530
0.04527	-0.01665
0.05539	-0.01799
0.06636	-0.01923
0.07738	-0.02028
0.08824	-0.02116
0.09893	-0.02191
0.10943	-0.02254
0.11973	-0.02306
0.12987	-0.02349
0.13988	-0.02385
0.14976	-0.02414
0.15954	-0.02436
0.16921	-0.02453
0.17878	-0.02466
0.18827	-0.02473
0.19768	-0.02477
0.20701	-0.02478
0.21627	-0.02475
0.22545	-0.02469
0.23454	-0.02461
0.24355	-0.02451
0.25249	-0.02438
0.26134	-0.02425
0.27013	-0.02409
0.27884	-0.02393
0.28749	-0.02376
0.60916	-0.02118
0.61852	-0.02134
0.62797	-0.02151
0.63754	-0.02169
0.64722	-0.02190
0.65703	-0.02212
0.66698	-0.02236
0.67707	-0.02262
0.68731	-0.02289
0.69768	-0.02318
0.70818	-0.02348
0.71879	-0.02380
0.72949	-0.02413
0.74030	-0.02448
0.75125	-0.02483
0.76236	-0.02520
0.77362	-0.02559
0.78502	-0.02598
0.79655	-0.02637
0.80818	-0.02678
0.81991	-0.02718
0.83172	-0.02759
0.84364	-0.02799
0.85566	-0.02839
0.86780	-0.02879
0.88005	-0.02918
0.89240	-0.02956
0.90487	-0.02993
0.91745	-0.03028
0.93014	-0.03061
0.94269	-0.03092
0.95401	-0.03117
0.96334	-0.03136
0.97100	-0.03150
0.97725	-0.03160
0.98234	-0.03170
0.98648	-0.03178
0.98987	-0.03182
0.99107	-0.02955
0.99348	-0.03027
0.99264	-0.03182
0.73457	0.03766
0.72760	0.03934
0.72064	0.04097
0.71366	0.04256
0.70663	0.04412
0.69951	0.04563
0.69227	0.04710
0.68489	0.04854
0.67734	0.04995
0.66963	0.05131
0.66176	0.05262
0.65373	0.05389
0.64549	0.05511
0.63701	0.05629
0.62832	0.05742
0.61942	0.05851
0.61033	0.05953
0.60101	0.06051
0.59146	0.06143
0.58167	0.06230
0.57164	0.06311
0.56138	0.06387
0.55088	0.06457
0.54011	0.06521
0.52910	0.06579
0.51787	0.06631
0.50645	0.06676
0.49485	0.06715
0.48310	0.06747
0.47120	0.06772
0.45916	0.06790
0.44696	0.06802
0.43459	0.06805
0.42204	0.06800
0.40932	0.06785
0.39646	0.06760
0.38348	0.06726
0.37040	0.06682
0.35722	0.06629
0.34396	0.06565
0.33065	0.06493

0.29607	-0.02359
0.30458	-0.02341
0.31303	-0.02322
0.32141	-0.02304
0.32974	-0.02286
0.33804	-0.02268
0.34630	-0.02251
0.35454	-0.02234
0.36275	-0.02217
0.37094	-0.02202
0.37913	-0.02188
0.38731	-0.02175
0.39548	-0.02162
0.40366	-0.02150
0.41183	-0.02139
0.42000	-0.02128
0.42819	-0.02117
0.43639	-0.02107
0.44463	-0.02097
0.45290	-0.02088
0.46120	-0.02080
0.46953	-0.02072
0.47789	-0.02066
0.48627	-0.02060
0.49467	-0.02056
0.50309	-0.02052
0.51154	-0.02050
0.52004	-0.02049
0.52862	-0.02050
0.53728	-0.02051
0.54602	-0.02054
0.55484	-0.02059
0.56372	-0.02065
0.57267	-0.02072
0.58168	-0.02081
0.59075	-0.02092
0.59991	-0.02104
0.99489	-0.03182
0.99544	-0.03085
0.99670	-0.03187
0.99706	-0.03127
0.99817	-0.03194
0.99838	-0.03159
0.99949	-0.03193
0.98804	-0.02869
0.98426	-0.02767
0.97960	-0.02638
0.97385	-0.02478
0.96674	-0.02280
0.95808	-0.02042
0.94783	-0.01762
0.93679	-0.01462
0.92579	-0.01167
0.91491	-0.00878
0.90413	-0.00595
0.89348	-0.00317
0.88305	-0.00047
0.87289	0.00215
0.86297	0.00470
0.85328	0.00718
0.84382	0.00960
0.83461	0.01195
0.82570	0.01424
0.81705	0.01646
0.80863	0.01864
0.80046	0.02075
0.79255	0.02281
0.78486	0.02481
0.77735	0.02676
0.76998	0.02868
0.76272	0.03056
0.75557	0.03240
0.74853	0.03419
0.74155	0.03594
0.31729	0.06410
0.30392	0.06318
0.29054	0.06217
0.27720	0.06107
0.26393	0.05988
0.25071	0.05860
0.23755	0.05723
0.22446	0.05578
0.21146	0.05424
0.19859	0.05263
0.18592	0.05094
0.17348	0.04919
0.16131	0.04738
0.14941	0.04551
0.13778	0.04359
0.12643	0.04161
0.11537	0.03957
0.10462	0.03749
0.09429	0.03537
0.08433	0.03322
0.07472	0.03101
0.06554	0.02877
0.05676	0.02650
0.04837	0.02417
0.04038	0.02179
0.03278	0.01934
0.02563	0.01681
0.01893	0.01417
0.01291	0.01145
0.00807	0.00885
0.00467	0.00659
0.00241	0.00464
0.00098	0.00289
0.00023	0.00135
0.00000	0.00000

e. BACXXX Short

Table 8-10 - List of the optimised coordinates for BACXXX (constant length)

BACXXX Short

x	y	x	y	x	y
0.00000	0.00000	0.68018	-0.01614	0.73954	0.04536
0.00020	-0.00099	0.69118	-0.01576	0.72861	0.04530
0.00055	-0.00194	0.70229	-0.01537	0.71764	0.04529
0.00106	-0.00281	0.71351	-0.01494	0.70662	0.04533
0.00172	-0.00360	0.72487	-0.01446	0.69553	0.04541
0.00249	-0.00431	0.73639	-0.01390	0.68437	0.04551
0.00342	-0.00495	0.74806	-0.01326	0.67313	0.04562
0.00453	-0.00557	0.75990	-0.01250	0.66178	0.04574
0.00597	-0.00627	0.77190	-0.01160	0.65032	0.04584
0.00787	-0.00713	0.78406	-0.01055	0.63875	0.04593
0.01031	-0.00812	0.79634	-0.00934	0.62711	0.04599
0.01349	-0.00923	0.80867	-0.00797	0.61541	0.04603
0.01774	-0.01044	0.82095	-0.00644	0.60366	0.04604
0.02334	-0.01178	0.83308	-0.00479	0.59189	0.04604
0.03054	-0.01327	0.84492	-0.00305	0.58013	0.04603
0.03971	-0.01475	0.85640	-0.00126	0.56838	0.04600
0.05086	-0.01615	0.86748	0.00050	0.55663	0.04595
0.06327	-0.01734	0.87803	0.00213	0.54487	0.04589
0.07575	-0.01824	0.88798	0.00356	0.53312	0.04581
0.08775	-0.01889	0.89729	0.00470	0.52141	0.04572
0.09919	-0.01936	0.90599	0.00550	0.50976	0.04560
0.11012	-0.01971	0.91415	0.00592	0.49820	0.04547
0.12063	-0.01995	0.92188	0.00598	0.48672	0.04532
0.13082	-0.02012	0.92930	0.00566	0.47536	0.04515
0.14072	-0.02026	0.93652	0.00489	0.46410	0.04498
0.15037	-0.02040	0.94372	0.00370	0.45297	0.04479
0.15979	-0.02055	0.95105	0.00219	0.44196	0.04458
0.16908	-0.02072	0.95865	0.00056	0.43108	0.04436
0.17834	-0.02089	0.96601	-0.00082	0.42033	0.04413
0.18755	-0.02106	0.97247	-0.00173	0.40970	0.04387
0.19670	-0.02124	0.97803	-0.00221	0.39920	0.04360
0.20582	-0.02142	0.98270	-0.00240	0.38882	0.04333
0.21494	-0.02160	0.98659	-0.00238	0.37856	0.04305
0.22408	-0.02179	0.98981	-0.00225	0.36840	0.04279
0.23323	-0.02196	0.99246	-0.00205	0.35834	0.04254
0.24241	-0.02214	0.99463	-0.00181	0.34836	0.04232
0.25163	-0.02230	0.99638	-0.00155	0.33846	0.04212
0.26089	-0.02246	0.99775	-0.00129	0.32863	0.04193
0.27020	-0.02260	0.99886	-0.00100	0.31887	0.04176
0.27954	-0.02273	0.99985	-0.00063	0.30917	0.04159
0.28892	-0.02284	1.00000	0.00082	0.29952	0.04143

0.29831	-0.02293
0.30772	-0.02301
0.31714	-0.02306
0.32656	-0.02309
0.33598	-0.02311
0.34540	-0.02311
0.35480	-0.02309
0.36422	-0.02306
0.37365	-0.02301
0.38310	-0.02295
0.39257	-0.02288
0.40205	-0.02279
0.41156	-0.02270
0.42108	-0.02258
0.43063	-0.02245
0.44019	-0.02230
0.44979	-0.02214
0.45941	-0.02196
0.46905	-0.02176
0.47871	-0.02155
0.48837	-0.02133
0.49803	-0.02110
0.50769	-0.02087
0.51737	-0.02063
0.52708	-0.02038
0.53682	-0.02013
0.54660	-0.01987
0.55642	-0.01961
0.56629	-0.01934
0.57623	-0.01907
0.58625	-0.01880
0.59636	-0.01853
0.60655	-0.01826
0.61680	-0.01800
0.62710	-0.01772
0.63746	-0.01744
0.64793	-0.01714
0.65855	-0.01683
0.66930	-0.01649

0.99909	0.00131
0.99809	0.00172
0.99682	0.00217
0.99515	0.00269
0.99308	0.00329
0.99053	0.00400
0.98742	0.00489
0.98364	0.00602
0.97914	0.00753
0.97385	0.00959
0.96787	0.01237
0.96139	0.01598
0.95501	0.02016
0.94913	0.02444
0.94355	0.02863
0.93808	0.03257
0.93256	0.03611
0.92688	0.03916
0.92097	0.04171
0.91478	0.04381
0.90824	0.04551
0.90136	0.04683
0.89415	0.04782
0.88657	0.04852
0.87861	0.04897
0.87031	0.04918
0.86170	0.04920
0.85276	0.04906
0.84353	0.04881
0.83400	0.04846
0.82416	0.04805
0.81409	0.04761
0.80382	0.04716
0.79342	0.04673
0.78286	0.04633
0.77215	0.04598
0.76133	0.04570
0.75046	0.04549

0.28992	0.04128
0.28039	0.04113
0.27090	0.04098
0.26148	0.04084
0.25211	0.04069
0.24280	0.04053
0.23354	0.04033
0.22436	0.04008
0.21526	0.03978
0.20623	0.03941
0.19729	0.03898
0.18840	0.03849
0.17950	0.03796
0.17052	0.03739
0.16142	0.03678
0.15220	0.03613
0.14276	0.03544
0.13294	0.03477
0.12255	0.03421
0.11143	0.03378
0.09958	0.03335
0.08699	0.03279
0.07369	0.03207
0.05992	0.03115
0.04624	0.02972
0.03387	0.02736
0.02394	0.02437
0.01687	0.02105
0.01190	0.01776
0.00830	0.01470
0.00569	0.01200
0.00376	0.00966
0.00237	0.00763
0.00141	0.00588
0.00076	0.00441
0.00034	0.00315
0.00008	0.00204
0.00000	0.00000

f. BACXXX Long

Table 8-11 - List of the optimised coordinates for BACXXX (variable length)

		BACXXX Long			
x	y	x	y	x	y
0.00000	0.00000	0.83430	0.04291	0.62362	-0.00380
0.00001	0.00042	0.84426	0.04271	0.61302	-0.00445
0.00020	0.00086	0.85402	0.04248	0.60256	-0.00508
0.00060	0.00138	0.86355	0.04221	0.59223	-0.00569
0.00129	0.00197	0.87286	0.04191	0.58197	-0.00630
0.00240	0.00267	0.88197	0.04157	0.57177	-0.00689
0.00409	0.00348	0.89086	0.04119	0.56166	-0.00748
0.00654	0.00443	0.89946	0.04077	0.55164	-0.00806
0.00997	0.00554	0.90776	0.04031	0.54172	-0.00863
0.01475	0.00681	0.91575	0.03982	0.53192	-0.00919
0.02122	0.00825	0.92344	0.03929	0.52223	-0.00973
0.02990	0.00983	0.93083	0.03872	0.51263	-0.01027
0.04114	0.01154	0.93793	0.03812	0.50312	-0.01078
0.05425	0.01332	0.94469	0.03747	0.49370	-0.01128
0.06804	0.01495	0.95109	0.03679	0.48436	-0.01177
0.08197	0.01649	0.95712	0.03608	0.47510	-0.01223
0.09592	0.01797	0.96280	0.03532	0.46590	-0.01268
0.10965	0.01933	0.96808	0.03451	0.45676	-0.01311
0.12304	0.02053	0.97289	0.03364	0.44766	-0.01352
0.13616	0.02164	0.97722	0.03271	0.43862	-0.01391
0.14906	0.02271	0.98100	0.03169	0.42964	-0.01427
0.16176	0.02377	0.98423	0.03059	0.42073	-0.01461
0.17424	0.02478	0.98705	0.02945	0.41189	-0.01491
0.18651	0.02571	0.98960	0.02831	0.40312	-0.01518
0.19863	0.02658	0.99201	0.02720	0.39441	-0.01542
0.21058	0.02740	0.99433	0.02613	0.38575	-0.01563
0.22235	0.02818	0.99633	0.02513	0.37713	-0.01581
0.23399	0.02894	0.99781	0.02427	0.36855	-0.01596
0.24556	0.02966	0.99880	0.02353	0.36001	-0.01608
0.25706	0.03035	0.99940	0.02289	0.35149	-0.01617
0.26849	0.03100	0.99970	0.02234	0.34301	-0.01624
0.27987	0.03161	0.99976	0.02186	0.33454	-0.01627
0.29119	0.03219	0.99964	0.02144	0.32607	-0.01628
0.30246	0.03274	0.99936	0.02106	0.31759	-0.01625
0.31369	0.03327	0.99897	0.02072	0.30910	-0.01620
0.32490	0.03379	0.99848	0.02042	0.30058	-0.01612
0.33609	0.03429	0.99788	0.02014	0.29205	-0.01601
0.34729	0.03478	0.99696	0.01981	0.28349	-0.01588
0.35848	0.03525	0.99489	0.01927	0.27490	-0.01571
0.36967	0.03570	0.99228	0.01881	0.26627	-0.01553
0.38089	0.03613	0.99043	0.01856	0.25762	-0.01531

0.39212	0.03655
0.40338	0.03695
0.41467	0.03734
0.42598	0.03773
0.43732	0.03811
0.44868	0.03849
0.46009	0.03887
0.47153	0.03924
0.48302	0.03959
0.49457	0.03992
0.50616	0.04024
0.51780	0.04054
0.52947	0.04083
0.54118	0.04110
0.55290	0.04136
0.56464	0.04162
0.57640	0.04186
0.58816	0.04209
0.59992	0.04231
0.61167	0.04251
0.62339	0.04270
0.63507	0.04287
0.64670	0.04303
0.65828	0.04318
0.66984	0.04332
0.68138	0.04345
0.69289	0.04356
0.70435	0.04365
0.71577	0.04372
0.72712	0.04376
0.73839	0.04378
0.74955	0.04377
0.76058	0.04374
0.77150	0.04368
0.78230	0.04361
0.79297	0.04351
0.80351	0.04339
0.81391	0.04325
0.82418	0.04309
0.98876	0.01837
0.98693	0.01818
0.98481	0.01799
0.98238	0.01780
0.97955	0.01760
0.97625	0.01740
0.97235	0.01718
0.96769	0.01696
0.96208	0.01671
0.95528	0.01642
0.94702	0.01607
0.93760	0.01562
0.92749	0.01507
0.91674	0.01441
0.90547	0.01367
0.89396	0.01289
0.88251	0.01214
0.87120	0.01143
0.85994	0.01074
0.84867	0.01007
0.83736	0.00940
0.82599	0.00872
0.81456	0.00803
0.80310	0.00733
0.79160	0.00662
0.78000	0.00590
0.76836	0.00516
0.75672	0.00443
0.74513	0.00369
0.73364	0.00296
0.72227	0.00225
0.71102	0.00155
0.69988	0.00087
0.68884	0.00020
0.67787	-0.00047
0.66695	-0.00113
0.65605	-0.00180
0.64517	-0.00247
0.63435	-0.00314
0.24893	-0.01508
0.24020	-0.01483
0.23143	-0.01456
0.22260	-0.01428
0.21369	-0.01398
0.20470	-0.01366
0.19563	-0.01333
0.18645	-0.01299
0.17714	-0.01263
0.16771	-0.01226
0.15818	-0.01187
0.14860	-0.01148
0.13892	-0.01107
0.12910	-0.01065
0.11920	-0.01024
0.10923	-0.00980
0.09922	-0.00934
0.08914	-0.00886
0.07898	-0.00836
0.06874	-0.00784
0.05842	-0.00729
0.04830	-0.00671
0.03912	-0.00613
0.03131	-0.00557
0.02483	-0.00501
0.01957	-0.00452
0.01538	-0.00407
0.01210	-0.00364
0.00949	-0.00326
0.00743	-0.00293
0.00583	-0.00265
0.00458	-0.00240
0.00350	-0.00215
0.00256	-0.00186
0.00174	-0.00155
0.00106	-0.00120
0.00052	-0.00081
0.00016	-0.00041
0.00000	0.00000

g. NASA SC(2)-0714 Short

Table 8-12 - List of the optimised coordinates for NASA SC(2)-0714 (constant length)

NASA SC(2)-0714 Short

x	y	x	y	x	y
0.00000	0.00000	0.71439	0.03964	0.67539	-0.03448
0.00088	0.00409	0.72383	0.03845	0.66593	-0.03648
0.00229	0.00798	0.73332	0.03727	0.65654	-0.03846
0.00421	0.01162	0.74289	0.03613	0.64719	-0.04049
0.00667	0.01500	0.75245	0.03495	0.63786	-0.04262
0.00970	0.01822	0.76205	0.03377	0.62852	-0.04472
0.01347	0.02139	0.77176	0.03262	0.61923	-0.04687
0.01820	0.02463	0.78151	0.03144	0.60991	-0.04898
0.02416	0.02806	0.79130	0.03029	0.60058	-0.05113
0.03166	0.03162	0.80110	0.02906	0.59125	-0.05322
0.04027	0.03496	0.81094	0.02779	0.58190	-0.05524
0.04934	0.03805	0.82086	0.02651	0.57251	-0.05716
0.05862	0.04077	0.83086	0.02512	0.56302	-0.05900
0.06796	0.04322	0.84086	0.02365	0.55343	-0.06074
0.07737	0.04546	0.85089	0.02215	0.54374	-0.06236
0.08691	0.04739	0.86096	0.02051	0.53401	-0.06386
0.09650	0.04918	0.87101	0.01876	0.52423	-0.06524
0.10615	0.05082	0.88104	0.01697	0.51436	-0.06648
0.11587	0.05223	0.89104	0.01503	0.50439	-0.06758
0.12566	0.05349	0.90104	0.01295	0.49436	-0.06856
0.13552	0.05462	0.91098	0.01075	0.48428	-0.06939
0.14544	0.05572	0.92085	0.00852	0.47418	-0.07008
0.15540	0.05672	0.93066	0.00617	0.46406	-0.07063
0.16537	0.05760	0.94040	0.00370	0.45394	-0.07114
0.17531	0.05836	0.95011	0.00112	0.44380	-0.07152
0.18521	0.05910	0.95956	-0.00152	0.43365	-0.07188
0.19515	0.05977	0.96816	-0.00391	0.42348	-0.07211
0.20509	0.06041	0.97523	-0.00602	0.41331	-0.07232
0.21503	0.06101	0.98088	-0.00763	0.40314	-0.07241
0.22497	0.06150	0.98552	-0.00887	0.39297	-0.07248
0.23489	0.06199	0.98918	-0.01012	0.38281	-0.07244
0.24481	0.06243	0.99197	-0.01129	0.37265	-0.07240
0.25473	0.06283	0.99413	-0.01233	0.36251	-0.07225
0.26464	0.06328	0.99584	-0.01324	0.35238	-0.07210
0.27459	0.06367	0.99720	-0.01404	0.34226	-0.07185
0.28458	0.06404	0.99828	-0.01475	0.33214	-0.07161
0.29462	0.06439	0.99914	-0.01541	0.32204	-0.07127
0.30472	0.06468	0.99987	-0.01613	0.31195	-0.07095
0.31486	0.06500	0.99955	-0.01772	0.30187	-0.07053
0.32505	0.06530	0.99854	-0.01799	0.29180	-0.07013
0.33529	0.06554	0.99744	-0.01813	0.28175	-0.06964

0.34557	0.06577
0.35588	0.06591
0.36623	0.06608
0.37660	0.06622
0.38699	0.06626
0.39739	0.06632
0.40781	0.06633
0.41822	0.06624
0.42864	0.06615
0.43904	0.06603
0.44944	0.06586
0.45983	0.06557
0.47019	0.06526
0.48053	0.06488
0.49083	0.06439
0.50109	0.06387
0.51132	0.06327
0.52149	0.06255
0.53160	0.06182
0.54167	0.06101
0.55170	0.06007
0.56169	0.05913
0.57164	0.05812
0.58152	0.05700
0.59133	0.05590
0.60108	0.05474
0.61073	0.05352
0.62030	0.05228
0.62979	0.05098
0.63929	0.04973
0.64877	0.04845
0.65822	0.04716
0.66764	0.04588
0.67700	0.04459
0.68635	0.04333
0.69567	0.04208
0.70502	0.04086

0.99615	-0.01820
0.99459	-0.01819
0.99265	-0.01809
0.99028	-0.01784
0.98742	-0.01733
0.98390	-0.01639
0.97946	-0.01506
0.97392	-0.01377
0.96706	-0.01242
0.95863	-0.01087
0.94903	-0.00942
0.93900	-0.00820
0.92889	-0.00729
0.91869	-0.00669
0.90842	-0.00638
0.89811	-0.00626
0.88777	-0.00634
0.87738	-0.00669
0.86695	-0.00720
0.85649	-0.00788
0.84603	-0.00870
0.83559	-0.00967
0.82520	-0.01077
0.81487	-0.01197
0.80457	-0.01321
0.79429	-0.01450
0.78402	-0.01592
0.77378	-0.01738
0.76364	-0.01890
0.75356	-0.02048
0.74351	-0.02206
0.73358	-0.02373
0.72375	-0.02540
0.71396	-0.02711
0.70421	-0.02887
0.69453	-0.03068
0.68494	-0.03255

0.27171	-0.06917
0.26167	-0.06862
0.25163	-0.06808
0.24156	-0.06745
0.23148	-0.06685
0.22139	-0.06616
0.21129	-0.06548
0.20121	-0.06473
0.19114	-0.06399
0.18112	-0.06318
0.17108	-0.06237
0.16099	-0.06149
0.15089	-0.06051
0.14081	-0.05946
0.13075	-0.05842
0.12072	-0.05730
0.11070	-0.05600
0.10067	-0.05462
0.09061	-0.05314
0.08054	-0.05156
0.07057	-0.04971
0.06064	-0.04766
0.05076	-0.04534
0.04117	-0.04260
0.03205	-0.03950
0.02421	-0.03614
0.01806	-0.03281
0.01326	-0.02958
0.00956	-0.02647
0.00660	-0.02333
0.00418	-0.01998
0.00227	-0.01635
0.00087	-0.01248
0.00000	-0.00841
0.00000	0.00000

h. NASA SC(2)-0714 Long

Table 8-13 - List of the optimised coordinates for NASA SC(2)-0714 (variable length)

NASA SC(2)-0714 Long					
x	y	x	y	x	y
0.00000	0.00000	0.71155	-0.01160	0.74804	0.03501
0.00023	-0.00204	0.72222	-0.01031	0.73844	0.03541
0.00150	-0.00413	0.73289	-0.00902	0.72883	0.03576
0.00377	-0.00620	0.74357	-0.00773	0.71920	0.03610
0.00691	-0.00819	0.75419	-0.00645	0.70953	0.03639
0.01077	-0.01008	0.76476	-0.00517	0.69982	0.03666
0.01521	-0.01184	0.77533	-0.00388	0.69005	0.03690
0.02012	-0.01346	0.78586	-0.00264	0.68025	0.03711
0.02552	-0.01498	0.79630	-0.00140	0.67043	0.03729
0.03172	-0.01649	0.80665	-0.00021	0.66061	0.03744
0.03913	-0.01804	0.81693	0.00093	0.65079	0.03760
0.04799	-0.01964	0.82707	0.00202	0.64095	0.03772
0.05851	-0.02127	0.83703	0.00300	0.63105	0.03782
0.06997	-0.02277	0.84684	0.00392	0.62110	0.03789
0.08153	-0.02412	0.85654	0.00479	0.61113	0.03792
0.09291	-0.02526	0.86613	0.00555	0.60111	0.03798
0.10404	-0.02628	0.87560	0.00620	0.59109	0.03798
0.11498	-0.02720	0.88493	0.00672	0.58106	0.03797
0.12572	-0.02798	0.89413	0.00713	0.57102	0.03798
0.13632	-0.02872	0.90317	0.00740	0.56098	0.03793
0.14677	-0.02942	0.91207	0.00755	0.55094	0.03787
0.15708	-0.03007	0.92079	0.00750	0.54085	0.03782
0.16724	-0.03062	0.92939	0.00731	0.53075	0.03773
0.17731	-0.03112	0.93785	0.00698	0.52062	0.03762
0.18738	-0.03163	0.94613	0.00644	0.51045	0.03753
0.19738	-0.03210	0.95420	0.00569	0.50026	0.03739
0.20727	-0.03251	0.96204	0.00473	0.49006	0.03723
0.21703	-0.03288	0.96936	0.00359	0.47981	0.03709
0.22675	-0.03325	0.97561	0.00239	0.46955	0.03691
0.23642	-0.03357	0.98059	0.00135	0.45925	0.03671
0.24609	-0.03390	0.98447	0.00037	0.44893	0.03652
0.25570	-0.03418	0.98739	-0.00063	0.43858	0.03635
0.26527	-0.03447	0.98973	-0.00134	0.42820	0.03614
0.27477	-0.03470	0.99174	-0.00174	0.41781	0.03590
0.28423	-0.03494	0.99348	-0.00195	0.40739	0.03570
0.29360	-0.03512	0.99494	-0.00206	0.39694	0.03544
0.30294	-0.03531	0.99615	-0.00208	0.38649	0.03517
0.31224	-0.03545	0.99718	-0.00205	0.37604	0.03493
0.32153	-0.03559	0.99811	-0.00197	0.36557	0.03464
0.33078	-0.03568	0.99902	-0.00179	0.35511	0.03435
0.34003	-0.03577	0.99988	-0.00132	0.34465	0.03402

0.34925	-0.03580	1.00000	-0.00067	0.33421	0.03367
0.35847	-0.03584	0.99975	-0.00016	0.32379	0.03336
0.36768	-0.03582	0.99935	0.00032	0.31338	0.03301
0.37689	-0.03580	0.99881	0.00083	0.30299	0.03264
0.38609	-0.03573	0.99805	0.00141	0.29261	0.03226
0.39531	-0.03566	0.99705	0.00206	0.28223	0.03185
0.40454	-0.03552	0.99573	0.00281	0.27184	0.03147
0.41378	-0.03540	0.99392	0.00365	0.26142	0.03107
0.42305	-0.03521	0.99142	0.00455	0.25096	0.03064
0.43234	-0.03502	0.98809	0.00544	0.24047	0.03021
0.44166	-0.03477	0.98400	0.00659	0.22992	0.02972
0.45100	-0.03453	0.97877	0.00808	0.21933	0.02920
0.46038	-0.03422	0.97223	0.00976	0.20874	0.02869
0.46978	-0.03392	0.96485	0.01157	0.19811	0.02814
0.47924	-0.03355	0.95709	0.01332	0.18744	0.02760
0.48874	-0.03319	0.94913	0.01496	0.17665	0.02700
0.49831	-0.03276	0.94098	0.01651	0.16578	0.02636
0.50795	-0.03226	0.93265	0.01796	0.15488	0.02568
0.51764	-0.03169	0.92415	0.01938	0.14399	0.02501
0.52738	-0.03106	0.91549	0.02072	0.13309	0.02430
0.53714	-0.03037	0.90675	0.02197	0.12214	0.02354
0.54694	-0.02962	0.89790	0.02314	0.11110	0.02268
0.55681	-0.02881	0.88892	0.02427	0.10002	0.02178
0.56677	-0.02793	0.87985	0.02535	0.08891	0.02084
0.57676	-0.02700	0.87073	0.02633	0.07778	0.01974
0.58677	-0.02602	0.86154	0.02729	0.06665	0.01856
0.59682	-0.02499	0.85226	0.02820	0.05553	0.01724
0.60694	-0.02391	0.84298	0.02902	0.04461	0.01573
0.61715	-0.02276	0.83368	0.02982	0.03426	0.01406
0.62746	-0.02158	0.82432	0.03057	0.02523	0.01226
0.63781	-0.02040	0.81491	0.03125	0.01812	0.01050
0.64821	-0.01918	0.80542	0.03191	0.01256	0.00883
0.65867	-0.01796	0.79591	0.03253	0.00824	0.00719
0.66919	-0.01669	0.78639	0.03311	0.00490	0.00554
0.67970	-0.01545	0.77684	0.03366	0.00240	0.00381
0.69025	-0.01420	0.76725	0.03414	0.00075	0.00196
0.70088	-0.01290	0.75764	0.03460	0.00000	0.00000

i. NASA SC(2)-0714 Faster Short

Table 8-14 - List of the optimised coordinates for NASA SC(2)-0714 at $M = 1.8$ (constant length)

NASA SC(2)-0714 Faster Short

x	y	x	y	x	y
0.00000	0.00000	0.68245	-0.02833	0.65568	0.03064
0.00006	-0.00839	0.68864	-0.02535	0.64735	0.03120
0.00088	-0.01247	0.69496	-0.02256	0.63895	0.03176
0.00228	-0.01634	0.70143	-0.01996	0.63045	0.03228
0.00421	-0.01996	0.70812	-0.01751	0.62187	0.03281
0.00671	-0.02332	0.71507	-0.01528	0.61323	0.03330
0.00993	-0.02645	0.72222	-0.01320	0.60455	0.03376
0.01427	-0.02955	0.72961	-0.01133	0.59585	0.03428
0.02057	-0.03273	0.73731	-0.00965	0.58708	0.03473
0.03006	-0.03599	0.74526	-0.00815	0.57823	0.03519
0.04476	-0.03931	0.75351	-0.00691	0.56930	0.03560
0.06516	-0.04252	0.76206	-0.00585	0.56031	0.03599
0.08832	-0.04571	0.77097	-0.00497	0.55128	0.03643
0.11104	-0.04901	0.78032	-0.00432	0.54221	0.03680
0.13087	-0.05274	0.79018	-0.00392	0.53312	0.03715
0.14734	-0.05702	0.80058	-0.00378	0.52402	0.03753
0.16079	-0.06166	0.81160	-0.00392	0.51490	0.03785
0.17157	-0.06671	0.82334	-0.00433	0.50576	0.03815
0.18013	-0.07198	0.83589	-0.00499	0.49660	0.03847
0.18695	-0.07727	0.84938	-0.00595	0.48740	0.03873
0.19254	-0.08229	0.86395	-0.00710	0.47819	0.03896
0.19729	-0.08702	0.87966	-0.00829	0.46896	0.03923
0.20146	-0.09149	0.89640	-0.00945	0.45970	0.03942
0.20546	-0.09547	0.91385	-0.01046	0.45043	0.03958
0.20958	-0.09891	0.93140	-0.01131	0.44115	0.03977
0.21398	-0.10177	0.94764	-0.01214	0.43185	0.03988
0.21869	-0.10421	0.96084	-0.01314	0.42254	0.03997
0.22394	-0.10613	0.97056	-0.01413	0.41321	0.04008
0.22965	-0.10767	0.97771	-0.01523	0.40387	0.04021
0.23592	-0.10873	0.98302	-0.01647	0.39452	0.04026
0.24266	-0.10946	0.98700	-0.01736	0.38515	0.04030
0.24991	-0.10978	0.99009	-0.01784	0.37578	0.04037
0.25755	-0.10984	0.99259	-0.01808	0.36639	0.04037
0.26564	-0.10956	0.99458	-0.01818	0.35700	0.04035
0.27407	-0.10910	0.99617	-0.01819	0.34760	0.04038
0.28289	-0.10838	0.99746	-0.01812	0.33819	0.04035
0.29201	-0.10757	0.99856	-0.01798	0.32879	0.04031
0.30148	-0.10658	0.99955	-0.01771	0.31940	0.04024
0.31121	-0.10558	0.99987	-0.01613	0.31001	0.04015
0.32123	-0.10449	0.99915	-0.01541	0.30062	0.04011
0.33150	-0.10346	0.99829	-0.01474	0.29123	0.04003

0.34201	-0.10240
0.35273	-0.10146
0.36366	-0.10053
0.37476	-0.09975
0.38602	-0.09900
0.39741	-0.09838
0.40891	-0.09780
0.42047	-0.09733
0.43207	-0.09686
0.44368	-0.09647
0.45527	-0.09604
0.46680	-0.09565
0.47825	-0.09516
0.48962	-0.09466
0.50088	-0.09401
0.51199	-0.09320
0.52291	-0.09220
0.53359	-0.09100
0.54398	-0.08957
0.55406	-0.08791
0.56385	-0.08598
0.57332	-0.08378
0.58240	-0.08131
0.59105	-0.07858
0.59927	-0.07559
0.60710	-0.07234
0.61454	-0.06884
0.62162	-0.06513
0.62839	-0.06136
0.63483	-0.05745
0.64107	-0.05355
0.64708	-0.04960
0.65298	-0.04577
0.65882	-0.04204
0.66462	-0.03835
0.67047	-0.03483
0.67640	-0.03149

0.99722	-0.01403
0.99585	-0.01323
0.99411	-0.01232
0.99187	-0.01129
0.98890	-0.01013
0.98484	-0.00893
0.97937	-0.00779
0.97214	-0.00638
0.96214	-0.00468
0.94857	-0.00299
0.93237	-0.00126
0.91518	0.00050
0.89821	0.00240
0.88207	0.00444
0.86703	0.00651
0.85314	0.00861
0.84032	0.01058
0.82851	0.01237
0.81747	0.01399
0.80711	0.01550
0.79727	0.01687
0.78787	0.01810
0.77885	0.01925
0.77008	0.02032
0.76154	0.02128
0.75323	0.02221
0.74502	0.02309
0.73686	0.02389
0.72875	0.02469
0.72067	0.02544
0.71265	0.02618
0.70462	0.02690
0.69651	0.02755
0.68840	0.02821
0.68027	0.02883
0.67213	0.02946
0.66393	0.03005

0.28185	0.03994
0.27247	0.03983
0.26307	0.03969
0.25362	0.03960
0.24413	0.03947
0.23458	0.03931
0.22496	0.03914
0.21532	0.03886
0.20562	0.03854
0.19588	0.03821
0.18612	0.03781
0.17625	0.03743
0.16629	0.03696
0.15626	0.03645
0.14618	0.03593
0.13601	0.03550
0.12578	0.03510
0.11548	0.03475
0.10513	0.03440
0.09467	0.03415
0.08414	0.03399
0.07370	0.03369
0.06328	0.03330
0.05291	0.03263
0.04284	0.03141
0.03332	0.02959
0.02515	0.02704
0.01874	0.02416
0.01373	0.02120
0.00980	0.01815
0.00669	0.01499
0.00420	0.01162
0.00228	0.00799
0.00089	0.00410
0.00000	0.00000

j. NASA SC(2)-0714 Faster Long

Table 8-15 - List of the optimised coordinates for NASA SC(2)-0714 at M=1.8 (variable length)

NASA SC(2)-0714 Faster Long

x	y	x	y	x	y
0.00000	0.00000	0.77353	0.00589	0.66045	-0.07691
0.00124	0.00292	0.78374	0.00432	0.65201	-0.07860
0.00452	0.00589	0.79391	0.00264	0.64353	-0.08029
0.00988	0.00862	0.80407	0.00090	0.63501	-0.08191
0.01650	0.01108	0.81426	-0.00085	0.62647	-0.08343
0.02411	0.01321	0.82451	-0.00272	0.61789	-0.08486
0.03228	0.01511	0.83472	-0.00465	0.60923	-0.08621
0.04087	0.01684	0.84492	-0.00659	0.60050	-0.08747
0.04990	0.01833	0.85516	-0.00863	0.59171	-0.08865
0.05915	0.01973	0.86537	-0.01074	0.58292	-0.08973
0.06857	0.02104	0.87554	-0.01285	0.57412	-0.09072
0.07819	0.02219	0.88571	-0.01505	0.56530	-0.09161
0.08796	0.02325	0.89591	-0.01735	0.55644	-0.09242
0.09786	0.02422	0.90607	-0.01971	0.54757	-0.09312
0.10786	0.02519	0.91618	-0.02207	0.53871	-0.09373
0.11796	0.02609	0.92626	-0.02450	0.52987	-0.09424
0.12810	0.02690	0.93632	-0.02702	0.52109	-0.09465
0.13826	0.02761	0.94637	-0.02962	0.51232	-0.09505
0.14843	0.02832	0.95618	-0.03224	0.50358	-0.09536
0.15866	0.02896	0.96508	-0.03461	0.49485	-0.09567
0.16894	0.02958	0.97241	-0.03667	0.48615	-0.09588
0.17924	0.03015	0.97823	-0.03823	0.47746	-0.09609
0.18957	0.03063	0.98296	-0.03942	0.46880	-0.09620
0.19991	0.03108	0.98676	-0.04061	0.46016	-0.09631
0.21026	0.03149	0.98970	-0.04173	0.45154	-0.09633
0.22061	0.03184	0.99200	-0.04271	0.44294	-0.09634
0.23098	0.03222	0.99384	-0.04357	0.43437	-0.09626
0.24138	0.03255	0.99532	-0.04433	0.42581	-0.09618
0.25183	0.03283	0.99650	-0.04499	0.41727	-0.09600
0.26233	0.03309	0.99748	-0.04562	0.40874	-0.09581
0.27286	0.03328	0.99833	-0.04630	0.40023	-0.09553
0.28344	0.03350	0.99889	-0.04716	0.39172	-0.09525
0.29405	0.03367	0.99838	-0.04778	0.38324	-0.09487
0.30469	0.03379	0.99749	-0.04804	0.37475	-0.09449
0.31534	0.03388	0.99648	-0.04817	0.36628	-0.09401
0.32601	0.03390	0.99528	-0.04823	0.35780	-0.09353
0.33668	0.03394	0.99381	-0.04823	0.34933	-0.09295
0.34736	0.03395	0.99196	-0.04813	0.34083	-0.09237
0.35802	0.03388	0.98966	-0.04789	0.33232	-0.09169
0.36868	0.03383	0.98684	-0.04740	0.32375	-0.09101
0.37932	0.03375	0.98329	-0.04651	0.31518	-0.09022

0.38994	0.03359	0.97877	-0.04522	0.30656	-0.08943
0.40056	0.03344	0.97320	-0.04397	0.29794	-0.08854
0.41115	0.03330	0.96634	-0.04263	0.28928	-0.08765
0.42173	0.03313	0.95791	-0.04105	0.28063	-0.08667
0.43229	0.03288	0.94839	-0.03954	0.27191	-0.08569
0.44283	0.03266	0.93854	-0.03823	0.26310	-0.08460
0.45336	0.03239	0.92872	-0.03719	0.25421	-0.08341
0.46386	0.03205	0.91895	-0.03644	0.24526	-0.08212
0.47435	0.03174	0.90924	-0.03596	0.23624	-0.08084
0.48482	0.03138	0.89960	-0.03567	0.22712	-0.07947
0.49527	0.03095	0.89005	-0.03557	0.21787	-0.07790
0.50570	0.03054	0.88061	-0.03576	0.20842	-0.07624
0.51612	0.03009	0.87123	-0.03613	0.19874	-0.07447
0.52655	0.02957	0.86195	-0.03669	0.18881	-0.07262
0.53699	0.02906	0.85278	-0.03743	0.17857	-0.07049
0.54744	0.02852	0.84373	-0.03835	0.16795	-0.06819
0.55788	0.02790	0.83484	-0.03946	0.15680	-0.06563
0.56830	0.02730	0.82610	-0.04073	0.14511	-0.06270
0.57870	0.02665	0.81744	-0.04207	0.13296	-0.05947
0.58905	0.02595	0.80885	-0.04352	0.12114	-0.05605
0.59936	0.02522	0.80036	-0.04514	0.11041	-0.05273
0.60965	0.02441	0.79192	-0.04684	0.10067	-0.04957
0.61996	0.02362	0.78360	-0.04863	0.09176	-0.04654
0.63029	0.02279	0.77535	-0.05048	0.08317	-0.04352
0.64061	0.02189	0.76710	-0.05235	0.07438	-0.04032
0.65090	0.02096	0.75894	-0.05429	0.06524	-0.03689
0.66115	0.01998	0.75083	-0.05621	0.05580	-0.03327
0.67137	0.01896	0.74271	-0.05814	0.04627	-0.02949
0.68156	0.01788	0.73455	-0.06008	0.03691	-0.02563
0.69175	0.01677	0.72640	-0.06201	0.02815	-0.02180
0.70194	0.01560	0.71825	-0.06395	0.02035	-0.01811
0.71215	0.01439	0.71007	-0.06588	0.01373	-0.01462
0.72239	0.01311	0.70190	-0.06782	0.00843	-0.01138
0.73264	0.01181	0.69369	-0.06967	0.00448	-0.00839
0.74285	0.01041	0.68546	-0.07151	0.00176	-0.00556
0.75305	0.00895	0.67718	-0.07337	0.00024	-0.00280
0.76330	0.00745	0.66882	-0.07514	0.00000	0.00000

Appendix D: Default Values

Table 8-16 - Default values of the mesh() function

Meshing Function			
Parameter	**Default Value**	**Type**	**Description**
nickname	“”	String	User-specified identifier
airfoil	*not_specified*	String	Airfoil designation
show_gui	*False*	Boolean	Shows the GUI if True
processor_count	*4*	Integer	Number of logical cores used
precision	*None*	Object	Single vs Double Precision. If unspecified, evaluates to double
chord_len	*1*	Float	Chord length in metres. Standard airfoil scale
te_len	*0.005*	Float	Trailing edge size, in metres
shut_down_when_done	*True*	Boolean	Closes Fluent after meshing is finished. Set to False if the user wishes to inspect the result before closing
has_te	*False*	Boolean	Does the airfoil have a trailing edge? False = closed
upper_name	*upper*	String	Upper surface boundary label name
lower_name	*lower*	String	Lower surface boundary label name
te_name	*te*	String	Trailing edge boundary label name
farfield_name	*farfield*	String	Farfield boundary label name
symmetry_1_name	*sym1*	String	Symmetry plane 1 label name
symmetry_2_name	*sym2*	String	Symmetry plane 2 label name
Boi_1	*True*	Boolean	Add local sizing to upper/lower surfaces
Boi_1_Control_Name	*airfoil_max*	String	Name of the local sizing task
Boi_1_Execution	*Face Size*	String	Sizing type for local sizing task
Boi_1_Size	*None*	Float	Local sizing size. Evaluates to chord_len / 100 if unspecified
Boi_2_Control_Name	*airfoil_te*	String	Name of the local sizing task for the trailing edge
Boi_2_Execution	*Proximity*	String	Sizing type for trailing edge task
Boi_2_Min_Size	*None*	Float	Minimum size for trailing edge task

Boi_2_Cells_Per_Gap	2	Integer	Number of cells per gap, if the trailing edge task is set to proximity
Boi_2_Scope_To	edges	String	Scope proximity sizing to edges
Surface_Rate	1.2	Float	Growth rate of surface mesh
Surface_Min_Size	None	Float	Surface minimum size. If unspecified, evaluates to te_len / 2 if has_te=True, else to Bl_First_Height * 10
Surface_Max_Size	None	Float	Surface maximum size. If unspecified, evaluates to chord_len / 2
Surface_Size_Function	Curvature	String	Curvature-based mesh sizing
Surface_Curvature_Normal_Angle	12	Integer	Normal angle for curvature, in degrees
Bl_Control_Name	uniform	String	Boundary layer type
Bl_First_Height	0.00002	Float	First boundary layer height, in metres
Bl_Number_Of_Layers	20	Integer	Number of boundary layers
Bl_Rate	1.2	Float	Boundary layer growth rate
Volume_Fill_Type	polyhedra	String	Volume fill type
Volume_Fill_Size	None	Float	Volume fill size. If unspecified, evaluates to chord_len / 2
Volume_Fill_Rate	1.2	Float	Volume mesh growth rate
Units	m	String	Metres (standard SI unit)

Table 8-17 - Default values of the solve() function

Solution Function

Parameter	Default Value	Type	Description
nickname	''	String	User-specified identifier
airfoil	not_specified	String	Airfoil identifier
show_gui	False	Boolean	Shows the GUI if True
processor_count	4	Integer	Number of logical cores used
precision	None	Object	Single vs Double Precision. If unspecified, evaluates to double
shut_down_when_done	True	Boolean	Closes Fluent after the solve is finished. Set to False if the user wishes to inspect the result before closing

$report\_file$	$True$	Boolean	Exports the drag and lift reports if True
$report\_plot$	$True$	Boolean	Generates a plot of the drag and lift reports if True
$report\_convergence$	0.00001	Float	Convergence threshold for drag and lift reports
$altitude$	0	Integer	Altitude above sea level, in metres
$mach\_num$	0.6	Float	Mach number of free stream air
aoa	0	Integer	Angle of attack, in degrees
$chord\_len$	1	Float	Chord length in metres. Standard airfoil scale
$mesh\_width$	0.1	Float	Depth of mesh (in the z direction) in metres, used for calculating area
$convergence\_criteria$	0.0000001	Float	Convergence threshold for residuals
$use\_convergence\_criteria$	$False$	Boolean	Overwrites the default convergence values if True
$use\_report\_convergence$	$True$	Boolean	Uses the report convergence if True
$iterations$	300	Integer	Maximum number of iterations
$time\_step\_scale$	5	Float	Pseudo-time step time scale factor
$report\_1\_name$	C_d	String	Drag coefficient report name
$report\_2\_name$	C_l	String	Lift coefficient report name
$operating\_pres$	0	Float	Operating gauge pressure, in Pascals
$solver\_type$	$pressure - based$	String	Solver algorithm
$visc\_method$	$sutherland$	String	Viscosity calculation method
$visc\_model$	$spalart - allmaras$	String	Turbulence model
k_omega_model	$geko$	String	Default model if $k - \omega$ is used
$curvature\_correction$	$False$	Boolean	Applies curvature correction if True
$compressibility\_effects$	$False$	Boolean	Applies compressibility effects if True
$farfield\_name$	$farfield$	String	Farfield boundary label name (to ensure compatibility with all meshes)

$upper\_name$	$upper$	String	Upper surface label name
$lower\_name$	$lower$	String	Lower surface label name
$flux\_report\_name$	$flux\_report$	String	Name of the mass flow report
$specification\_method$	$Intensity\ and\ Viscosity\ Ratio$	String	Turbulence specification method
$turb\_int$	1	Integer	Turbulence intensity (%)
$turb\_visc\_ratio$	1	Integer	Turbulent viscosity ratio
$flow\_scheme$	$Coupled$	String	Pressure-velocity coupling method
$gradient\_scheme$	$least - square - cell - based$	String	Gradient computation method
$density\_scheme$	$second - order - upwind$	String	Density discretisation method
$turb\_kin\_e\_scheme$	$second - order - upwind$	String	Turbulent kinetic energy discretisation method
mom_scheme	$second - order - upwind$	String	Momentum discretisation method
$pres\_scheme$	$second - order$	String	Pressure discretisation method
$energy\_scheme$	$second - order - upwind$	String	Energy equation discretisation method
$spec\_diss\_rate\_scheme$	$second - order - upwind$	String	Specific dissipation rate discretisation method
mod_turb_visc	$second - order - upwind$	String	Modified turbulent viscosity discretisation method
$pseudo\_time\_method$	$global - time - step$	String	Pseudo time method
$hybrid\_initialize$	$True$	Boolean	Uses hybrid initialisation if True
$high\_order\_term\_relax$	$False$	Boolean	Applies high order term relaxation if True
$relaxation\_factor$	0.75	Float	Relaxation factor if high order term relaxation is used
$courant\_number$	200	Integer	Courant number for transient solves
$warped\_face\_gradient$	$False$	Boolean	Uses warped face gradient correction if True
$symmetry\_1\_name$	$sym1$	String	Symmetry plane 1 label name
$surface\_name$	$surface$	String	Surface boundary label name for 2D meshes
$contour\_lines$	$False$	Boolean	Shows contour lines on Mach contours and

			pressure contours if True
smooth_or_banded	*banded*	String	Smooth or banded contours
generate_mach_cont	*True*	Boolean	Generates Mach contours if True
generate_pres_cont	*True*	Boolean	Generates pressure contours if True
generate_yplus	*True*	Boolean	Generates the y^+ plot if True
generate_pres_plot	*True*	Boolean	Generates the pressure distribution plot if True
transient	*False*	Boolean	Specifies whether to use steady-state or transient solver
two_d_space	*planar*	String	2D solution formulation
time_step_count	*10*	Integer	Number of transient time steps
time_step_size	*0.0001*	Float	Transient time step size
iters_per_time_step	*20*	Integer	Iterations per time step
pause_before_solve	*False*	Boolean	Pauses before starting the solve. Useful for inspecting boundary conditions before starting
under_relaxation	*0.75*	Float	Under-relaxation factor for transient solvers

Table 8-18 - Default values of the optimise() function

Optimisation Function			
Parameter	**Default Value**	**Type**	**Description**
nickname	''	String	User-specified identifier
airfoil	*not_specified*	String	Airfoil identifier
contour_lines	*False*	Boolean	Shows contour lines on Mach contours and pressure contours if True
smooth_or_banded	*banded*	String	Smooth or banded contours
save_screenshots	*True*	Boolean	Saves screenshots between each adjoint iteration if True
transient	*False*	Boolean	Uses a transient solver if set to True
solver_type	$pressure - based$	String	Solver type for the final solution (adjoint uses pressure-based)
visc_model	$k - omega$	String	Final solution viscous model (adjoint uses $k - \omega$ GEKO)
show_gui	*True*	Boolean	Shows the GUI if True
processor_count	*4*	Integer	Number of logical cores used
upper_name	*upper*	String	Upper surface label name
lower_name	*lower*	String	Lower surface label name

aoa	0	Integer	Angle of attack, in degrees
$convergence\_criteria$	0.00001	Float	Convergence threshold for residuals
$use\_convergence\_criteria$	$True$	Boolean	Overwrites the default convergence values if True
use_energy	$False$	Boolean	Enables energy in the adjoint solver if True
$use\_turbulence$	$False$	Boolean	Enables turbulence effects in the adjoint solver if True
use_best_match	$False$	Boolean	Matches the adjoint flow settings to the flow settings from the solve
$apply\_preconditioning$	$False$	Boolean	Applies preconditioning if True
$target\_change$	10	Integer	Target design change per adjoint iteration
$use\_percentage$	$True$	Boolean	If True, target_change is treated as a percentage
$optimize\_target$	$lift - to - drag$	String	Observable to optimise
$mach\_num$	0.5	Float	Mach number of free stream air
$time\_step\_scale$	5	Float	Pseudo-time step time scale factor
$time\_step\_size$	0.001	Float	Transient time step size
$time\_step\_count$	100	Integer	Number of transient time steps
$iters\_per\_time\_step$	20	Integer	Iterations per time step
$iterations$	300	Integer	Iterations for the final solve
$infinite\_mode$	$False$	Boolean	Enables infinite mode if True. Infinite mode overrides optimization_loop_count and runs for an infinite number of adjoint iterations, but stops when mesh quality falls below threshold
$min\_orth\_quality\_limit$	0.06	Float	Minimum orthogonal quality for infinite mode. Stops when the mesh quality is below this number
$maintain\_len$	$True$	Boolean	Keeps the airfoil length constant if True. Otherwise, allows adjoint to change length
$adjoint\_iterations$	20	Integer	Adjoint solver iterations
$optimization\_loop\_count$	5	Integer	Number of times adjoint will iterate
$pseudo\_time\_method$	off	String	Pseudo time method
$temp\_iterations$	200	Integer	Temporary maximum number of iterations, in effect while adjoint is running
$temp\_time\_step\_scale$	$None$	Float	Temporary pseudo-time step time scale factor, in effect while the adjoint solver is running. If unspecified, evaluates to 0.001
$temp\_time\_step\_size$	$None$	Float	Temporary transient time step size, in effect while the adjoint solver is running. If unspecified, evaluates to 0.001
$temp\_time\_step\_count$	10	Integer	Temporary number of transient time steps, in effect while the adjoint solver is running

$temp\_iters\_per\_time\_step$	20	Integer	Temporary number of iterations per time step, in effect while the adjoint solver is running
$farfield\_name$	$farfield$	String	Farfield boundary label name
$shut\_down\_when\_done$	$True$	Boolean	Closes Fluent after optimisation is finished. Set to False if the user wishes to inspect the result before closing
$generate\_yplus$	$True$	Boolean	Generates the y^+ plot if True
$generate\_pres\_plot$	$True$	Boolean	Generates the pressure distribution plot if True
$report\_1\_name$	C_d	String	Drag report name
$report\_2\_name$	C_l	String	Lift report name
$report\_plot$	$True$	Boolean	Exports the drag and lift reports if True
$report\_file$	$True$	Boolean	Generates a plot of the drag and lift reports if True
$make\_gifs$	$True$	Boolean	Creates animated GIFs for the mesh, Mach contour, and pressure contour screenshots if True
$gif\_duration$	100	Integer	Duration of each frame of the GIF
$change\_boundary\_conditions$	$False$	Boolean	If True, changes the boundary conditions and re-solves
$altitude$	0	Integer	Altitude above sea level, in metres, for the changed boundary conditions
$mesh\_width$	0.1	Float	Depth of mesh (in the z direction) in metres, used for calculating area
$infinite\_mode\_max$	999	Integer	Maximum iterations in infinite mode. Useful if the mesh quality never falls below the threshold

Appendix E: Sample Code

All PyFluent and Python code used was uploaded to a GitHub repository, available at: https://github.com/Dzermann/pyfluent-airfoil-optimisation. The repository contains all scripts required to run the solver, as well as an example outputs and an input file.